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IETS TCP Task XXIV by IEA

Coordination of the Task "Process Integration for Industry Decarbonization" of the Technology Collaboration Programme (TCP) on Industrial Energy-Related Technologies and Systems (IETS).



Source: AI generated picture in Claude of a “collaborative framework of a shared database of ex-ante digital models”







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Task XXIV “Process integration for industry decarbonization” - Subtask 1 “Enabling a shared data-base of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics”

https://iea-industry.org/tasks/process-integration-for-industry-decarbonization/

The authors bear the entire responsibility for the content of this report and for the conclusions drawn therefrom.

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Summary

Task XXIV “Process Integration for Industry Decarbonization” of the IEA Industrial Energy-Related Technologies and Systems (IETS) Technology Collaboration Programme (TCP) started its first year of coordination activities in 2025. The Swiss Federal Office of Energy (SFOE) funds the coordination of this international initiative, which aims to develop a documented, open-source database of ex-ante models covering industrial processes, decarbonization technologies, and energy systems. This interim report documents the progress achieved during Year 1 across five main areas: ontology and procedure development, model interface definition, process integration documentation, sustainability metrics, and dissemination. The coordination activities began with the kick-off meeting in March 2025, followed by participant presentations in April and a working session on ontologies and procedures in May. These meetings established the rules for Task participation, distributed leading roles among contributors, and initiated discussions on model documentation requirements. Representatives from 13 countries engaged in Task XXIV activities, namely ten participating countries (Canada, Denmark, France, Germany, Italy, the Netherlands, Norway, Portugal, Sweden and Switzerland, which manages the Task) and three observer countries (Belgium, Chile and Colombia). During the second semester of 2025, bilateral meetings with individual participant countries addressed specific technical questions and facilitated alignment between Task objectives and ongoing national research projects.

A central achievement of Year 1 was the development of a simplified model documentation template. The template architecture balances comprehensiveness with practicality, capturing essential information for process integration analyses including metadata, mass and energy flows, documented parameters, decision variables, block flow diagrams, and supplementary materials. The framework adopts a software-agnostic approach that accommodates diverse modelling paradigms, from linear programming to metaheuristic optimization, and from sequential modular to equation-oriented structures. The survey distributed to participating countries provided critical insights that informed the database architecture. The analysis revealed a heterogeneous software landscape dominated by Python and commercial process simulators such as Aspen®, complemented by open-source alternatives and specialized in-house platforms. Confidential industrial data, governed by non-disclosure agreements, patents and co-owned models, emerged as the primary barrier to model sharing, followed by unpublished research data. The survey identified strong support for tiered access mechanisms and revealed country-specific priorities regarding open access, confidentiality, anonymization, and validation protocols. These inputs shaped the seven-tier layered disclosure framework, where Layers 0-2 are publicly accessible, Layers 3-4 are available on a case-by-case basis, and Layers 5-6 require bilateral agreements with embargo periods and co-authorship arrangements.

Dissemination activities established Task XXIV’s visibility within the international research community. Task managers delivered a keynote lecture at ESCAPE 2025; presented the Task activities to participants of Task XI Biorefineries at EUBCE 2025; and co-organized a joint workshop on industrial symbiosis with Task XI, as well as a Task XXIV special session on industrial decarbonization at IBBCE 2025. These exchanges with Task XI identified synergies in data standardization and technology assessment methodologies. The first annual in-person intra-task meeting of Task XXIV participants was also held during the IBBCE 2025 conference. Task XXIV also participated in a joint IETS-ETSAP workshop that addressed energy systems modelling. Task coordinators participated in the 40th and 41st IETS Executive Committee meetings and prepared annual reports for the IETS Secretariat and SFOE. Student semester and master projects and continuing education programs provided feedback on knowledge transfer mechanisms. Strategic partnerships with professional societies, including EFCE and ECOS, will support future dissemination. The data management plan addresses findability, accessibility, interoperability, and reusability through git-based version control likely hosted on Swiss servers. Anonymization techniques transform sensitive industrial data into shareable formats compatible with data protection regulations across participating countries. The project remains on track with no major deviations from schedule or budget. Year 1 delivered D1 (ontologies and procedures) and advanced D5 (confidentiality analysis) and D7 (dissemination). Year 2 will focus on completing the documentation tool (D2) and populating the database (D3), and will start demonstrating impact through participants’ case studies.

Zusammenfassung

Der Task XXIV „Prozessintegration zur Dekarbonisierung der Industrie“ des Technologie-Kooperationsprogramms (TCP) „Industrielle energiebezogene Technologien und Systeme“ (IETS) der Internationalen Energieagentur (IEA) hat 2025 sein erstes Koordinationsjahr begonnen. Das Bundesamt für Energie (BFE) finanziert die Koordination dieser internationalen Initiative, die eine dokumentierte, frei zugängliche Datenbank mit Ex-ante-Modellen industrieller Prozesse, Dekarbonisierungstechnologien und Energiesysteme aufbauen will. Dieser Zwischenbericht stellt die Fortschritte des ersten Jahres in den Bereichen Ontologien und Verfahren, Modellschnittstellen, Dokumentation der Prozessintegration, Nachhaltigkeitskennzahlen und Verbreitung dar. Auf das Auftakttreffen im März 2025 folgten Präsentationen der Teilnehmenden im April und eine Arbeitssitzung zu Ontologien und Verfahren im Mai. Diese Treffen legten die Teilnahmeregeln fest, verteilten die Führungsrollen und eröffneten die Diskussion über die Modelldokumentation. Beteiligt waren 13 Länder: zehn teilnehmende Länder (Kanada, Dänemark, Frankreich, Deutschland, Italien, Niederlande, Norwegen, Portugal, Schweden und die Schweiz, die die Koordination übernimmt) sowie drei Beobachterländer (Belgien, Chile und Kolumbien). In der zweiten Jahreshälfte 2025 behandelten bilaterale Treffen spezifische technische Fragen und stimmten die Ziele des Tasks auf die laufenden nationalen Forschungsprojekte ab.

Ein wesentliches Ergebnis des ersten Jahres ist eine vereinfachte Vorlage für die Modelldokumentation. Sie verbindet Vollständigkeit mit Praxistauglichkeit und erfasst die für Analysen der Prozessintegration wesentlichen Informationen, etwa Metadaten, Stoff- und Energieströme, dokumentierte Parameter, Entscheidungsvariablen, Blockfließbilder und ergänzende Unterlagen. Die Vorlage ist softwareunabhängig und eignet sich für verschiedene Modellierungsparadigmen, von der linearen Programmierung bis zur metaheuristischen Optimierung und von sequenziell-modularen bis zu gleichungsorientierten Ansätzen. Die Umfrage bei den teilnehmenden Ländern war für die Gestaltung der Datenbankarchitektur entscheidend. Sie zeigte eine heterogene Softwarelandschaft, die von kommerziellen Prozesssimulatoren dominiert und durch Open-Source-Lösungen und interne Plattformen ergänzt wird. Vorbehalte der Industriepartner und Geheimhaltungsvereinbarungen bilden das Haupthindernis für den Austausch von Modellen, gefolgt von zeitlichen und finanziellen Einschränkungen. Die Umfrage ergab zudem eine deutliche Unterstützung für gestufte Zugangsmechanismen sowie länderspezifische Prioritäten in Bezug auf Open Access, Vertraulichkeit, Anonymisierung und Validierungsprotokolle. Daraus entstand ein siebenstufiges Offenlegungskonzept: Die Stufen 0 bis 2 sind öffentlich zugänglich, die Stufen 3 und 4 auf Einzelfallbasis verfügbar, und die Stufen 5 und 6 erfordern bilaterale Vereinbarungen mit Embargofristen und Regelungen zur Ko-Autorenschaft.

Um die internationale Sichtbarkeit des Tasks XXIV zu sichern, hielten die Task-Verantwortlichen einen Hauptvortrag auf der ESCAPE 2025, stellten ihre Aktivitäten dem Task XI (Bioraffinerien) auf der EUBCE 2025 vor und organisierten auf der IBBCE 2025 gemeinsam einen Workshop zur industriellen Symbiose und eine Sondersitzung, in deren Rahmen auch das erste jährliche Präsenztreffen der Teilnehmenden stattfand. Der Austausch mit dem Task XI zeigte Synergien bei der Datenstandardisierung und den Methoden zur Technologiebewertung auf. Der Task XXIV beteiligte sich außerdem an einem gemeinsamen Workshop von IETS und ETSAP zur Modellierung von Energiesystemen. Die Koordinatoren nahmen an der 40. und 41. Sitzung des IETS-Exekutivkomitees teil und erstellten die Jahresberichte für das IETS-Sekretariat und das BFE. Semester- und Masterarbeiten sowie Weiterbildungsprogramme dienten der Evaluation des Wissenstransfers. Partnerschaften mit EFCE und ECOS werden die künftige Verbreitung unterstützen. Der Datenmanagementplan zielt mit einer Git-basierten Versionskontrolle auf Auffindbarkeit, Zugänglichkeit, Interoperabilität und Reproduzierbarkeit ab. Anonymisierungsverfahren machen sensible Industriedaten unter Einhaltung der Datenschutzbestimmungen austauschbar. Das Projekt verläuft planmäßig und im Rahmen des Budgets. Im ersten Jahr wurde das Ergebnis D1 (Ontologien) erarbeitet, und die Ergebnisse D5 (Analyse der Vertraulichkeit) und D7 (Verbreitung) schritten voran. Das zweite Jahr ist der Befüllung der Datenbank (D3), der Fertigstellung des Dokumentationstools (D2) und dem Beginn der Fallstudien der Teilnehmenden gewidmet.

Résumé

La « Task XXIV », intitulée « Intégration des procédés pour la décarbonation industrielle » et relevant du programme de collaboration technologique (TCP) « Technologies et systèmes liés à l'énergie industrielle » (IETS) de l'Agence internationale de l'énergie (AIE), a entamé sa première année de coordination en 2025. L'Office fédéral de l'énergie (OFEN) finance la coordination de cette initiative internationale, qui vise à développer une base de données documentée et en accès libre regroupant des modèles ex ante de procédés industriels, de technologies de décarbonation et de systèmes énergétiques. Ce rapport intermédiaire présente les progrès de la première année en matière d'ontologies et de procédures, d'interfaces de modèles, de documentation de l'intégration des procédés, d'indicateurs de durabilité et de diffusion. Les activités ont débuté par une réunion de lancement en mars 2025, suivie de présentations des participants en avril et d'une session de travail consacrée aux ontologies et aux procédures en mai. Ces réunions ont établi les règles de participation, réparti les rôles de pilotage et engagé les discussions sur la documentation des modèles. Des représentants de 13 pays ont pris part aux activités : dix pays participants (Canada, Danemark, France, Allemagne, Italie, Pays-Bas, Norvège, Portugal, Suède et Suisse, qui assure la coordination) et trois pays observateurs (Belgique, Chili et Colombie). Au second semestre 2025, des réunions bilatérales ont traité des questions techniques spécifiques et aligné les objectifs de la « Task » sur les projets de recherche nationaux.

Une réalisation majeure de la première année est l'élaboration d'un modèle simplifié de documentation. Ce modèle concilie exhaustivité et praticité et recueille les informations essentielles aux analyses d'intégration des procédés, telles que les métadonnées, les flux de matière et d'énergie, les paramètres documentés, les variables de décision, les diagrammes de blocs et la documentation complémentaire. Indépendant des logiciels utilisés, il s'adapte à divers paradigmes de modélisation, de la programmation linéaire à l'optimisation par métaheuristiques, et des structures modulaires séquentielles aux approches fondées sur des équations. L'enquête menée auprès des pays participants, déterminante pour concevoir l'architecture de la base de données, a révélé un paysage logiciel hétérogène, dominé par les simulateurs de procédés commerciaux et complété par des solutions en accès libre et des plateformes internes. Les objections des partenaires industriels et les accords de non-divulgation constituent le principal obstacle au partage des modèles, suivis par les contraintes de temps et de ressources financières. L'enquête a également mis en évidence un soutien marqué aux mécanismes d'accès gradués, ainsi que des priorités propres à chaque pays concernant le libre accès, la confidentialité, l'anonymisation et les protocoles de validation. Ces éléments ont permis d'élaborer un cadre de diffusion structuré en sept niveaux : les niveaux 0 à 2 sont accessibles au public, les niveaux 3 et 4 au cas par cas, et les niveaux 5 et 6 nécessitent des accords bilatéraux assortis de périodes d'embargo et de dispositions relatives à la co-rédaction.

Pour assurer la visibilité internationale de la « Task XXIV », ses responsables ont prononcé une conférence plénière lors d'ESCAPE 2025, présenté leurs activités à la « Task XI » (bioraffineries) lors de l'EUBCE 2025, et co-organisé un atelier sur la symbiose industrielle ainsi qu'une session spéciale lors de l'IBBCE 2025, où s'est également tenue la première réunion annuelle en présentiel des participants. Ces échanges avec la « Task XI » ont mis en évidence des synergies en matière de normalisation des données et de méthodologies d'évaluation technologique. La « Task XXIV » a participé à un atelier conjoint de l'IETS et de l'ETSAP sur la modélisation des systèmes énergétiques. Les coordinateurs ont participé aux 40e et 41e réunions du comité exécutif de l'IETS et préparé les rapports annuels pour le secrétariat de l'IETS et l'OFEN. Des projets semestriels et de master, ainsi que des programmes de formation continue, ont permis d'évaluer les mécanismes de transfert de connaissances. Des partenariats avec l'EFCE et l'ECOS soutiendront les futures actions de diffusion. Le plan de gestion des données vise la repérabilité, l'accessibilité, l'interopérabilité et la reproductibilité grâce à un contrôle de version. Des techniques d'anonymisation rendent les données industrielles sensibles partageables d'après la protection des données. Le projet respecte le calendrier et le budget. La 1ere année a produit le livrable D1 (ontologies) et fait progresser les livrables D5 (analyse de la confidentialité) et D7 (diffusion). La 2ème année sera consacrée au peuplement de la base de données (D3), à la finalisation de l'outil de documentation (D2) et au début des études de cas menées par les participants.

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List of abbreviations

ACT	Activity
AMPL	A Mathematical Programming Language
ARC	Advanced Recycling Conference
BFE	Bundesamt für Energie (Swiss Federal Office of Energy)
CAPE-OPEN	Computer-Aided Process Engineering - Open Simulation Environment
CBC	COIN-OR Branch and Cut (solver)
CIRAIG	International Reference Centre for the Life Cycle of Products, Processes and Services
CPLEX	IBM ILOG CPLEX Optimization Studio
D1-D7	Deliverables 1 through 7
DEXPI	Data Exchange in the Process Industry
DTU	Technical University of Denmark
DWSIM	Open-source chemical process simulator
EC	European Commission
ECOS	Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems
EFCE	European Federation of Chemical Engineering
EPFL	École Polytechnique Fédérale de Lausanne
ESCAPE	European Symposium on Computer Aided Process Engineering
ETSAP	Energy Technology Systems Analysis Programme
EU	European Union
EUBCE	European Biomass Conference and Exhibition
ExCo	Executive Committee
FAIR	Findability, Accessibility, Interoperability, Reproducibility
FMI	Functional Mock-up Interface
FOSSEE	Free and Open Source Software for Education
GAMS	General Algebraic Modelling System
GHGT	Greenhouse Gas Control Technologies Conference
GLPK	GNU Linear Programming Kit
IBBCE	International Conference on Bioenergy and Bioeconomy
IEA	International Energy Agency
IETS	Industrial Energy-related Technologies and Systems
IP	Intellectual Property
IPESE	Industrial Process and Energy Systems Engineering

ISO	International Organization for Standardization
JSON	JavaScript Object Notation
LAPSE	Library of Academic Process Systems Engineering
LLM	Large Language Model
MATLAB	Matrix Laboratory software
MILP	Mixed-Integer Linear Programming
OntoCAPE	Ontology for Computer-Aided Process Engineering
OSMOSE	OptimiSation Multi-Objectifs de Systèmes Énergétiques intégrés
PCCC	Post-Combustion Capture Conference
PSE	Process Systems Engineering
Q1-Q4	Quarter 1 through Quarter 4
rSOC	Reversible Solid Oxide Cell
SFOE	Swiss Federal Office of Energy
TCCS	Trondheim Carbon Capture and Storage Conference
TCP	Technology Collaboration Programme
TRNSYS	Transient System Simulation Tool
VBA	Visual Basic for Applications

Nomenclature

Symbol	Description	Unit
Latin symbols		
$\dot{E}E_{u,i}^{+}$, $\dot{E}E_{u,i}^{-}$	Electricity input (+) and output (−) of unit u at interface i	kW
f_u	Load factor (capacity utilization) of unit u	–
$f_{u,i}$	Load factor of interface i of unit u	–
$F_{M,u,i}$, $F_{Q,u,i}$, $F_{EE,u,i}$, $F_{Eq,u,i}$	Thermo-physical model (constitutive equations) returning mass flows, heat demands, electricity demands and equipment sizes, respectively	–
$\dot{M}_{m,u,i}^{+}$, $\dot{M}_{m,u,i}^{-}$	Inlet (+) and outlet (−) flow of material m of unit u at interface i	kg/h
$\dot{M}_{F,i}^{+}$	Inlet feedstock flow at interface i	kg/h
$\dot{M}_{r,i}^{+}$, $\dot{M}_{r,i}^{-}$	Inlet (+) and outlet (−) support material, energy and waste flows at interface i	kg/h
$\dot{M}_{P,i}^{-}$	Outlet product flow at interface i	kg/h
$\dot{M}_{e,i}^{-}$	Elementary flow to the environment at interface i	kg/h
$\dot{m}_{Eq,e,i}^{-C}$, $\dot{m}_{Eq,e,i}^{-U}$, $\dot{m}_{Eq,e,i}^{-EoL}$	Elementary flows per unit of equipment size related to construction, lifetime auxiliary flows (use) and end of life	kg/h per unit of equipment size
$\dot{Q}_{T,u,i}^{+}$, $\dot{Q}_{T,u,i}^{-}$	Heating and cooling demands of unit u at temperature level T (in K) and interface i (heat cascade contributions)	kW
S_u^{*}	Nominal (reference) size of unit u	model-specific (e.g. kg//h, kW)
$S_{Eq,u,i}$	Size of the equipment Eq of unit u at interface i	equipment-specific (e.g. m², kW, m³)
x	Composition of the input flows; $(x \cdot \dot{M})_{m,u,i}^{+}$ denotes the input flows with their composition	kg/kg or mol/mol
y_u	Binary variable defining the existence of unit u	–
$y_{u,i}$	Binary variable defining the existence of interface i of unit u	–
Greek symbols		
$\pi_{u,i}$	Physical constants and operating conditions of unit u at interface i	various
Indices, subscripts and superscripts		
u	Process unit	–
i	Interface (level of detail) of a unit	–
m	Material (stream)	kg/h
T	Temperature level	K
e	Elementary flow to environment	kg/h, kW
Eq	Equipment	–
F, r, P	Feedstock, support resources (material, energy, waste), products	–
+, −	Inlet (input) and outlet (output) flows	–
C, U, EoL	Construction, use (lifetime auxiliary flows), end of life of the equipment	–

Symbols used in the formulation of the ex-ante model (Appendix B, Figure B2).

1 Introduction

1.1 Context and motivation

Process integration has long been a key approach for improving the energy performance of industrial systems. Most recently, its classical focus on heat recovery needed to be expanded toward a systemic dimension encompassing energy efficiency and conversion, carbon capture and sequestration, industrial symbiosis, urban systems integration, waste management, power-to-gas pathways, and renewable energy grids. Yet, many industries remain unaware of the improvement potential in their own plants, mainly due to the limited access to information on process integration techniques and decarbonization technologies, insufficient financial resources and lack of skilled personnel to conduct thorough energy audits. Existing tools for energy auditing and process integration analysis are often proprietary or, when open-source, they lack validated information about technological solutions. This fact hinders the assessment of viable decarbonization roadmaps and delays the energy transition. On top of this, the results from collaborative research projects between academic institutes and industries are typically presented as poorly documented, static data in journals and reports, which makes them difficult to compare, reproduce, and validate. Accordingly, the challenge lies not merely in gathering and sharing data, but in developing a structured framework to facilitate the knowledge transfer and build trust with industrial partners.

To address these challenges, Task XXIV "Process Integration for Industry Decarbonization" was launched in 2025 within the Technology Collaboration Programme on Industrial Energy-Related Technologies and Systems (IETS TCP) of the International Energy Agency, with its coordination by EPFL funded by the Swiss Federal Office of Energy (SFOE). The project directly addresses key priorities outlined in the SFOE research concept for industrial processes, aligning with three themes of strategic importance: process integration and waste heat utilization, decarbonization of industrial value chains, and renewable energy integration in industrial processes. These priorities are critical to supporting Switzerland's transition toward a low-carbon, energy-efficient industrial sector.

1.2 Task XXIV and Subtask 1 structure description

Task XXIV "Process Integration for Industry Decarbonization" of the IEA Technology Collaboration Programme on Industrial Energy-Related Technologies and Systems (IETS TCP) runs from January 2025 to December 2027 and is managed by Dr. Daniel Flórez-Orrego and François Maréchal (EPFL). Its overall objective is to create a model-database sharing framework that leverages the role of process integration for decarbonizing the industrial sectors and to assess its impact on the energy transition [1]. The Task comprises a single subtask, Subtask 1, "Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics", which is led by EPFL.

The Subtask 1 fosters international collaboration and aims to elucidate symbiotic operations, identify opportunities for waste heat and material upgrade in industrial and urban systems, and clarify the technical and environmental constraints that prevent integration strategies in the practice. In the context of growing industrial decarbonization efforts, such a validated database, along with rigorous process integration frameworks featuring sustainable metrics, will allow the assessment of innovative integrated concepts involving circularity, CO_2 capture, electrification, and renewable energy management.

Subtask 1 is structured in the following activities (Figure 1):

- **ACT1 Specifications, ontologies and procedures:** rules, templates and procedures for building, sharing, validating and maintaining the database of ex-ante models.
- **ACT2 Industrial sectors:** sector-based ex-ante process integration models, including their data, documentation and validation.

- **ACT3.1 Decarbonization technologies:** ex-ante technology integration models, including their data, documentation and validation.
- **ACT3.2 Energy systems:** integrated assessment and energy system models that provide the energy infrastructure interfaces.
- **ACT4 Process integration (heat and mass):** integration of process, technology and energy infrastructure models through their interfaces.
- **ACT5 Sustainability metrics:** indicators used to assess and compare integration options and decarbonization pathways.
- **ACT6 Roadmaps, use cases and dissemination:** roadmaps and recommendations, use cases per sector and per technology, knowledge transfer, dissemination and continuing education.

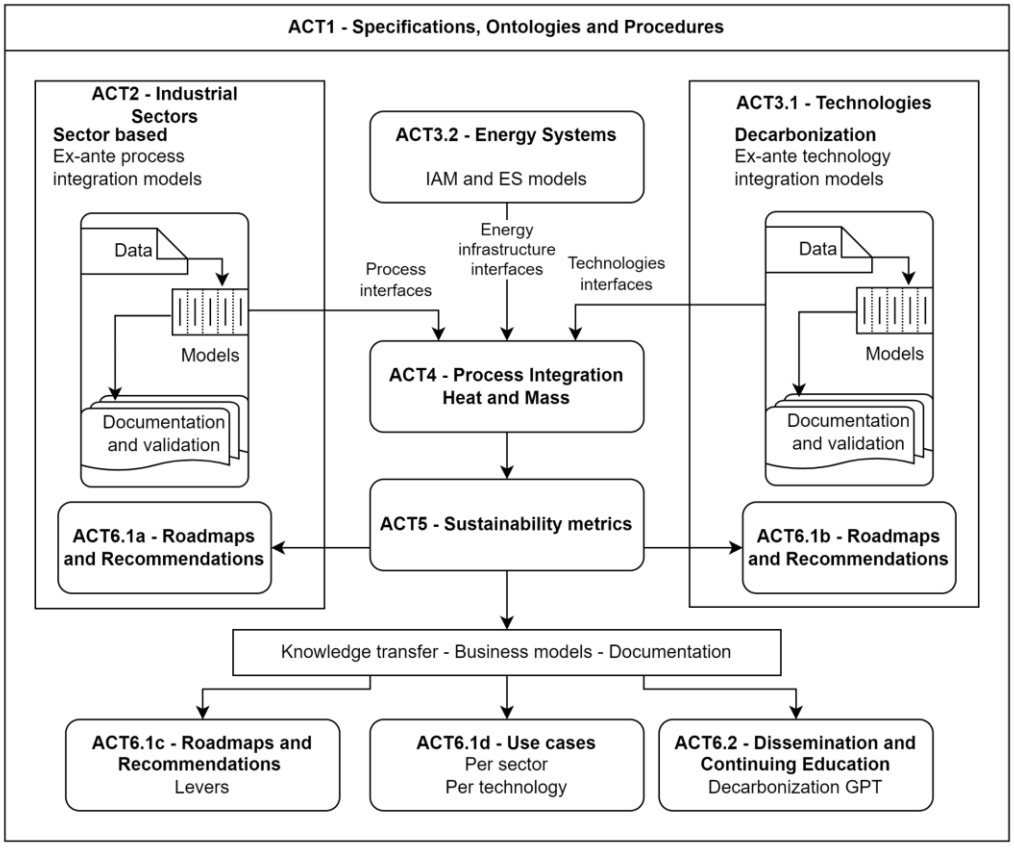


Figure 1: Structure of the Activities (formerly, Working Packages) of the IETS Task XXIV “Process integration for industry decarbonization” - Subtask 1 “Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics”.

The Task managers at EPFL build on their experience from previous international collaborative projects in energy auditing, process integration and optimization (Table 1), which also provided insights into the challenges of model sharing, documentation, validation and reporting. The accessibility, reproducibility and interoperability of reliable documented models of existing and emerging technological solutions and industrial sectors were central to the success of these projects. Building on these lessons, Task XXIV draws on the collective experience of researchers and industry partners across the participating countries. Transparent protocols for confidentiality and model validation protect sensitive information while promoting industrial symbiosis and delivering actionable decarbonization roadmaps.

Table 1: International collaborative projects that provided experience in process integration and optimization, models sharing, documentation, validation, and reporting.

Project	Description
AIDRES - Advancing industrial decarbonization by assessing the future use of renewable energies in industrial processes [2].	Generation of a database to support EU long-term goals of achieving a fully integrated industrial strategy. It is a valuable resource for industries that offers insights into effectiveness, efficiency, and cost of potential innovation pathways to achieve carbon neutrality in key sectors, such as steel, chemical, cement, glass, fertilizer, and refinery by 2050.
SCCER EIP – Efficiency of industrial processes [3].	Comprehensive techno-economic information of existing and new technologies, new integration methods and tools on company and/or site level to facilitate the application of energy efficiency measures and novel technologies.
SCCER BIOSWEET – Biomass for Swiss energy future [4]	Development and implementation of concepts of biomass valorization technologies that promote alternative energy carriers for mobility or heat and power applications in order to support the energy policy and market development.
EPOS – Enhanced energy and resource efficiency and performance in process industry operations via onsite and cross-sectorial symbiosis [5].	Development of a cross-sectorial Industrial Symbiosis (IS) platform for identifying cluster opportunities, bringing together global process industries from five industrial sectors, namely minerals, steel, cement, chemicals, and engineering. The development of blueprint models allowed to characterize in a simplified way the material and energy needs of the energy conversion systems.
METHAREN - Innovative biomethane system integration boosting production while managing renewable energies intermittency [6,7]	Optimization of biomethane production from biogenic CO_2 to increase the overall production capacity of biomethane, while reducing overall production costs. Demonstrated the system efficiency to manage renewable energy systems intermittency by transforming electrons into biomethane as a flexible renewable energy carrier, using rSOC systems.
NETZEROLAB VALAIS – Enabling carbon neutral production at Novelis aluminium plant in Sierre Switzerland [8,9]	A systemic study has been launched aiming at increasing the inherent efficiency of the process units and reusing waste heat of aluminium plants, as well as increasing the use of renewable energy resources. Different key performance indicators, such as energy efficiency, CO_2 emissions, costs, and technology readiness levels, are used to evaluate various decarbonization technologies.
IEA ANNEX 58 HTHP – Integration of High-Temperature Heat Pumps in the Swiss Industry [10]	Overview of the technological possibilities and applications as well as to develop concepts and strategies for the transition towards heat pump process heat supply. Improve the understanding of the technology potential among manufacturers, potential end-users, consultants, energy planners and policy makers.
MONTHEY - Decarbonization of the Industrial Ecosystem of Chablais [11]	Application of advanced process modeling and optimization methods to the industrial ecosystem in Monthey (VS), Switzerland, revealing untapped industrial symbiosis opportunities arising from complex interactions between the chemical site, incinerator, utility plant, and urban systems. These entities can form a local energy community with coordinated self-consumption targets.
MASBOOK - Continuous training program: process engineering for decarbonization of the Swiss industry [12]	Development of modeling, reporting and documentation tool to facilitate the definition, validation and sharing processes of ex ante models used in the Master of Advanced Studies in Sustainable Energy Systems Engineering. This approach facilitates reporting and reproducibility of the pinch and energy integration analyses applied to real industrial case studies.

1.3 Project objectives

The main objectives of the Task XXIV managed by the Swiss team at EPFL are:

1) Develop specifications and procedures for generating, validating, and sharing process and energy integration models.
2) Share flexible ex-ante (blueprint) models of key industrial sectors.
3) Share energy technology models to promote renewable energy use in industry.
4) Share energy systems models, including distribution networks and seasonal demand data.
5) Compare and document process integration methods/tools used by the participants for assessing the decarbonization pathways in industry.
6) Compare process integration results to outline roadmaps of industrial decarbonization and symbiosis.
7) Share findings and experiences through workshops, scientific events, and educational programs.

The project targets and achievement indicators are described below:

- Coordination of Task XXIV - Process Integration for Industry Decarbonization: Coordination and preparation of the bi-annual, annual and final reports; organization of Task meetings and internal workshops; presentation of results in conferences, IETS workshops and executive committees.
- Development of the specifications and procedures for creating, validating, and sharing process and energy technology integration models for key industrial sectors: Open-access database of key industrial processes and energy technology models featuring performance indicators. Detailed description of their potential for advancing industrial decarbonization.
- Comparison and documentation of process integration methods and tools used by Task participants to assess decarbonization pathways in industry: Documented list of the available open-source tools and methods for analyzing industrial process integration and decarbonization pathways.
- Share findings and experiences through workshops, scientific events, and educational programs: Reporting sessions to Swiss Federal Office of Energy (SFOE), as well as dissemination in conferences, IETS workshops, bi-annual, annual and final reports, and continuing education programs (Master of Advanced Studies of EPFL).
- Interaction with ongoing projects to facilitate sharing models and experiences on decarbonization roadmaps of Swiss industry and energy systems: Structured strategy for attracting collaborators to share models and experiences resulting of ongoing Swiss decarbonization projects.

2 Approach, method, results and discussion

2.1 Approach

According to Figure 2, industrial processes and technologies operate within a complex context, in which they (i) draw resources from and discharge emissions to the ecosphere; (ii) connect to shared infrastructure, such as electrical grids, district heating networks, and supply chains within the technosphere; and (iii) function under regulatory frameworks that impose constraints on emissions, impact and resource utilization. The *background* characteristics of the broader energy system (e.g. grid carbon intensity, ambient conditions, infrastructure, market prices, regulations) influence the *foreground* operational decisions within each industrial process.

An ex-ante model is a mathematical representation of a physical industrial unit, technology or system, where material and energy transformations occur. Unlike an ex-post model, which is derived from measured operational data; an ex-ante model is built from first principles, design specifications, and engineering correlations to predict system behaviour prior to implementation or during early design phases. This predictive character makes an ex-ante model essential for technology assessment, process synthesis, and strategic planning in industrial decarbonization contexts.

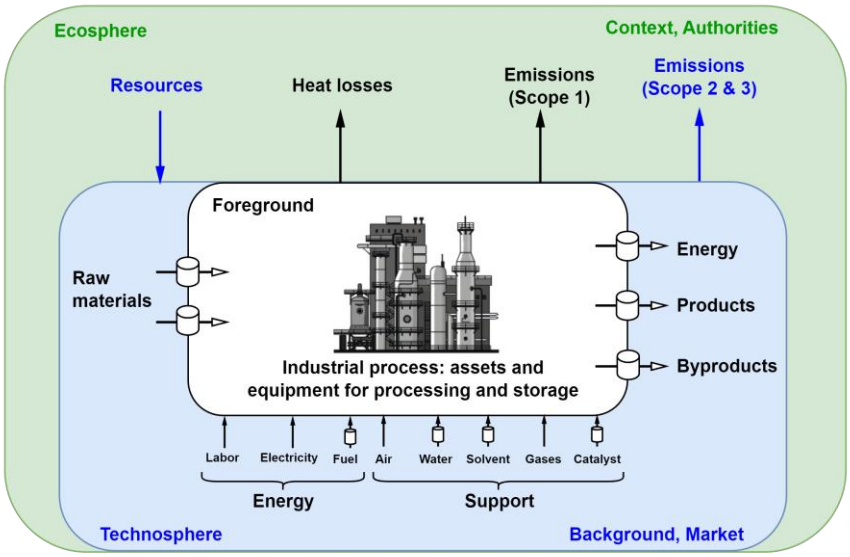


Figure 2: Representation of the industrial process and its relationship with technosphere and ecosphere. The background influences the selection of operating conditions in the foreground, which works under uncertain market conditions and regulatory contexts imposed by authorities [13].

An ex-ante model is internally governed by conservation laws (mass, energy, and component), constitutive equations (property correlations, reaction kinetics, transport phenomena), and decision parameters (design specifications, operating ranges, and set points). Yet, its practical value for process integration stems from its ability to interface with other models, infrastructure systems and the environment. In this sense, an ex-ante model can be also understood as a bounded entity that exchanges mass, energy, heat, and information flows with its surroundings through *connectors*, which serve as interface points between the model internal domain and the broader industrial ecosystem. In other words, to enable a systematic and consistent analysis of its interactions with the technosphere (anthropogenic systems) and the ecosphere (natural environment), an ex-ante model must capture not only the internal transformations, but also the quality attributes of its boundary flows (i.e. temperature, pressure, composition, temporal profile, and impact).

Hence, a formal definition of ex-ante model structure is crucial to enable stakeholders (e.g., researchers, industry practitioners, and policy makers) to share and communicate unambiguously about process engineering models. An ontological framework provides this structure by defining classes of entities, and their attributes, relations and axioms (i.e., logical constraints) [14]. By specifying what constitutes a model unit, stream, or engineering variable, a standardized definition enables participants to map their internal representations to a shared framework. It facilitates information exchange and discoverability, as well as automation of model quality check and validation across organizations. This is particularly critical for international collaborative efforts in which participants with distinct modelling paradigms and tools must contribute to a common database. This is a key goal of IETS Task XXIV "Process Integration for Industry Decarbonization", whose structure is shown in Figure 1. The Activity 1 (ACT 1) in Subtask 1, primarily conducted in 2025, focused on establishing simplified ontologies[1], specifications[2], and procedures[3] required for building, sharing, validating and maintaining a collaborative database of ex-ante models of industrial processes, and energy technologies and systems.

Selecting an ontology requires a balance between expressiveness, needed to capture the richness of process integration applications, and tractability, needed for practical implementation and long-term adoption. Existing ontologies (OWL-based frameworks, OntoCAPE, PROCO) and software interoperability standards (FMI, CAPE-OPEN, DEXPI) were therefore reviewed (Table 2; the detailed review and its references are given in Appendix A). These frameworks offer comprehensive coverage but entail significant adoption costs, and several of their features, such as compiled code, simulation APIs, dynamic co-simulation or P&ID-level detail, are not required for process integration. Task XXIV therefore adopts selected concepts from each of them in a simplified template that can be edited in human-readable formats such as spreadsheets or JSON files.

Table 2: Ontologies and interoperability standards reviewed, and elements adopted by Task XXIV (details and references in Appendix A).

Framework	Type	Main scope	Elements adopted by Task XXIV
OWL	Ontology language	Machine-readable classes, properties and relations	Class-attribute logic only; not adopted as a language (steep learning curve, complex n-ary relations)
OntoCAPE	Process engineering ontology (800+ entities)	Chemical processes, plant structure, materials, mathematical models	Definitions of process units, streams and thermodynamic properties
PROCO	Community ontology for process chemistry	Product quality and compliance	Principles of community governance and iterative refinement
FMI	Co-simulation and model exchange standard	Dynamic simulation of energy systems	Description of interfaces and parameters; compiled code, APIs and binaries not required
CAPE-OPEN	Software interface standard (CO-LaN)	Steady state flowsheeting components	Material, energy and information ports; public and private parameters
DEXPI	Plant data exchange standard	Plant breakdown, equipment taxonomy, P&ID topology	Plant breakdown structure (ISO 10209) and relevant classes

2.2 Method

Formal ontologies and heavyweight standards offer comprehensive coverage but impose significant adoption costs, requiring extensive training and infrastructure that may not be justified for process integration applications. Although lightweight ontologies may lack the precision needed for complex modelling scenarios, such as dynamic simulation or control-based modelling, a pragmatic approach can draw

[1] Terminology, Concepts, Variables, Objects, Attributes and Relations thereof.
[2] Syntax, Metadata format, Data structure, Model layout, and Connector consistency.
[3] Sharing, Documentation, Maintenance (Housing/Validation/Versioning), Authorship and Confidentiality protocols.

on existing ontological and standardization work, adapting it to the specific requirements of process integration and industrial decarbonization assessments. In practice, domain ontologies for specific applications tend to adopt simpler purpose-driven approaches, provided that they effectively follow FAIR principles (Findability, Accessibility, Interoperability, and Reusability) for scientific and industrial data management [15–18]. Accordingly, IETS Task XXIV adopts a simplified model structure template designed to capture essential information for process integration analyses while remaining accessible to partners with diverse modelling backgrounds and software tools. This approach favours clarity and implementability over ontological completeness, while maintaining sufficient rigour for meaningful inter-institutional model exchange.

The resulting template is organized as a class-attribute tree with six primary entities: METADATA (status, attribution, reference, documentation, confidentiality and disclosure layer), CHANGELOG (version tracking), SUPPL MATERIAL (auxiliary files), VARIABLES (engineering parameters classified by type and user grade), UNIT (connectors, heat streams, equipment and operating costs of each process unit) and CALCULATIONS (derived quantities). Each process unit is characterized by a nominal size, a binary variable defining its existence and a continuous load factor, and it exchanges mass, electricity, heat and information with the external superstructure through connectors that reference a global connectors database. This structure supports black-, grey- and white-box representations with consistent mass and energy balances, and it provides the heat cascade, cost and life cycle information required for process integration analyses. The mathematical formulation, the complete class-attribute tree, worked examples of each template section and the planned extensions of the framework (multi-time and pseudo-transient operation, energy infrastructure models and automated reporting) are presented in Appendix B; the editable template is supplied as Annex 4.

In 2026, Task participants shall populate the proposed ex-ante model template with their own data, documenting models to comply with requirements for reproducibility, validation, and publication. A centralized repository of ex-ante models facilitates data discoverability, visualization, and maintenance, engaging practitioners to contribute to exchange with validation protocols. In this way, industries can mitigate the lack of open-access technical information. Naturally, as agreed in bilateral and inter-task meetings, data confidentiality must be assessed case-by-case, enabling models to be shared across participants and industry at different levels of detail. Beyond industrial processes, ex-ante models of technologies spanning solar and biomass energy conversion, power-to-gas, heat electrification, carbon capture, storage and utilization, and thermal cycles can be shared and validated together with technology suppliers, who may use the model database as a showcasing platform for existing and emerging technologies. Validation through previous case studies and industrial experiences of Task participants will add robustness to the collection of industrial process and energy technology models.

2.3 Key results and achievements in Y2025

This section summarizes the measurable outcomes of the first year of Task XXIV coordination (March to December 2025). The coordination activities behind these results are described in Section 2.4 and in Chapters 4 and 5, while the technical background of the ontology review and of the ex-ante model template is provided in Appendices A and B. Table 3 summarizes the key performance indicators and Table 4 the status of deliverables D1 to D7.

2.3.1. Main outcomes and contributions to deliverables

- **Simplified ex-ante model template or schema, and documentation formats (D1, D2):** Software-agnostic template or schema with six primary entities (metadata, change log, supplementary material, variables, units and calculations) accommodates black-, grey- and white-box models and captures data required for process integration (see Section 2.2, Appendix B and Annex 4). Documentation formats have been pre-screened.
- **Model inventory (D3):** Ten models proposed by participants as starting points for the shared database (Table 7), and an EPFL IPESE list of 43 industrial process and 33 technology model entries to be formatted with the template (Figure 4).

- **Layered disclosure framework (D5):** Seven access layers (L0 to L6), ranging from public metadata to full model details, with case-by-case or bilateral access for the most sensitive layers (Table 8 and Figure 5).
- **Evidence base from the participants (D5, D6):** a 57-question survey with 10 responses from institutions in nine countries and bilateral meetings with ten country consortia. Industrial confidentiality (non-disclosure agreements, patents, co-owned models) was identified as the main barrier to model sharing (Figure C9).
- **Reporting, network and visibility (D2, D7):** One keynote lecture (ESCAPE 35), two events co-organized at IBBCE 2025, eight journal articles and 22 conference contributions by Task participants. Annual reports to SFOE and IETS.

2.3.2. Summary of key performance indicators Y2025

Table 3: Key performance indicators of Task XXIV at Y2025.

Indicator	Products	Source in this report
Participating countries	10 (CA, CH as Task manager, DE, DK, FR, IT, NL, NO, PT, SE)	Chapter 4
Observer and interested countries	3 observer countries (BE, CL, CO); participation of Austria not defined	Chapter 4
Registered contributors	54 researchers from 25 institutions, including observers and interested partners	Table 10
Active contributors in 2025	15 participant presenters at the first in-person meeting	Table C1, Appendix C
Intra-task plenary meetings	4 meetings: kick-off Task structure and rules (14 March 2025), participant presentations (11 April 2025), ontologies and procedures (23 May 2025), first in-person meeting in Lisbon (5 December 2025)	Section 2.4; Figure 3
Bilateral meetings	With 10 country consortia (September to November 2025); address specific technical questions and coordination needs; alignment between Task objectives and participants' ongoing national projects, which fund their contributions	Section 2.4
IETS Executive Committee meetings	2 (ExCo 40th, 15 May 2025; ExCo 41st, 12 to 13 November 2025)	End of Chapter 5
Inter-task and joint events	5 events: IETS-ETSAP workshop; Task XI special session and Task XI intra-task meeting (EUBCE 2025); Task XV workshop; Joint Task XI and XXIV workshop on industrial symbiosis (IBBCE 2025) demonstrated capacity to lead inter-task initiatives at the intersection of bioeconomy, energy systems and industrial decarbonization	Chapter 4
Events (co-)organized by Task XXIV	3 at IBBCE 2025: joint workshop (full day), Task XXIV special session (6 presentations), first in-person meeting	Chapter 5
Keynote lectures	1 (ESCAPE 35, Ghent, one hour)	Chapter 5
Participants' survey	57 questions in 11 sections; 10 responses from institutions in 9 countries	Section 2.4
Models identified	10 starting-point models from participants; 43 process and 33 technology model entries from EPFL IPESE	Table 7; Figure 4
Template and sharing framework	1 simplified ex-ante model template; 7 disclosure layers (L0 to L6)	Section 2.2; Table 8; Appendix B
Journal articles	8 manuscripts in 2025	Chapter 5

Indicator	Products	Source in this report
Conference technical contributions	22 (ESCAPE 35 2025: 10; ECOS 2025: 1; IBBCE 2025: 6; and 1 in EUBCE2026; GHGT-17, TCCS-13, ARC 2025, and PCCC8 each.	Chapter 5
Student projects	3 (1 master thesis, 2 semester projects)	Chapter 5
Public recordings	3 (kick-off meeting, first in-person meeting, Task XV workshop) available in YouTube	Appendix C; Chapter 4
Committed effort	At least 13 person-weeks per year per participating organization	Section 2.4

2.3.3. Status of deliverables of Task XXIV Subtask 1

Table 4: Deliverables of Task XXIV coordination: expected outcomes, due dates, description and completion status.

ID	Expected outcome	Due	Description	Completion
D1	Ontologies and procedures for sharing ex-ante models	Q4 2025	Simplified template or schema, class-attribute tree, layered disclosure and sharing procedures (completed)	Completed 100 %
D2	Documentation formats and webapp tool for managing the files storage system	Q4 2026	Model documentation formats (JSON, Quarto, Excel, HTML, etc.). Web-based tool with hyperlinks to supplementary data repositories, and visualization prototypes.	Started 10%
D3	Open-source repository of documented ex-ante models with supplementary data	Q1, Q2 2027	Models inventory. Version-control repository (git-like concept). Editorial roles and edition permissions. Database population with participants' models.	Started 10%
D4	Annual and final reports to SFOE and the IETS Secretariat	Recurring (Q2, Q4)	Progress reported at the 40th and 41st ExCo meetings; contributions to the IETS annual report; and interim report Y2025 (completed)	On schedule 33%
D5	Findings on hosting and maintaining an open database, including confidentiality and intellectual property	Q4 2025, Q4 2026, Q4 2027	Survey and bilateral meetings (completed); Seven-layered disclosure framework (established). Models' validation and sensitive data anonymization protocols.	On schedule 33%
D6	Experiences using others' models in own process integration frameworks, and roadmaps discussions	Q2, Q3 2027	Software and modelling paradigms (already mapped by the survey). Sharing experiences on using others' models in own integration tools. Sustainability metrics and roadmaps discussion	Started 10%
D7	Dissemination to SFOE, IETS workshops, EFCE and ECOS events, and continuing education	Recurring	Y2025 dissemination activities completed (see Table 3). Dissemination activities for Y2026 in preparation.	On schedule 33%

Table 5 shows the schedule of the Subtask 1 activities and deliverables over the three years of the Task.

Table 5: Schedule of the Subtask 1 activities (ACT) and deliverables (D1 to D7) for 2025 to 2027.

Year	2025				2026				2027			
Activity	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4

1				D1								
2				D5				D2				
3.1								D5				
3.2									D3			
4										D3		
5.1										D6		
5.2											D6	D5
6	D7	D4	D7	D4	D7	D4	D7	D4	D7	D4	D7	D7

2.4 Management activities and participants' survey

Task XXIV participating countries joined efforts through plenary meetings, bilateral exchanges, a survey and a first in-person meeting to develop a standardized ex-ante model template and the procedures for model sharing. The coordination was led by the Task managers Daniel Florez-Orrego and François Maréchal, with the support of the EPFL team members Oktay Boztas, Meire Ribeiro-Domingos and Pullah Bhatnagar. Figure 3 shows the chronology of the activities. This section summarizes their main outcomes, while the detailed account of each meeting and the complete survey results are given in Appendix C.

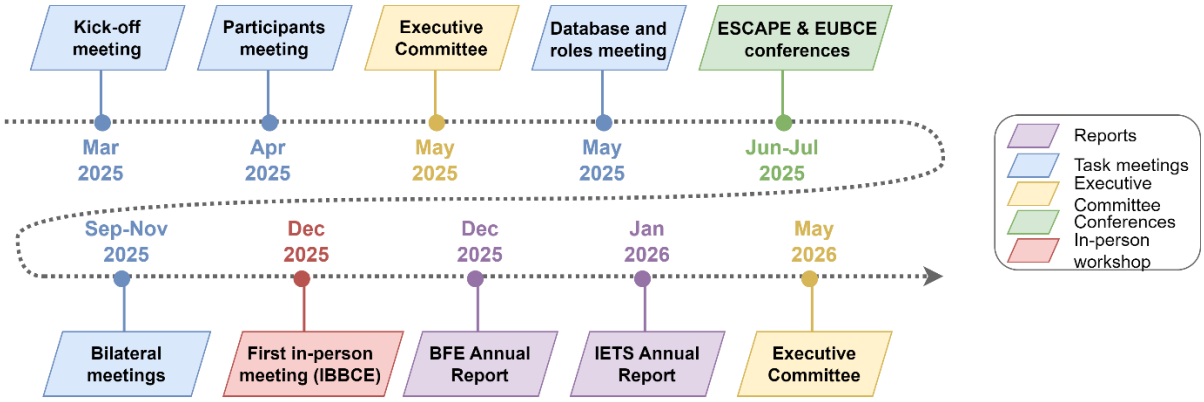


Figure 3: Chronology of the kick-off, intratask (plenary), bilateral and executive committee meetings, as well as report deadlines and dissemination in conferences and workshops.

2.4.1. Meetings and engagement

Table 6 summarizes the coordination meetings held in 2025. The plenary meetings defined the rules for Task participation, mapped the expertise of the participants and identified the minimum information required to document ex-ante models: metadata (authorship, software, sharing layer), mass, energy, emission and information flows, documented parameters and assumptions, decision and calculated variables, simplified block flow diagrams and supplementary material. The bilateral meetings with the country consortia clarified the remaining questions on procedures, template and confidentiality, and revealed country-specific priorities (Table 9). Their recurring themes were confidentiality and intellectual property protection, validation and quality assurance of shared models, technical interoperability across software tools, and the educational value of a shared database.

Table 6: Coordination meetings of Task XXIV held in 2025 and their main outcomes.

Date	Meeting	Main outcomes
14 March 2025	Kick-off meeting (online)	Task structure and contribution areas; IETS reporting rules; engagement procedure and Letter of Commitment; commitment of at least 13 person-weeks per year per participating organization

Date	Meeting	Main outcomes
11 April 2025	Participant presentations (online)	Expertise of the participants and contribution areas; inventory of existing databases (e.g., AIDRES, EPOS, MASBOOK, LAPSE, etc.)
15 May 2025	40th IETS Executive Committee	Progress report on Task XXIV and alignment with the IETS strategic objectives
23 May 2025	Ontologies and procedures (online)	Minimum information required to document and validate ex-ante models; translation effort and starting-point models
September to November 2025	Bilateral meetings with 10 country consortia (online)	Country-specific expectations and concerns (Table 9); input to the participants' survey
12 to 13 November 2025	41st IETS Executive Committee	Progress report on Task XXIV and discussion of communication strategies
5 December 2025	First in-person meeting, Lisbon (IBBCE 2025)	Discussion of the survey results; common understanding of the "rules of the game"; Annual Report contributions (Section 2.4.3)

2.4.2. Main findings of the participants' survey

The survey (57 questions in eleven thematic sections) received ten responses from institutions in nine countries (Switzerland, Denmark, Belgium, Italy, Sweden, Colombia, Chile, Portugal and Germany) before 4 December 2025. Its main findings, detailed in Appendix C, are the following:

- **Expertise and models:** industrial sector and process models are the dominant research focus of the participants, followed by decarbonization technologies and process integration frameworks. Participants proposed ten starting-point models (Table 7), complemented by the EPFL IPESE list of anonymized process and technology models (Figure 4).
- **Software and practices:** Python and Aspen® suite dominate the software landscape; 70% of the respondents combine licensed and open-source software and 30% rely mainly on licensed tools. Documentation is mostly embedded as plain text in flowsheets or code, and OneDrive is the most used storage platform, which supports a centralized registry linked to institutional repositories.
- **Barriers:** confidential industrial data (non-disclosure agreements, patents, co-owned models) is the main barrier to open model sharing, followed by unpublished research data, whereas the fear of exposing errors and the loss of competitive advantage ranked lowest.
- **Recognition and intellectual property:** citation in publications is the primary recognition expectation; Creative Commons licensing, embargo periods, tiered access and patent protection were considered important, while full open access received limited support.
- **Disclosure layers:** Layers 0 to 2 of the proposed framework (Table 8) received nearly unconditional support, Layers 3 and 4 conditional support, and Layers 5 and 6 were the most sensitive (Figure 5).
- **Contributions and validation:** sharing published models shows the strongest commitment; literature benchmarking is the main validation method used today, and model quality certification procedures received strong support.

Table 7: Models to be used as a starting point by the participant countries.

Model / Sector	Software	Status
Ammonia plant / Chemicals	Aspen® Plus	Open
Zeotropic mixture heat pump	Python	Design/off-design
Quicklime production	OSMOSE	To be published

Model / Sector	Software	Status
Energy storage / reactor modelling	MATLAB® + Python	Published
Electricity market + H_2 production	N/A	Confidential
Dual fluidized bed for plastic waste	Aspen® Plus	Published by June 2026
Agroindustry (Sugar/Ethanol)	Aspen® Plus	Published
Solar energy integration to industrial	TRNSYS®	Unpublished
CO_2 to Methanol / CCU	Aspen® Plus	Unpublished
Cement plant scheduling optimization	Python + Gurobi	Paper Q1 2026

In addition to the models proposed by the participants, the EPFL IPESE group has compiled a list of anonymized industrial process and technology models from previous collaborative projects, which will be formatted and shared using the simplified ex-ante model template (Figure 4).

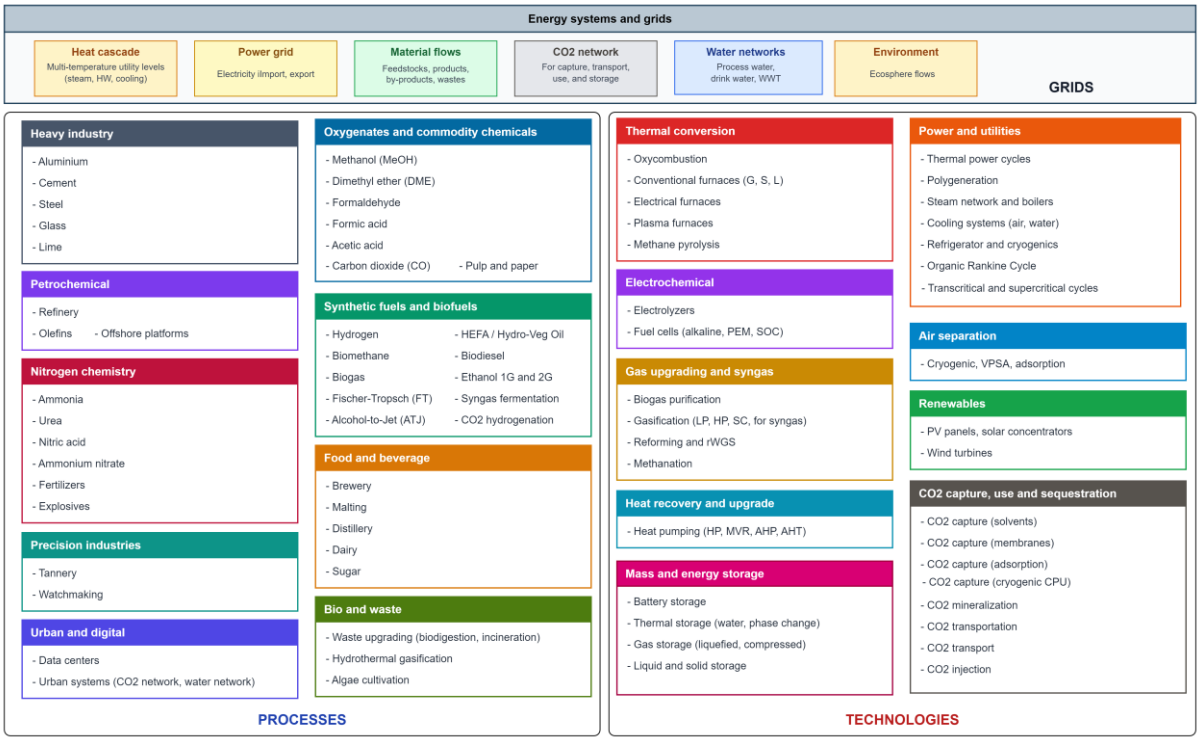


Figure 4: List of anonymized models developed by the EPFL IPESE group to be formatted and shared using the simplified ex-ante model template.

Table 8: Proposed multi-layered sharing framework.

Layer	Description
L0	Public Metadata
L1	Structure & Topology
L2	Mass & Energy Balances
L3	Thermodynamic Data
L4	Kinetic & Reaction Data
L5	Economic Data
L6	Full Model Details

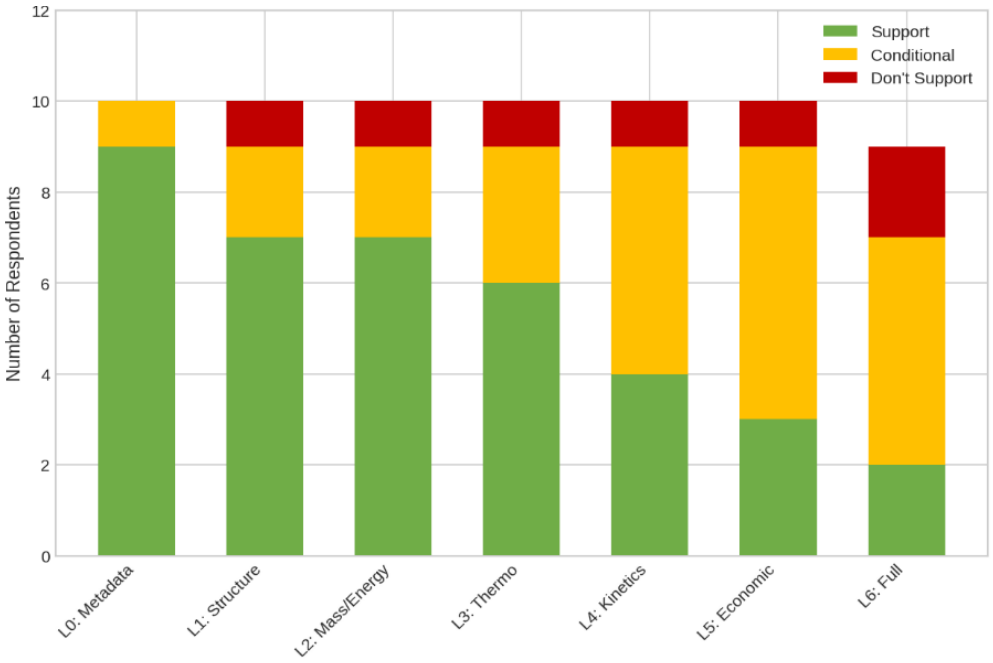


Figure 5: Support for multi-layered sharing approach. One respondent was unsure of L6.

2.4.3. First in-person meeting and next steps

The first in-person meeting took place on 5 December 2025 at ISEC Lisbon, in conjunction with IBBCE 2025. Its agenda (Table C1) included the key findings of the bilateral meetings, the survey results and short presentations by 15 participants from the countries' consortia on their shareable models, their feedback on the simplified template and their concerns regarding confidentiality, citation, layered sharing and anonymization. The meeting aimed at a common understanding of the "rules of the game", encompassing simplified ontologies, model structure, sharing, maintenance and peer-review procedures, and confidentiality protocols. Populating the template with detailed modelling data is planned for 2026. Each participant country was requested a contribution of max. 5 pages to the Task XXIV Annual Report 2025 by 19 December 2025 (Annex 3).

3 Conclusions and outlook

The first year of Task XXIV demonstrated the ability of the management and operative teams to lead initiatives that bridge academic rigor with practical industrial application. These activities positioned the Task as a central node in the global industrial decarbonization network, with capabilities in database implementation, analytical methodology development, and facilitation of international collaboration to accelerate knowledge transfer and technology deployment. The coordination activities established the foundational framework for model sharing with clear objectives, engaged participants, and concrete mechanisms for achieving the Task's goals. Representatives from 10 participating countries (Canada, Denmark, France, Germany, Italy, the Netherlands, Norway, Portugal, Sweden and Switzerland) and three observer countries (Belgium, Chile and Colombia) engaged through meetings, surveys, bilateral discussions, and the first in-person intra-task meeting.

The Task demonstrates that genuine international collaboration is possible when intellectual property, confidentiality, and attribution concerns are addressed transparently through structured dialogue. Participants moved from cautious scepticism about model sharing to actively proposing layered access frameworks and validation protocols, creating not merely a data repository but a collaborative ecosystem. The survey distributed to participating countries yielded critical insights regarding diverse legal and operational frameworks for proprietary information management across academic and industrial contexts. Industrial confidentiality emerged as the primary barrier to sharing, yet the need to balance transparency for reproducible science with legitimate intellectual property protection was broadly acknowledged.

These insights informed the seven-layer disclosure framework, where Layers 0-2 are publicly accessible, Layers 3-4 are available on a case-by-case basis, and Layers 5-6 require bilateral agreements. The proposed model documentation template captures essential information for process integration analyses, while remaining software-agnostic and maintaining physical consistency. The framework (Deliverable D1) supports approaches ranging from black-box to white-box representations and accommodates proprietary software alongside surrogate models and open-source alternatives. Anonymization techniques, including generalization, thermodynamic consistency verification, obfuscation, and aggregation, enable the transformation of sensitive industrial data into shareable formats (D5).

Dissemination activities (D7) established Task XXIV's visibility through presentations at ESCAPE 2025, EUBCE 2025, ECOS 2025, IBBCE 2025, and a joint IETS-ETSAP workshop, reaching diverse audiences of researchers, practitioners, and policymakers. The co-organization of the Industrial Symbiosis workshop with Task XI Biorefineries and the Special Session of Task XXIV in IBBCE demonstrated capacity to lead inter-task initiatives at the intersection of bioeconomy and industrial decarbonization. Participation in Executive Committee meetings and preparation of annual reports ensured alignment with IETS and SFOE strategic objectives (D4). Three student projects contributed to database development and documentation tools, while the Pinch Bot application within the parallel Pinch Small project illustrates how structured ontologies can enable domain-specific AI assistants for policy support, engineering assessments, and continuing education.

With the foundational framework established, the second year will focus on database population and demonstrated impact through implemented case studies. The priority is transitioning from defining rules of engagement to seeing shared models used across countries to inform decarbonization strategies. Key deliverables will include the documentation tool (D2), the initial database repository (D3), and further findings on confidentiality and intellectual property challenges (D5). Engagement with industrial stakeholders must be strengthened to provide validation using real plant data, and automated tools for metadata extraction should be developed to minimize documentation burden (D6). The Task remains on track to deliver a collaborative platform that accelerates knowledge transfer toward climate neutrality goals in energy-intensive industries.

Table 9: Key points raised by each participant country during the bilateral meetings.

Country	Key Point
Switzerland	Maximization of open access data for public information
Sweden	Impact of project confidentiality on sharing capabilities
Netherlands	Requirement for open sharing of proprietary software models
Italy	Model portability across scales (engineering, chemistry, control)
Germany	Data anonymization while maintaining real-world relevance
Denmark	Added value of documentation beyond journals publication
Chile	Data exchange policies between researchers and industry
Canada	Sustainability metrics for industrial symbiosis
Portugal	Validation by sub-teams then by the broader community
Belgium	Different interfaces for various stakeholder types
Colombia	Educational platform as a model database dissemination

4 National and international cooperation

The goal of Task XXIV is to make models database and experiences on process integration approaches accessible to policymakers, industry leaders, and government officials, ensuring that the knowledge and insights generated are not confined to academic or technical circles. The project will support stakeholders in identifying and implementing the most suitable technological solutions for their specific geographical contexts, enabling transparent and reproducible assessment of decarbonization pathways. In Switzerland, the evolving framework proves valuable for large industries and, particularly, for small and medium-sized enterprises (SMEs), which often lack the technical resources to evaluate complex energy efficiency and decarbonization strategies. The national and international cooperation underpinning Task XXIV builds upon an extensive network of collaborations established through previous and ongoing projects. In Switzerland and abroad, partnerships with industrial, academic, and governmental stakeholders have provided a solid foundation for developing and validating process integration models across diverse sectors. Past and current industrial partners include:

- Novelis Sierre (aluminium),
- Richemont (luxury goods),
- Hermes (luxury goods),
- CREMO and ELSA (dairy),
- Morand (distillery),
- Spiess (food processing),
- SATOM (waste incineration),
- CIMO (utilities),
- Monthey Energies (electricity supply),
- Syngenta (chemicals),
- Buhler (machinery),
- CSD Engineers (consulting), and
- Louis Dreyfus Company (agri-food).
- TreaTech (wastewater treatment)

Academic partnerships with the University of Applied Sciences and Arts of Western Switzerland (HES-SO), the Eastern Switzerland University of Applied Sciences (OST), and the School of Business and Engineering Vaud (HEIG-VD), together with the support of the Swiss Federal Office of Energy (SFOE), have enabled the development of models databases and energy planning tools that, while not fully documented or formatted to IETS standards, will constitute valuable assets for the shared ex-ante models repository. This national cooperation was complemented by close connections with leading academic and industrial partners abroad, including Flemish Institute for Technological Research (VITO, Belgium),

Universidad Nacional de Colombia Medellín, Universidade de São Paulo (Brazil), KU Leuven (Belgium), Polytechnique Montréal (Canada), the International Research Consortium on Life Cycle Assessment and Sustainable Transition (CIRAIG), Ghent University (Belgium), University of Liège (Belgium), Instituto Superior Técnico (IST, Portugal), Politecnico di Milano (Italy), Sapienza Università di Roma (Italy), and Universidad Industrial de Santander (Colombia), Universidad Politécnica de Madrid (Spain), Ecole Nationale Supérieure des Ingénieurs en Arts Chimiques et Technologiques (France), Newcastle University (England), and RWTH Aachen University (Germany).

Accordingly, this established network of collaborations paved the way for broader international engagement and the coordination of Task XXIV by the École Polytechnique Fédérale de Lausanne (EPFL) and its Industrial Process and Energy Systems Engineering (IPESE) group. Currently, Task XXIV has expanded to include participants from Canada, Denmark, France, Germany, Italy, the Netherlands, Norway, Portugal and Sweden, together with Switzerland as Task manager, observers from Belgium, Chile and Colombia, and interested partners from Austria, whose participation is not yet defined (Figure 6). Altogether, 54 researchers from 25 institutions, including observers and interested partners, contribute to the shared model database initiative (Table 10).



Figure 6: Countries of the participants, observers and other interested partners of Task XXIV.

Table 10: List of the participants, observers, collaborators and other interested partners of Task XXIV as of December 2025.

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Belgium	BE	Rafailia Mitraki	rafailia.mitraki@uliege.be	University of Liège
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		Paul Stuart	paul.stuart@polymtl.ca	Polytechnique Montréal
		Sara Russo Garrido	russo_garrido.sara@uqam.ca	Université du Québec à Montréal
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Task XXIV actively engages with other Tasks within the IEA IETS Technology Collaboration Programme to translate technical insights into actionable recommendations and support ongoing international efforts in industrial decarbonization. Collaborations facilitate knowledge exchange across complementary domains, ensuring that process integration methodologies benefit from and contribute to broader developments in energy systems modelling, biorefineries, waste heat utilization, digitalization, and electrification.

- *Presentation in the Joint workshop co-organized by IETS TCP and the Energy Technology Systems Analysis Programme ETSAP*

On April 10th 2025, Task XXIV participated in a joint online workshop co-organized by IETS TCP and the Energy Technology Systems Analysis Programme (ETSAP), focusing on industrial sector modelling in energy system models [4]. The workshop, moderated by Etienne Bernier (Natural Resources Canada) and Anna Krook-Riekkola (ETSAP), explored the role of energy systems in transitioning towards decarbonized industry and promoted collaboration on modelling data collection. Task Manager Daniel Florez-Orrego, together with Task XXIV participants from Switzerland (Oktay Boztas, Meire Ribeiro-Domingos, Pullah Bhatnagar) and Belgium (Muhammad Salman), presented a talk titled "Process Integration for Industry Decarbonization: Enabling a Shared Database of Ex-Ante Models" (Figure 7). The presentation showcased selected case studies on aluminium, olefins, watchmaking, sugar, and synthetic natural gas production, demonstrating how ex-ante models can support system-wide analyses including energy security, carbon management, electrical grid balancing, and advanced decarbonization pathways. In the aluminium industry, carbon capture can only halve CO_2 emissions (100 kg_{CO2}/t_{Al}), while combined renewable electricity and biomass conversion achieve -200 kg_{CO2}/t_{Al}. Power-to-gas alone cuts CO_2 emissions by only 80%, failing net-zero targets. Electrification plus biomass conversion consumes 27% less energy and 40% less biomass than biomass-only configurations. In the energy industry, biomass conversion reduces the dependence on fossil resources and minimizes CO_2 emissions. The integration of anaerobic digestion, electrolysis, and digestate gasification maximizes CH_4 yield per unit of feedstock, while a flexible storage strategy manages renewable electricity fluctuations. In the plastics industry, typically dependent on naphtha-cracking, thermochemical recycling via plastic-biomass co-gasification and methanol-to-olefins conversion promotes carbon circularity. Carbon neutrality is achieved using CO_2 capture, methanation, and rWGS, as well as integrating solid oxide electrolyzers. Liquid CO_2 storage cuts electricity costs by 28%. In the sugar Industry, a bio-waste utilization strategy integrates CO_2 capture, methanation, and heat pumps. Biogas upgrading to biomethane and heat recovery improves energy efficiency and costs while reducing CO_2 emissions; whereas biogenic carbon utilization for methanation or injection enables negative emissions. Finally, in the watch-making industry, an energy audit analyzing the technologies and seasonal energy demands proposes combining photovoltaic, solar-thermal, and photovoltaic thermal (PVT) systems to reduce CO_2 emissions. Considering a seasonal optimization approach, PVT plus thermal storage achieves a reduction of 17% in annual energy consumption. PV panels reduce energy demand by 37% in summer and 5% in winter.

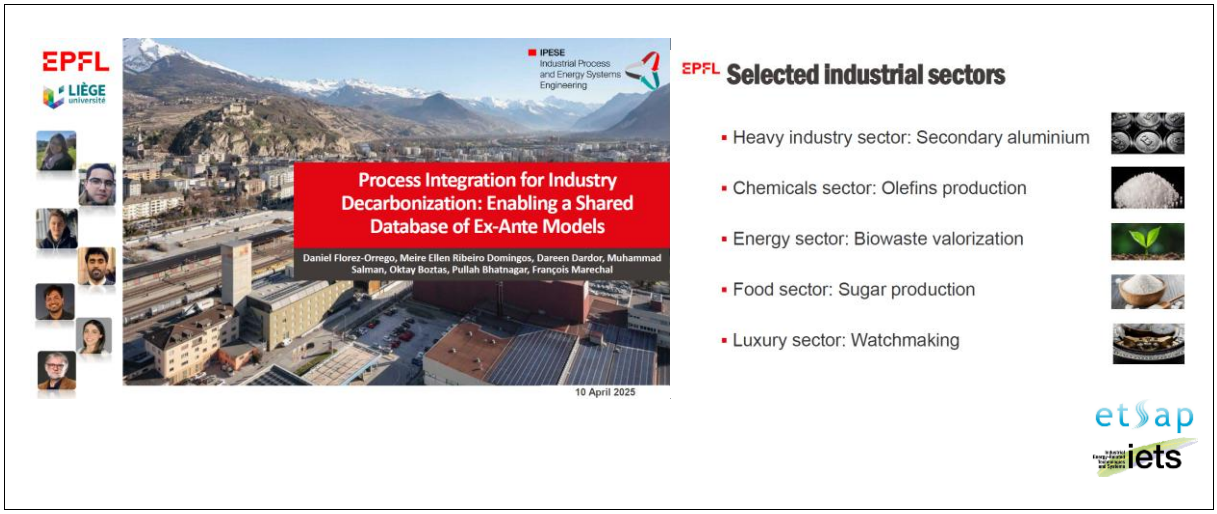


Figure 7: Flyer of the Task XXIV presentation in the joint online workshop co-organized by IETS TCP and the Energy Technology Systems Analysis Programme (ETSAP).

[4] https://iea-industry.org/events/event/joint-workshop-with-etsap/

From the presenters Q&A session, it was identified that ETSAP's top-down energy system frameworks can provide macroeconomic context, policy scenario analysis, and cross-sectoral interactions, while IETS's bottom-up process integration models offer detailed representations of unit operations, mass and energy balances, and technology-specific performance characteristics. By feeding validated process-level data, such as energy consumption profiles, emissions factors, flexibility constraints, and technology readiness levels, into economy-wide models, analysts can improve the accuracy of sectoral projections and propose more realistic decarbonization pathways. Conversely, energy system models can inform process-level studies by providing boundary conditions on energy prices, grid carbon intensity, and resource availability. Particular attention must be given to temporal resolution (capturing seasonal and diurnal variability), spatial granularity (accounting for regional infrastructure constraints), and the representation of emerging technologies, such as industrial carbon capture and storage, power-to-X pathways, and demand-side flexibility to accurately assess industry potential for carbon removal, electrical grid balancing, and other system-wide interactions. The workshop concluded that collaborative data collection and model harmonization between ETSAP's economy-wide energy system modelling frameworks and IETS's process-level models could significantly enhance the robustness of industrial decarbonization scenarios and support policymaking.

- *Collaboration with Task XI "Industry-based Biorefineries", coordinated by Paul Stuart and Marzouk Benali in the European Biomass Conference and Exhibition (EUBCE 2025)*

The collaboration with Task XI "Industry-based Biorefineries", coordinated by Paul Stuart and Marzouk Benali, was first established during the European Biomass Conference and Exhibition (EUBCE 2025) held in Valencia on June 9th -12th 2025, EUBCE, Room 3. Task XXIV participated in two Task XI activities: A special session on Biorefineries (Figure 8) and the Third intra-task meeting (Figure 9), held in June 12th and 13th 2025, respectively. Key synergies between the two Tasks have been identified (Figure 10):

- The assessment of sustainable biomass utilization pathways requires industrially validated models of processes and energy technologies, which can be ranked using process integration methods based on rational sustainability metrics.
- Models should accurately reproduce the performance of biomass and other renewable energy conversion processes, supported by clear and well-referenced documentation.
- Roadmaps must account for the trade-offs inherent in biomass utilization, including supply chain logistics, sizing and economies of scale, business-model feasibility and resilience, renewables seasonality, and competition among users across different chemical, energy and urban sectors.
- More attention is given to biogenic carbon management integrated with co-electrolysis, as well as biological (e.g., digestion, fermentation) and thermal conversion of waste biomass; crucial approaches for reducing the burden on biomass as a strained material and energy resource.
- Knowledge sharing through the dissemination of industrial applications and comparative case studies among researchers, industries, and suppliers can shed light on the best practices and process integration strategies for maximizing the potential of the biomass resources.
- Synergies with Task XXI on circular economy, Task XV on industrial excess heat recovery, Task XVIII on digitalization and Task XIX on electrification were also identified, recognizing the potential of circularity, heat integration, and electrification strategies to enhance process integration methodologies and digital tools.



Figure 8: Participation of Task XXIV management in the special session on Biorefineries organized by IETS TCP Task XI in EUBCE conference, Valencia, Spain in June 9th -12th 2025.



Figure 9: Presentation of the Task XXIV in the Third intra-task meeting of the IETS Task XI, held in EUBCE conference, Valencia, Spain, contributing to (i) a shared models inventory for biomass conversion technologies; (ii) integrating biorefinery concepts within industrial symbiosis systems; (iii) benchmarking multi-criteria sustainability metrics for comparative pathway assessment; and (iv) documenting design tools and roadmaps for biomass-based industrial systems

- *Collaboration in workshop of Task XV Industrial Waste Heat Utilization*

Collaboration with Task XV provided an opportunity to showcase energy planning tools developed at Industrial Process and Energy Systems Engineering group at EPFL and to explore synergies in industrial waste heat recovery and utilization. Task XV aims to initiate the concrete implementation of waste heat utilization projects in industry by creating the necessary framework conditions, establishing an international network and information infrastructure for key stakeholders to exchange knowledge, developing expertise on industrial waste heat, and accelerating the deployment of best practices in the process industry. As part of this collaboration, Dorsan Lepour presented the Renewable Energy Hub Optimizer (REHO), a decision-support tool that supports sustainable energy system planning in energy communities and neighbourhood-scale building stocks through multi-criteria, multi-objective optimization. The tool employs GIS-based holistic modelling of energy demands and flows at both building and neighbourhood levels, enabling planners to balance economic, ecological, and technical parameters. A key feature is the coordination of energy and device exchanges between buildings and industrial sites, directly aligning with Task XXIV's objectives on industrial symbiosis, waste heat valorization, and urban systems integration. The presentation underscored the potential for combining Task XV's focus on waste heat utilization with Task XXIV's process integration methodologies to identify and implement cost-effective heat recovery solutions across industrial and urban energy systems. The recording is available in the Task XV YouTube channel [5]

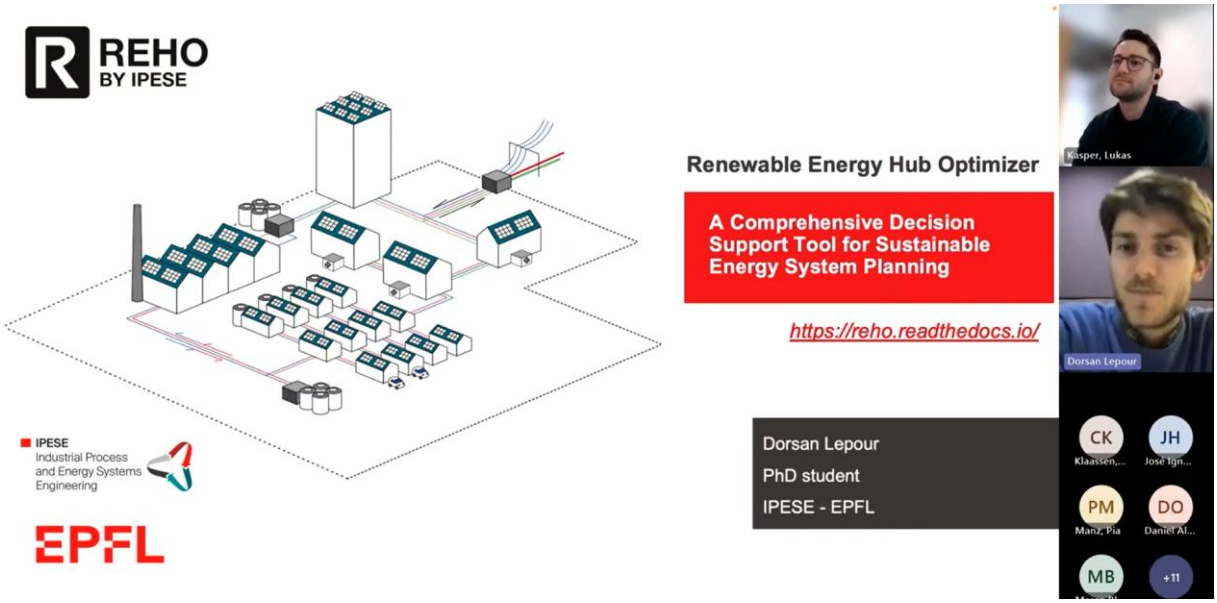


Figure 10: Participation of Task XXIV in the workshop of Task XV Industrial Waste Heat Utilization.

- *Integration into EPFL's Master of Advanced Studies and Continuing Education Programs.*

Continuing education programs are part to the dissemination strategy of Task XXIV, as they help training skilled engineers that contribute to the evolution of the Subtask. This creates a feedback loop that enlarges and refines the project's database and models, ensuring the initiative remains relevant and up-to-date while continuously adapting to emerging challenges and opportunities in the industrial decarbonization landscape. In turn, insights gained from Task XXIV will inform and shape the curriculum of modern degree programs, while generating revenue to support the academic business model. Prof. François Maréchal and Dr. Daniel Florez-Orrego, delivered lectures as part of the Master of Advanced Studies in Sustainable Energy Systems Engineering: Industry Decarbonization Module, established through the continuing education programme jointly offered by EPFL and HES-SO in spring 2025 [6]. The

[5] https://www.youtube.com/watch?v=I9dwnsJweaA&list=PLuitK2C8j5-v_uD9ST-QvY4t-DcE5lZPj&index=8
[6] https://www.formation-continue-unil-epfl.ch/formation/sustainable-energy-cas-industry-decarbonisation/

integration of existing ex-ante models[7] into the pedagogical programme enabled effective knowledge transfer while providing valuable feedback for improving model documentation and sharing practices. The IETS Task XXIV database will also be utilized in extension courses and other training programmes, which can in turn contribute additional models (following BFE certification through industrial audit and a pinch analysis) or assist in validating existing models.



Figure 11: Master of Advanced Studies of the Sustainable Energy Systems Engineering: Industry Decarbonization - CAS course. Task XXIV databases, refined through pedagogical application and feedback from course participants, contribute to the IETS Task XXIV shared repository.

5 Publications and other communications

Dissemination strategies aim to make the models database and process integration approaches widely accessible, ensuring that the knowledge and insights generated extend beyond academic and technical circles to reach policymakers, stakeholders, industry leaders, and government officials. A key dissemination channel is the presentation of findings at workshops and international conferences. These events amplified the reach and engaged a global audience to encourage adoption of process integration and decarbonization practices across other Tasks and collaboration programs.

[7] https://ipese-web.epfl.ch/mas/book-website

- *33rd European Biomass Conference and Exhibition (EUBCE 2025).*

https://www.eubce.com

The 33rd European Biomass Conference and Exhibition (EUBC E2025) in Valencia, Spain, June 9th - 12th 2025 provided a platform for establishing collaboration between Task XXIV and other IETS Tasks, particularly Task XI on Industry-based Biorefineries. As previously mentioned, in June 12th and 13th 2025, participants from Tasks XI, XVIII (Digitalization), XIX (Electrification), and XXIV convened to discuss synergies in assessing sustainable biomass utilization pathways. The discussions emphasized that models should accurately reproduce the performance of biomass and renewable energy conversion processes with clear, well-referenced documentation. The meeting highlighted the importance of sharing knowledge, disseminating successful industrial application stories, and discussing case studies between researchers, industries, and suppliers to maximize the potential of biomass resources. This inter-task collaboration established a foundation for joint activities on industrial symbiosis and decarbonization roadmaps.

Additionally, Task co-manager Daniel Florez-Orrego presented "Sustainable Biofuels Production via Industrial Symbiosis and Waste Heat Utilization in Heavy Industry" (Florez-Orrego, D., Dardor, D., Ribeiro Domingos, M., Germanier, R., Maréchal, F.), a collaborative effort with the Net Zero Lab Valais consortium (Novelis, Oiken, EPFL, HES-SO). The presentation addressed the importance of adequately modelling biogenic carbon management (carbon circularity) integrated with co-electrolysis (electrification), alongside biological and thermal waste biomass valorization. These modelling capabilities are essential for quantifying the impacts on supply chains and evaluating competing biomass utilization pathways to produce sustainable aviation fuels, methanol, aluminium, and syngas.

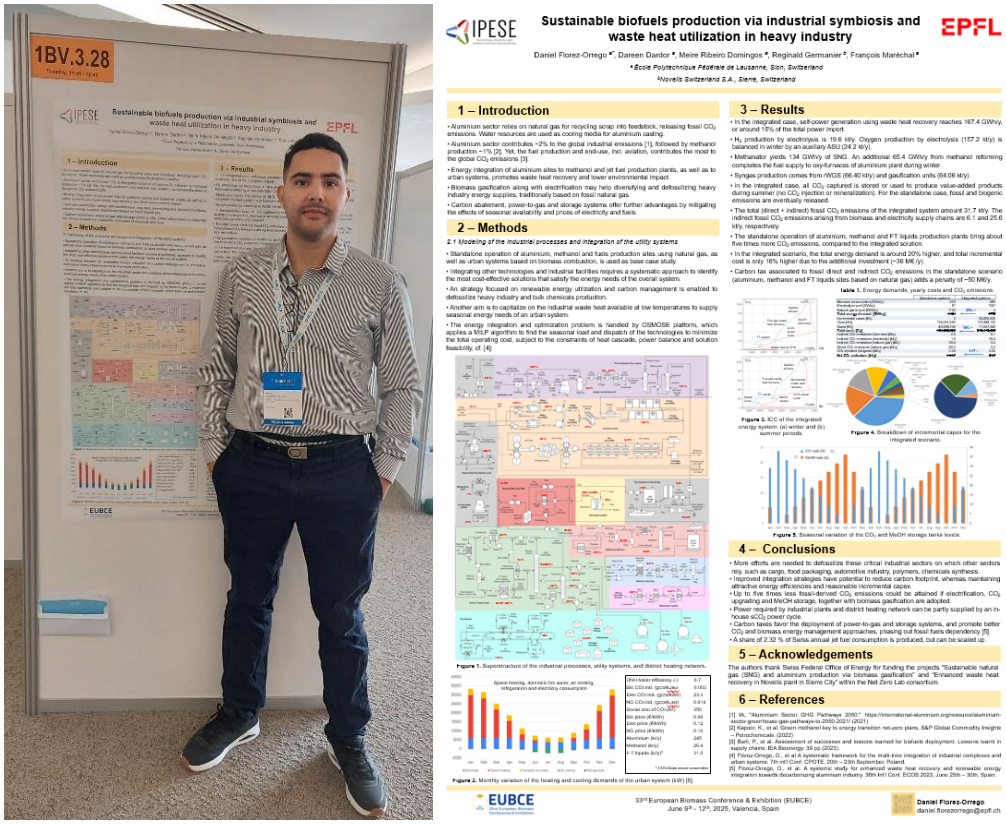


Figure 12: Task XXIV presentation titled "Sustainable Biofuels Production via Industrial Symbiosis and Waste Heat Utilization in Heavy Industry" in the 33rd European Biomass Conference and Exhibition (EUBC E2025) in Valencia, Spain in June 9th - 12th 2025.

- *European Symposium on Computer Aided Process Engineering (ESCAPE 2025): IETS Task XXIV Keynote Lecture and Technical Contributions*

https://escape35-belgium.eu

The 35th European Symposium on Computer Aided Process Engineering (ESCAPE 2025) in Ghent, Belgium, July 5th and 9th 2025, featured a keynote lecture presenting the advances, challenges, and opportunities of Task XXIV. The one-hour presentation “Process Integration for Industry Decarbonization - Subtask 1 Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics”, chaired by Grégoire Léonard (University of Liège) and Henrique Matos (IST Lisbon), attracted a full-house audience of researchers, industry representatives, and policy makers.

The keynote detailed the Task's objectives in enabling a shared database of ex-ante models and outlined the collaborative framework for sharing experiences on decarbonization roadmaps. The discussion identified additional action points and exciting collaboration opportunities, reinforcing Task XXIV visibility within the computer-aided process engineering (CAPE) community.

Additionally, several Task XXIV participants presented technical contributions on relevant topics including modelling of synthetic natural gas production from waste biomass valorization, reversible solid oxide cell integration in heavy industry, and renewable electricity utilization for secondary aluminium decarbonization, circular olefins production, and enhanced biodiesel production, demonstrating the practical applications of the process integration approaches and the ex-ante modelling capabilities, promoted by the Task.



Figure 13: Task XXIV presentation in the 35th European Symposium on Computer Aided Process Engineering (ESCAPE 2025) in Ghent, Belgium, July 5th and 9th 2025.



Systems & Control Transactions

Research Article - Peer Reviewed Conference Proceeding
ESCAPE 35 - European Symposium on Computer Aided Process Engineering
Ghent, Belgium. 6-9 July 2025
Jan F.M. Van Impe, Grégoire Léonard, Satyajeet S. Bhonsale, Monika E. Polańska, Filip Logist (Eds.)

PSE PRESS

Process integration for industry decarbonization: Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics

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ABSTRACT

The mass adoption of process simulators has accelerated data generation for the

Figure 14: Task XXIV keynote lecture in the 35th European Symposium on Computer Aided Process Engineering (ESCAPE 2025) in Ghent, Belgium, July 5th - 9th 2025.

Other presentations of Task XXIV participants in ESCAPE 2025 conference:

1. Florez-Orrego, D., Dardor, D., Ribeiro Domingos, M., Germanier, R., Maréchal, F., Integration of renewable energy and reversible solid oxide cells to decarbonize secondary aluminium production and urban systems. 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35**. July 6 - 9, 2025. Ghent, Belgium
2. Ribeiro Domingos, M., Florez-Orrego, D., Sfeir, S., Maréchal, F., Impact of the share of renewable energy integration in the selection of sustainable natural gas production pathways. 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35.** July 6 - 9, 2025. Ghent, Belgium
3. Bhatnagar, P., Florez-Orrego, D., Margni, M., Maréchal, F., Assessing the impact of circularity and energy system evolution on scope 3 emissions: a case study of the watch industry. 38th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems - **ECOS 2025**, 29 June - 4 July 2025, Paris, France
4. Corre, S., Ribeiro Domingos, M., Florez-Orrego, D., Magnaval, G., Margni, M., Maréchal, F., Techno-Economic and Environmental Analysis of Biomethane Production from Sewage Sludge in Hydrothermal Gasification Process, 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35**. July 6 - 9, 2025. Ghent, Belgium
5. Dardor, D., Florez-Orrego, D., Germanier, R., Margni, M., Maréchal, F. On the Economic Uncertainty and Crisis Resiliency of Decarbonization Solutions for the Aluminium Industry. 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35**. July 6 - 9, 2025. Ghent, Belgium
6. Baibhav, V., Florez-Orrego, D., Bhatnagar, P., Maréchal, F., Process Integration and Waste Valorization for Sustainable Biodiesel Production Towards a Transportation Sector Energy Transition. 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35**. July 6 - 9, 2025. Ghent, Belgium
7. Limpach, J, Salman, M., Florez-Orrego, D., Maréchal, F., Leonard, G. Evaluating Energy Transition Pathways for CO2 Reduction in Industries with Low-Temperature Heat Demand: A Superstructure Optimization Approach. 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35**. July 6 - 9, 2025. Ghent, Belgium
8. Boztas, O., Ribeiro Domingos, M., Florez-Orrego, D., Maréchal, F. Olefins production through sustainable pathways: techno-economic and environmental assessment. 35th European Symposium on Computer Aided Process Engineering **– ESCAPE 35**. July 6 - 9, 2025. Ghent, Belgium
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- *International Conference on Bioenergy & Bioeconomy (IBBCE 2025), Joint workshop on Industrial Symbiosis co-organized by Tasks XI and XXIV, Special Session Task XXIV, and First in-person intra-task meeting.*

https://ibbce.com

Task XXIV co-organized three major activities during the International Conference on Bioenergy & Bioeconomy (IBBCE 2025), December 3rd - 5th 2025, ISEC Lisbon, Portugal. The full-day joint workshop on "Industrial Symbiosis as a Catalyst for Biorefinery Value Chains" co-organized with Task XI established a comprehensive framework for understanding how industrial symbiosis can evolve beyond classical waste-to-resource paradigms into purposeful integration of diverse industries exchanging materials, energy, water, data and services. Decision-making metrics for policy, economic, technical, and social dimensions were assessed, identifying drivers such as climate and social targets, shared utilities, as well as barriers including regulatory fragmentation and technology risks. Through structured morning presentations by Tasks XI and XXIV managers Paul Stuart, Marzouk Benali, Daniel Florez-Orrego and François Maréchal, and afternoon breakout sessions focusing on data frameworks and case studies, participants approached analytical methods and tools for maximizing the value of industrial symbiosis in bioeconomy value chains. The presentation of Task XXIV representatives highlighted the importance of using modelling frameworks and databases, as well as systematic approaches and tools to identify opportunities and implementation barriers through quantitative energy, environmental, and financial risk assessment via scenario analyses.



Figure 15: Joint workshop on Industrial Symbiosis co-organized by Task XI and Task XXIV, held on December 2nd 2025 at the ISEC Lisbon, Portugal, at the International Conference on Bioenergy & Bioeconomy (IBBCE 2025).

In addition, the 90-minute Task XXIV Special Session showcased special research results spanning industrial sectors from aluminium production to chemical and food manufacturing, demonstrating the versatility and practical applicability of process integration methodologies for industry decarbonization. Six presentations illustrated how to achieve low- to carbon-negative operations through (i) biomass energy and renewable electricity integration for aluminium production, (ii) transformation of whiskey distilleries into energy-flexible systems via heat pumping, (iii) conversion of municipal solid waste into chemical solvents, and (iv) bio-waste valorization in combination with solid oxide cells for biomethane production. The Belgian industrial symbiosis case study and the Colombian energy systems assessment collectively demonstrated that rigorous thermodynamic analysis combined with life cycle assessment provides robust evidence for decision-making across diverse geographical and industrial contexts, from European clusters to Latin American national energy transitions. The presentations illustrated how systematic process integration approaches enable quantification of decarbonization potential, economic viability assessment, and identification of synergies within industrial ecosystems. This session reinforced Task XXIV's important role in advancing decision-support tools and methodologies for design and optimization of integrated energy systems.





Figure 16: Special session of the Task XXIV on Process Integration for Industry decarbonization, held in December 4th 2025 within the International Conference on Bioenergy & Bioeconomy (IBBCE 2025), December 3rd – 5th 2025, ISEC Lisbon, Portugal.

Presentations in the Special Session of the Task XXIV International Conference on Bioenergy & Bioeconomy – IBBCE 2025, December 3-5, 2025 XXIV in IBBCE conference:

1. Mitraki, R., Salman, M., Flórez-Orrego, D., Maréchal, F., Léonard, G., Accelerating Industrial Decarbonization through Symbiosis: Insights from the Belgian Case. International Conference on Bioenergy & Bioeconomy – **IBBCE 2025**, December 3-5, 2025. Lisbon, Portugal.
2. Ribeiro Domingos, M., Boztas, O., Flórez-Orrego, D., Dardor, D., Maréchal, F., Integrated Biowaste and Biogenic CO2 Valorization, and Solid Oxide Cells for Enhanced Sustainable Natural Gas Production. International Conference on Bioenergy & Bioeconomy **– IBBCE 2025**, December 3-5, 2025. Lisbon, Portugal.
3. Boztas, O., Flórez-Orrego, D., Ribeiro Domingos, M., Maréchal, F., Municipal Solid Waste Valorization into Chemical Solvents for Industrial Symbiosis: Techno-Economic and Environmental Assessment. International Conference on Bioenergy & Bioeconomy – **IBBCE 2025**, December 3-5, 2025. Lisbon, Portugal.
4. Dardor, D., Ribeiro Domingos, M., Flórez-Orrego, D., Germanier, R., Margni, M., Maréchal, F., Integration of Biomass Gasification and Power-to-Hydrogen for Enabling Carbon-Negative Aluminium Production. International Conference on Bioenergy & Bioeconomy – **IBBCE 2025**, December 3-5, 2025. Lisbon, Portugal.
5. Bhatnagar, P., Flórez-Orrego, D., Boztas, O., Ribeiro Domingos, M., Maréchal, F., Transforming Whiskey Production into a Carbon-Negative and Energy-Flexible System via Heat Pumping and Renewable Energy Integration, International Conference on Bioenergy & Bioeconomy **– IBBCE 2025**, December 3-5, 2025. Lisbon, Portugal
6. Silva Ortiz, P. Assessment of the Impact of Alternative Energy Resources in Colombia and their Relationship with Climate Change from a Thermodynamic Perspective and the Sustainable Development Goals, International Conference on Bioenergy & Bioeconomy – **IBBCE 2025**, December 3-5, 2025. Lisbon, Portugal

Finally, the two-hour First in-person intra-task meeting provided Task XXIV participants with necessary face-to-face collaboration opportunities for strategic coordination, relationship building among international members, and discussion of the Task's ongoing activities and future directions. The meeting's timing immediately following the Joint Workshop and Special Sessions allowed Task participants to build upon the technical discussions to address governance, coordination mechanisms, and strategic priorities for the ongoing work in model-sharing frameworks, industrial case study development, and collaborative research activities. Further details regarding the specific outcomes and decisions from this meeting are documented in the previous sections.

- *Additional technical contributions to international conferences by Task XXIV Participants*

Other presentations to international conferences and events are listed below:

1. Fortet Casabella, J. Greenhouse Gas Control Conference (GHGT-17), October 2024.
2. Harvey, S., Fortet Casabella, J., Comparison of RED III Compliance for Synthetic Fuel Production Routes for the Aviation Sector Using Exergoeconomic Analysis Tools. Trondheim Carbon Capture and Storage (TCCS-13), June 2025. Trondheim, Norway.
3. Fortet Casabella, J., Towards Full Carbon Recovery in a Dual Fluidized Bed Steam Cracker (DFB-SC) for Thermochemical Recycling of Plastic Waste. Advanced Recycling Conference (ARC 2025), November 2025.
4. Salman, M., Decision Support Framework for Industrial Energy Transition and Decarbonization: A Techno-Economic Approach. Energy Summer School 2025: Navigating the Energy Transition in an Insecure World, August 31 – September 5, 2025. Ascona, Switzerland. (Best Poster Award)
5. Salman, M., Costa, A., Mitraki, R., Kim, S.M., Fakhraddinfakhriazar, S., Beguin, B., Florez-Orrego, D., Weireld, G., Maréchal, F., Léonard, G., Electrifying Post-Combustion Carbon Capture: How Future Energy Scenarios Reshape Techno-Economic Performance. Post-Combustion Capture Conference (PCCC8), September 16-18, 2025. Marseille, France.

- *Publications in international journals*

Multiple publications in international journals have been published by Task XXIV participants under the

supervision of Task XXIV managers, highlighting the importance of the modelling and simulation approaches leveraging the process integration methods and tools.

1. Florez-Orrego, D., Dardor, D., Germanier, R., Margni, M., Maréchal, F., A systemic study for decarbonizing secondary aluminium production via waste heat recovery, carbon management and renewable energy integration. **Energy Conversion and Management**, v. 341, 2025, pp. 120021. https://doi.org/10.1016/j.enconman.2025.120021
2. Corre, S., Ribeiro Domingos, Florez-Orrego, D., M., Magnaval, G., Margni, M., Maréchal, F., Life cycle assessment and techno-economic comparison of methane production routes from sewage sludge: Incineration vs. Hydrothermal Gasification and Anaerobic Digestion. **Energy Conversion and Management**. v. 341, 2025, pp. 120015. https://doi.org/10.1016/j.enconman.2025.120015
3. Salman, M., Flórez-Orrego, D., Coppitters, D., Mitraki, R., Maréchal, F., Léonard, G. Decarbonizing the Glass Industry: A Comprehensive Techno-Economic Assessment of Low-Emission Pathways. **Computers & Chemical Engineering**. 8 August 2025, 109329 https://doi.org/10.1016/j.compchemeng.2025.109329
4. Dardor, D., Florez-Orrego, D., Germanier, R., Margni, M., Maréchal, F., Decarbonizing the aluminium industry: A comprehensive review of pathways and process integration perspectives. **Energy Strategy Reviews**. Volume 61, September 2025, 101853 https://doi.org/10.1016/j.esr.2025.101853
5. Boztas, O., Ribeiro-Domingos, M., Florez-Orrego, D., Maréchal, F., Techno-economic and environmental assessment of sustainable thermochemical pathways for olefins production promoting carbon circularity. **Renewable Energy**, 21 August 2025, 124303. https://doi.org/10.1016/j.renene.2025.124303
6. Dardor, D., Florez-Orrego, D., Germanier, R., Margni, M., Maréchal, F., Modelling the impact of future uncertainty in energy prices on aluminium decarbonization pathways. **Energy Conversion and Management**, v 348, Part A, 15 January 2026, 120579. https://doi.org/10.1016/j.enconman.2025.120579
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8. Cortvriendt, L., Florez-Orrego, D., Maréchal, F., Bongartz, D., Data-driven systematic methodology for predicting optimal heat pump integration based on temperature levels and refrigerants. **Energy Conversion and Management**, Volume 326, 15 February 2025, 119495. https://doi.org/10.1016/j.enconman.2025.119495.

- *Other dissemination, reporting and knowledge-transfer activities*

Strategic partnerships with international professional societies enhance dissemination efforts. The Energy Section of the European Federation of Chemical Engineering[8] (EFCE) and the ECOS[9] (Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems) society are non-profit organizations chaired or assisted by Task XXIV managers and several Task participants. Both will contribute as non-funded/non-profit partners through their dissemination and networking missions. EFCE specializes in organizing events, publishing proceedings, and facilitating cooperation activities among researchers through its regional and international networks. ECOS, with its focus on energy systems analysis and environmental impact assessment, provides an ideal platform for sharing process integration methodologies and sustainability metrics developed within the Task XXIV.

The Task XXIV managers also participated in the IETS Executive Committee meetings held on May 15th 2025 (40th ExCo Meeting) and November 12-13, 2025 (41st ExCo Meeting). These meetings provided

[8] https://efce.info/Section_Energy.html
[9] https://ecosinternationalsociety.org/index.html

opportunities to report on Task XXIV progress, align collaboration with the broader IETS strategic objectives, as well as discussions on communication strategies. Task XXIV managers also contributed to several IETS communication channels coordinated by the Secretariat. These include the IETS Newsletter (winter edition 2026), which highlights Task activities and key achievements; the Annual Brief to the IEA, which reports key indicators such as publications, events, and participant numbers; and the IETS Annual Report, which documents major achievements, participating countries, and planned activities. These reporting mechanisms allow Task outcomes reach a broader audience of policymakers, industry leaders, and researchers beyond the immediate Task participants.

In addition to IETS-level reporting, the Task coordination activities funded by the Swiss Federal Office of Energy (SFOE) are documented in this interim report submitted to the Swiss IETS contracting party. The report details the progress of coordination activities, budget status, and alignment with SFOE research priorities in process integration, industrial decarbonization, and renewable energy integration. A parallel SFOE-funded project, PinchSmall, aims to deliver a reference Pinch Analysis tool for the Swiss Confederation. This project has embedded a Pinch Bot application that draws on the structured information and standardized ontologies proposed in IETS Task XXIV. By feeding this curated data to Large Language Models, domain-specific AI assistants for process integration and industrial decarbonization can be developed. Such applications have the potential to support policymakers with science-based recommendations, assist engineers in preliminary feasibility assessments, and provide interactive learning tools for continuing education programs.

- *Supervision of semester project and master thesis related to the IETS Task XXIV*

- Khalil, Celia. Enabling a Shared Database of Ex-Ante Models of Industrial Processes, Decarbonization Technologies and Energy Systems Featuring Sustainability Metrics. Semester Project, École Polytechnique Fédérale de Lausanne, Switzerland. IETS TCP IEA, Supervisors: D. Flórez-Orrego, O. Boztas. January 2026.

- Wiart, Hermione. Data and Ex-Ante Models Sharing to Enable Knowledge Transfer of Industrial Decarbonization Technologies. Semester Project, École Polytechnique Fédérale de Lausanne, Switzerland. IETS TCP IEA, Supervisors: Daniel Flórez-Orrego, Oktay Boztas. January 2026.

- Mannik, Cecilia. Enabling a Database of Open-Source Blueprint Models for Process Integration: Definitions and Procedures. Master Thesis, École Polytechnique Fédérale de Lausanne, Switzerland. IETS TCP IEA, Supervisors: Daniel Flórez-Orrego, Oktay Boztas. June 2025.

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7 Appendices and Annexes

This chapter contains three appendices, which are part of this report,

- Appendix A. Review of ontologies and interoperability standards: comparison of OWL, Onto-CAPE, PROCO, FMI, CAPE-OPEN and DEXPI, supporting Section 2.1 and Table 2.
- Appendix B. Ex-ante model template: mathematical formulation, class-attribute tree, examples of each template section and planned extensions of the framework, supporting Section 2.2.
- Appendix C. Coordination meetings and participants' survey: detailed account of the plenary and bilateral meetings, complete survey results and first in-person meeting, supporting Section 2.4.

Four annexes are supplied as separate files:

- Annex 1. Participants' survey form (Microsoft Forms).
- Annex 2. Participants' survey answers (Microsoft Forms).
- Annex 3. Contributions of the participant countries to the Task XXIV reporting (PDF files).
- Annex 4. Spreadsheet of the ex-ante model template (Excel file).

Appendix A. Review of ontologies and interoperability standards

This appendix details the review of ontologies and software interoperability standards summarized in Table 2. The challenge lies in selecting an ontology that is sufficiently expressive to capture the richness of process integration applications, while remaining tractable for practical implementation and long-term adoption [19]. Heavyweight ontologies like OWL, OntoCAPE, PROCO with thousands of entities may offer comprehensive coverage, but entail significant adoption costs. OWL (Web Ontology Language), a standard language for creating ontologies, provides computer-readable definitions of classes, properties, and relationships that enable automatic integration across computational environments. Notwithstanding, OWL's steep learning curve, complex modelling of n-ary relationships, resource intensity or limited expressivity (depending on the version) make it unsuitable for process engineering applications. OntoCAPE[10] (Ontology for Computer-Aided Process Engineering) is an ontology that comprises 62 OWL, over 800 entities and 5000+ axioms. It provides a comprehensive formalization of process engineering concepts and covers a broad scope including chemical processes, plant structures, materials, and mathematical models. OntoCAPE has a monolithic, self-contained architecture that defines its own upper-level concepts and meta-models, rather than building on foundational ontologies [20]. Although it ensures internal consistency, it also presents challenges for interfacing with other domain ontologies (environmental impact databases, economic assessment frameworks, etc.). Anyhow, OntoCAPE offers valuable reference material for defining process units, streams, and thermodynamic properties to be adapted in a more flexible, simplified ontology framework. On the other hand, PROCO (Process Chemistry Ontology) is a process chemistry ontology that promotes community-based development following shared principles of openness, orthogonality, and interoperability [21]. While it focuses on product quality and compliance aspects of process chemistry, it can offer lessons about how process engineering ontologies might evolve to support cross-domain applications, community governance and iterative refinement [22].

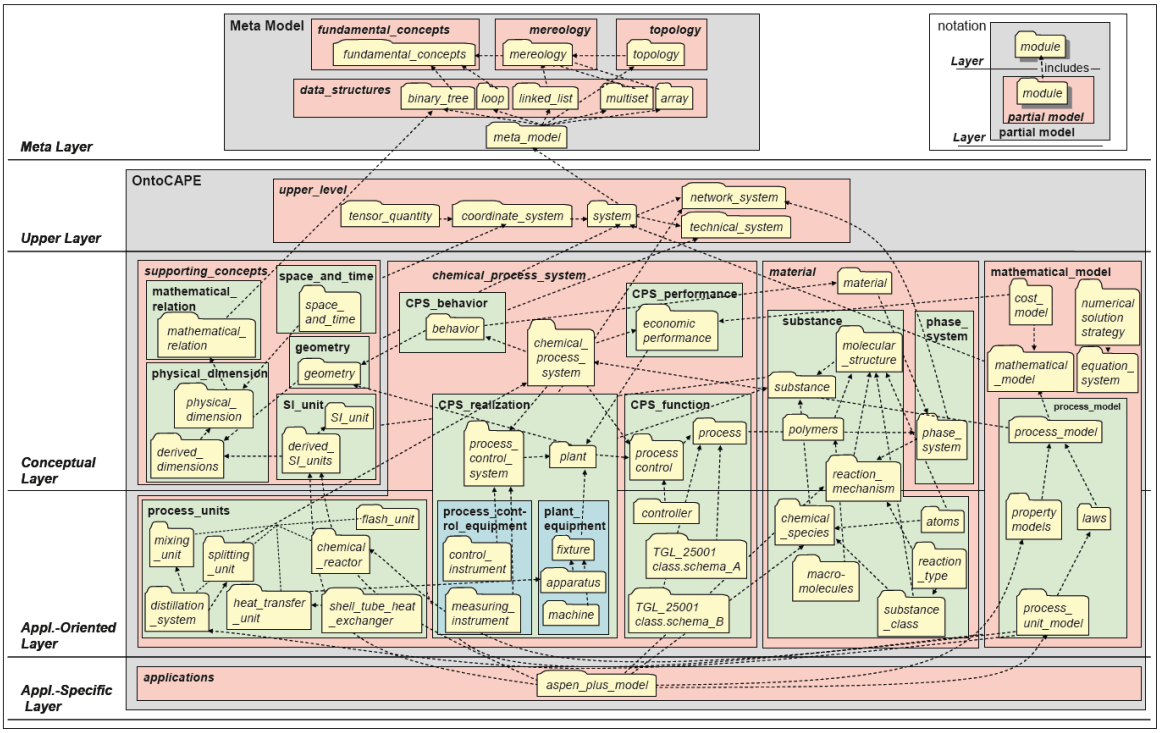


Figure A1: Structure of OntoCAPE as an ontology for chemical process engineering [20].

[10] OntoCAPE is an ontology. CAPE-Open is an interface standard for software interoperability.

It is worth noticing that, while ontologies define *what* concepts exist and *how* they relate; open standards, like FMI, Cape-OPEN and DEXPI specify technical protocols, file formats, APIs, and validation procedures for software systems interfaces to enable data exchange, model coupling and interoperability between different tools. The Functional Mock-up Interface (FMI) by Modelica [23] is a co-simulation and model exchange standard adopted in energy systems domains. An FMU (Functional Mock-up Unit) packages several standardized components, such as a model description (XML file defining interfaces, parameters, and capabilities), compiled C code implementing the mathematical model, documentation, auxiliary resources (e.g., lookup tables, configuration files), a standardized API for communication with simulation tools, and platform-specific binaries for different operating systems. Its primary strength lies in dynamic simulation, with mechanisms for time-stepping coordination, event handling, and state management. The focus on dynamic behaviour limits its applicability to steady-state models and hinders adoption in the process integration practice. In fact, for process integration purposes, infrastructure such as compiled code, simulation APIs, and platform binaries to support executable model coupling is, in principle, unnecessary. Figure A2 illustrates the FMU structure and interface architecture.

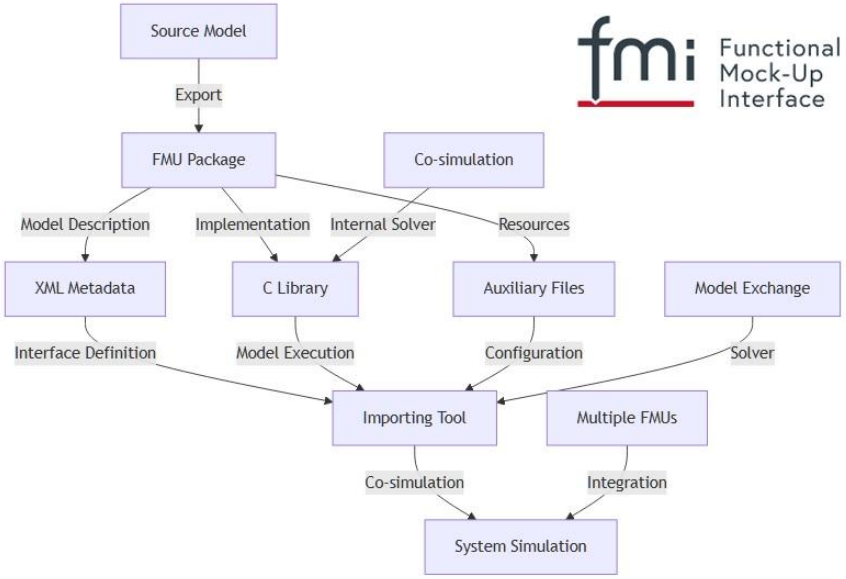
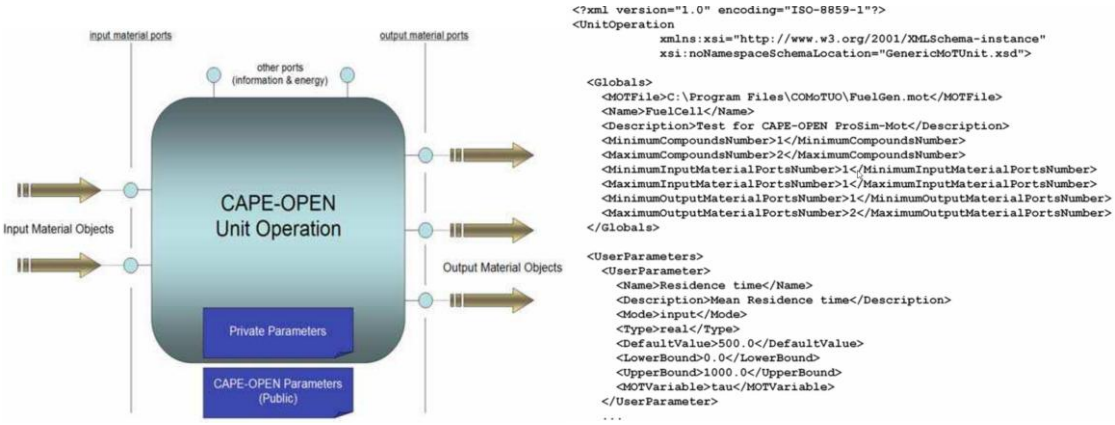


Figure A2: Functional mock-up unit and interface structure [23].

CAPE-Open specifications, developed by CO-LaN, define interfaces for unit operations, thermodynamic property packages, reaction kinetics, and numerical solvers [24,25]. Unlike FMI, CAPE-Open targets steady-state simulations and the component-based architecture of flowsheeting software. The interface distinguishes between material, energy, and information ports, with parameters classified as public or private (Figure A3a). Process Modelling Components (PMCs), such as CoolProp, interface with Process Modelling Environments (PMEs) like DWSIM through middleware, with data exchange via DLL files and XML configuration. The XML configuration maps variables through three blocks: "Globals" (unit operation metadata), "UserParameters" (modifiable parameters with bounds), and "MaterialPorts" (connection definitions for data exchange). However, direct XML editing is not user-friendly (Figure A3b). For IETS Task XXIV, it is therefore more practical to adopt a model template that incorporates CAPE-Open's essential characteristics in a more readable format, such as spreadsheets or JSON.



(a) (b)

Figure A3: CAPE-OPEN standard: (a) Unit operation ports, sockets, and parameters (b) Structure of XML configuration file [25].

Lastly, DEXPI (Data Exchange in the Process Industry) reportedly addresses challenges of plant data exchange, extending existing standards like ISO15926 to better specify classes. This specification provides comprehensive data models, covering plant breakdown structure (site, plant, area, process unit), equipment taxonomies, tag classifications, piping components, instrumentation objects, and topology relationships (Figure A4). It encompasses DEXPI P&ID for piping and instrumentation diagram exchange, and DEXPI Process for top-down design of ex-ante models (Figure A4) [26]. Notably, this infrastructure is more comprehensive than required for process integration purposes. Thus, as with FMI and CAPE-Open, relevant classes and hierarchies from DEXPI standard can be adopted to propose a simplified template focused on essential process integration data for IETS Task XXIV model exchange.

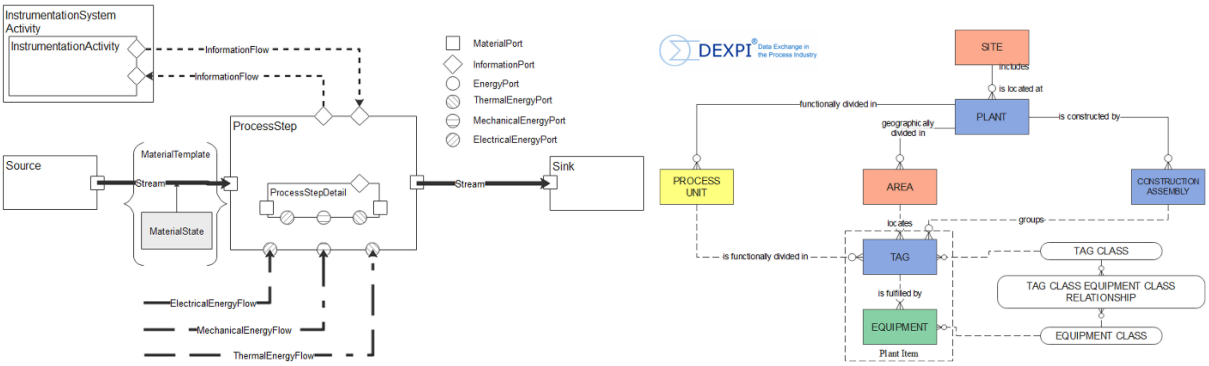


Figure A4: Definition of classes in the DEXPI Process Data Model [26,27].

Appendix B. Ex-ante model template: formulation, structure and planned extensions

This appendix presents the mathematical formulation of the ex-ante model, the class-attribute tree of the exchange template, example entries for each template section (Figures B1 to B12) and the planned extensions of the framework (Figures B13 and B14).

The plant breakdown structure based on ISO 10209, shown in Figure B1, provides a guideline for outlining the different scopes and subdivisions of industrial process and energy technology models. An enterprise (i.e., economic sector) undertaking a process integration project typically encompasses multiple firms within an industrial complex, each featuring distinct industrial clusters. These clusters contain process plants that can be represented by ex-ante models, which are the core entities of process integration analysis. Models are composed of process units corresponding to different plant sections, responsible for processing, treatment, utility supply, storage, and other productive activities at either fixed or variable operating regimes. Each plant item (e.g., filter, column, rotary equipment, vessel, reactor, heat exchanger, or pipeline) is linked to uniquely identified unit operations (e.g., separation, mechanical operation, storage, conversion, heat and mass transport) within the process plant. These items contribute to the mass, heat, power, and equipment-related information flows (see Figure B2) that define the process unit integration opportunities.

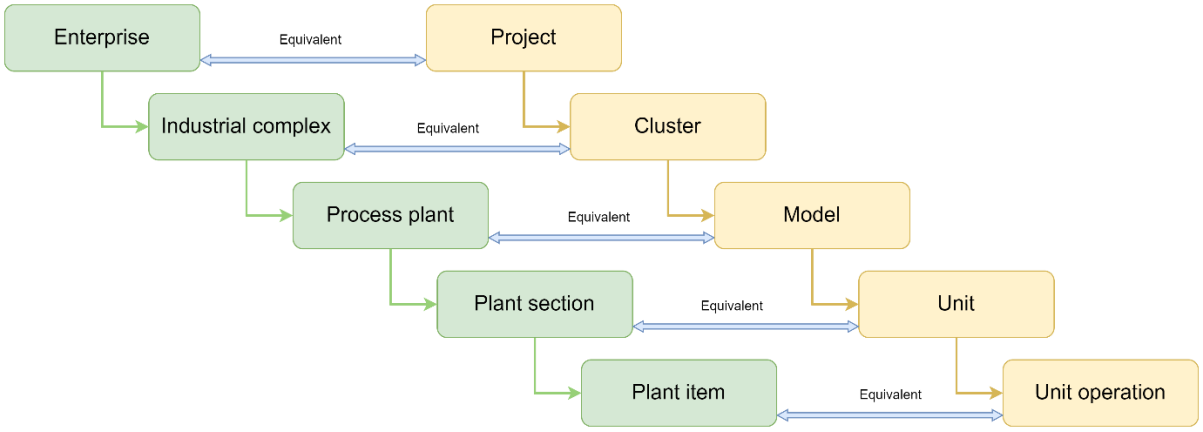


Figure B1: Plant breakdown structure based on ISO 10209 (adopted by DEXPI) [28].

Figure B2 illustrates the structure of an ex-ante model that contains a process unit U with a nominal size S^*_u. The unit has an associated binary variable y_u (defining whether unit U exists) and a continuous load factor f_u (setting the capacity utilization factor of unit U). It is worth noticing that, depending on the level of detail (white, grey, or black box modelling), different representations may exist with corresponding binary variables $y_{u,i}$ and load factors $f_{u,i}$ for each interface i. For instance, a white box model for unit U may involve detailed mass and heat flows at the deepest level of process knowledge, including heating/cooling demands and molar compositions. On the other hand, a black box model may provide only knowledge about the overall fuel and power consumption at the plant battery limits. In any case, both approaches should lead to consistent conclusions regarding overall material and energy balances, regardless of the internal detail adopted.

Internal transformations are governed by thermo-physical models and constitutive equations, which determine the mass flows ($\dot{M}^-_{m,u,i} = F_{M,u,i}[\pi_{u,i}, S^*_u, (x\cdot\dot{M})^+_{m,u,i}]$), thermal demands ($\dot{Q}_{T,u,i} = F_{Q,u,i}[\pi_{u,i}, S^*_u, (x\cdot\dot{M})^+_{m,u,i}]$), electricity demands ($\dot{E}E_{u,i} = F_{EE,u,i}[\pi_{u,i}, S^*_u, (x\cdot\dot{M})^+_{m,u,i}]$), and equipment-related correlations ($S_{Eq,u,i} = F_{Eq,u,i}[\pi_{u,i}, S^*_u, (x\cdot\dot{M})^+_{m,u,i}]$). These equations are parametrized by physical constants and operating conditions ($\pi_{u,i}$) and input compositions $(x\cdot\dot{M})_{m,u,i}$. Equipment impact accounting encompasses the construction ($S_{Eq,u,i} \cdot \dot{m}^{-C}_{Eq,e,i}$), lifetime auxiliary flows ($S_{Eq,u,i} \cdot \dot{m}^{-U}_{Eq,e,i}$), and end of life ($S_{Eq,u,i} \cdot$

$\dot{m}^{-EoL}_{Eq,e,i}$). These relationships define the ex-ante model as a bounded entity with clearly defined connections with an ex-ante model superstructure and the industrial infrastructure through mass, energy, heat, and information flows. Those connections include inlet material flows of feedstock ($\dot{M}^{+}_{F,i}$) and support material, energy, and waste streams ($\dot{M}^{+}_{r,i}$); outlet material flows as products ($\dot{M}^{-}_{P,i}$) and elementary flows to environment ($\dot{M}^{-}_{e,i}$); as well as interfaces for electricity input/output ($\dot{E}E^{+-}_{u,i}$) and heat cascade contributions (heating/cooling demands $\dot{Q}^{+-}_{T,u,i}$). In this way, the model supports process integration analysis by characterizing heat cascades and mass balances, while capturing lifecycle impacts from equipment construction through end of life.

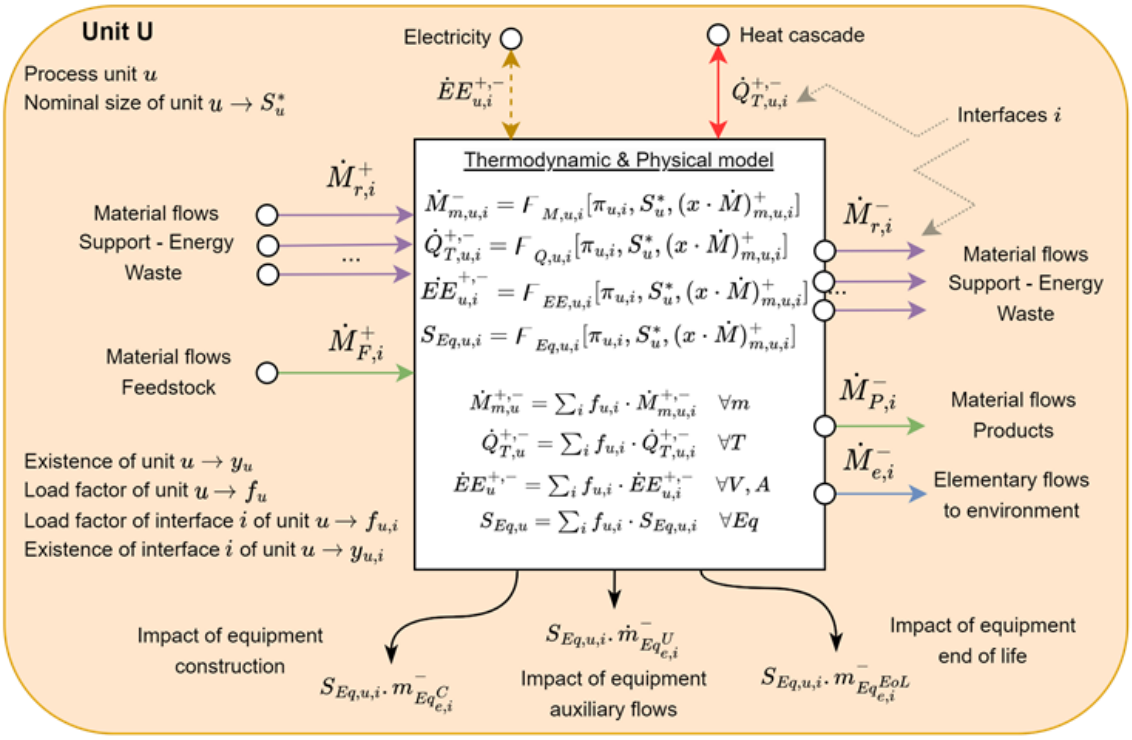


Figure B2: General representation of the mass, heat, power, and equipment-related information flows that define the process unit integration opportunities [13].

Once the mathematical formulation of the ex-ante model is established, a class-attribute tree can provide a formal representation of the information architecture (metadata, data, and specifications) underlying the ex-ante model exchange template (Figure B3). At the root level, a MODEL entity branches into primary sub-entities, namely METADATA (status, attribution, reference, and documentation), CHANGELOG (version and changes tracking), SUPPL MATERIAL (auxiliary files indexation), VARIABLES (specification of engineering parameters and model variables), UNIT (streams and equipment tables definition to account for process units' transformations), and CALCULATIONS (derived quantities in an equation-oriented approach observing model constraints). The tree structure ensures that each information element has a defined location, type, and relationship to other elements in the proposed simplified ex-ante model template (Figures B4–B12), enabling consistent interpretation and operationalization across Task XXIV participating institutions. Beyond technical model content, establishing a shared database of ex-ante models requires additional dimensions for metadata covering hosting, storage, versioning, intellectual property and confidentiality management, software information, and previsualization and referencing capabilities. This cooperative effort entailed assigning persistent identifiers to models and datasets, developing metadata schemas aligned with process engineering conventions, establishing accessible repositories with validation mechanisms, and supporting layered disclosure protocols for handling authorship and confidentiality concerns.

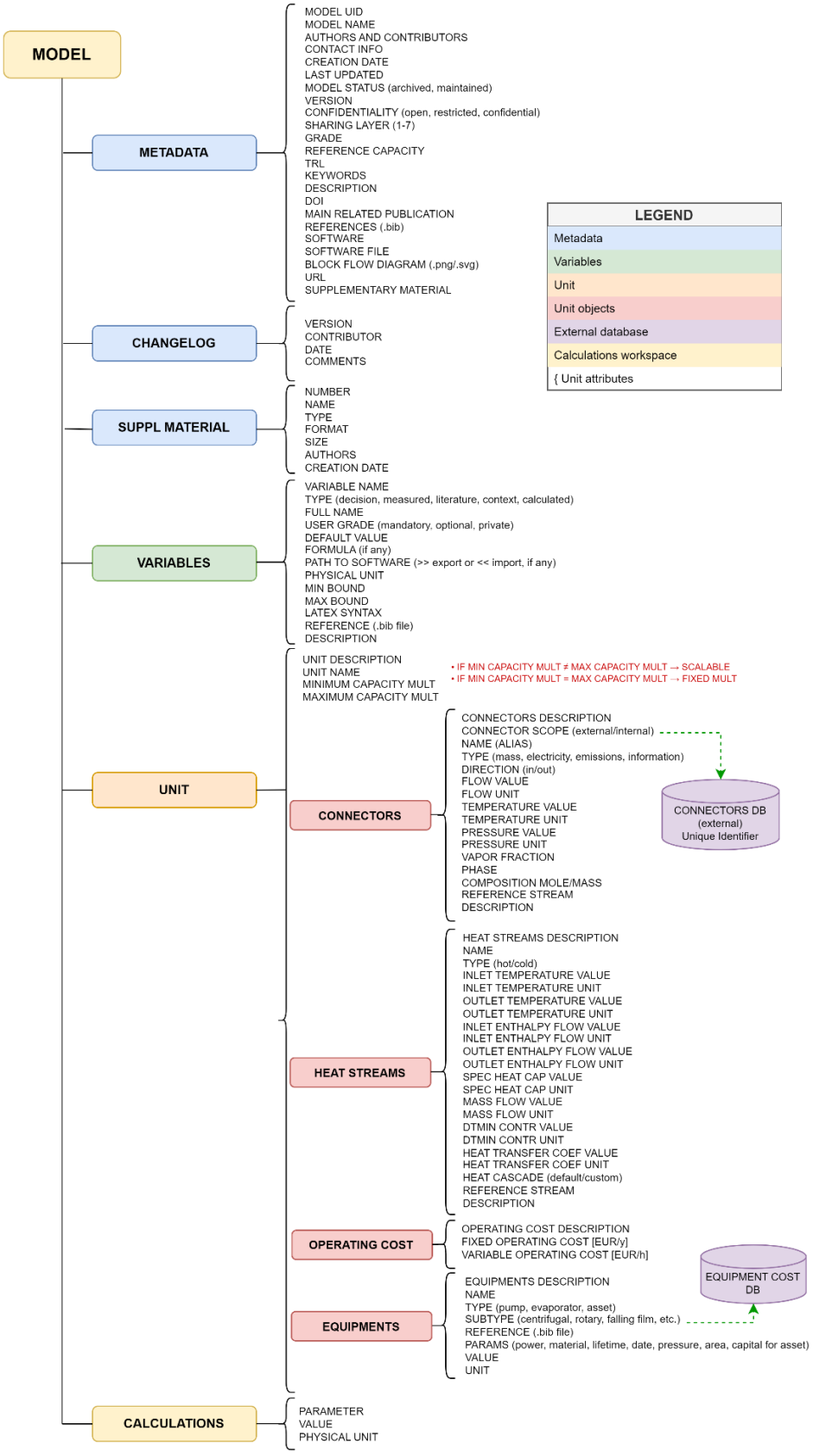


Figure B3: Ex-ante model class-attribute tree proposed for the model exchange template.

METADATA implements these attributes; CHANGELOG and SUPPL MATERIAL provide version tracking and auxiliary file indexation; VARIABLES captures parameter specifications with type classification (decision, measured, literature, context, calculated) and user grade (mandatory, optional, private) controlling access and modification rights. To support reproducibility and documentation, the template includes descriptions of key variables, assumptions, modelling tools, and block flow diagrams, accommodating both analytical formulas expressing relationships between engineering variables and paths to flowsheeting software for automated data extraction (e.g., Aspen Plus variables).

The UNIT section constitutes the technical core of the template, encompassing four sub-components essential for process integration analysis. CONNECTORS specify boundary flows (mass, electricity, emissions, information) with quality attributes (composition, temperature, pressure, vapor fraction, phase) and direction, enabling explicit identification of model interfaces to broader infrastructure and superstructure. External database references for CONNECTORS ensure standardization and reuse across models. HEAT STREAMS document the heating and cooling demands arising from waste heat recovery, chemical reactions, phase changes, and dissipative devices (compressors, pumps, motors), providing temperature levels, enthalpy flows, heat capacity, mass flow, minimum temperature difference contributions, and heat transfer coefficients. These parameters enable the construction of composite curves and identification of heat integration opportunities across model boundaries. OPERATING COST captures fixed and variable cost components for economic assessment, while EQUIPMENTS provides equipment-related specifications linking to external (investment) cost databases. The capacity multiplier logic (MINIMUM/MAXIMUM CAPACITY MULT) distinguishes between fixed process units and scalable utilities, supporting optimization formulations. This structure ensures that models are technically complete, discoverable, traceable, and comparable between participating institutions.

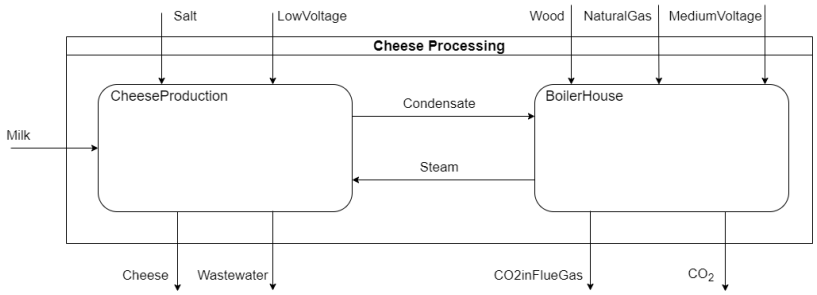


Figure B4: Block flow diagram of an example of ex-ante model comprising two units. Connectors are classified as internal (exchanges between units within the model boundary) or external (interfaces to the broader superstructure and infrastructure).

MODEL UID	123-456-789
MODEL NAME	CheeseProcessing
AUTHORS AND CONTRIBUTORS	Daniel Florez-Orrego, Oktay Boztas
CONTACT INFO	daniel.florezorrego@epfl.ch, oktay.boztas@epfl.ch
CREATION DATE	01.06.2025
LAST UPDATED	18.06.2025
MODEL STATUS	Maintained
VERSION	1.0.1
CONFIDENTIALITY	Open
SHARING LAYER	4
GRADE	GRAY-BOX
REFERENCE CAPACITY (MAIN PRODUCT/FEEDSTOCK)	10 kg/h Cheese
TRL	9
KEYWORDS	Cheese, Food
DESCRIPTION	This model represent a dairy process for cheese production from fresh milk, using a wood and natural gas powered boiler house for utility supply. The boiler house unit also includes an amine-based post-combustion carbon capture system.
DOI	https://doi.org/10.0/9887.663A
MAIN RELATED PUBLICATION	https://linktomypublication.com
REFERENCES	cheesefactory.bib
SOFTWARE	ASPEN
SOFTWARE FILE	CheesePlant.bkp
BLOCK FLOW DIAGRAM (.png or .svg)	local/path/to/CheeseProcessing.svg
URL	www.gitlab.epfl.ch/cheeseprocessing
SUPPLEMENTARY MATERIAL	See SUPPL MATERIAL tab

Figure B5: Metadata definition for an example of ex-ante model, specifying version, authorship, confidentiality, disclosure layer, grade, reference capacity, modelling paradigm, and software information.

EXTERNAL CONNECTORS

UID	NAME (ALIAS)	TYPE	FLOW UNIT	TEMPERATURE	TEMPERATURE UNIT	PRESSURE	PRESSURE UNIT	VAPOR FRACTION [MASS]	PHASE	COMPOSITION [MOLE/MASS]	DESCRIPTION
MILK-FAT3	MILK	MASS	kg/h	25	C	1	bar	-1	(L)	MASSFRAC:FAT[0.03]&H2O[0.97]	Infeed milk flow into the cheese production unit
CHEESE-SOFT	CHEESE	MASS	kg/h	25	C	1	bar	-1	(S)	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	Produced cheese flow from the cheese production unit
SALT-NACL	SALT	MASS	kg/h	25	C	1	bar	-1	(S)	MASSFRAC:NACL[1]	Consumed salt flow into the cheese production unit
WASTEWATER-SEWAGE1	WASTEWATER	MASS	kg/h	25	C	1	bar	-1	(L)	MASSFRAC:H2O[0.99]&FAT[0.01]	Produced waste water flow from the cheese production unit
TAP-WATER	WATER	MASS	kg/h	25	C	1	bar	-1	(L)	MASSFRAC:H2O[1]	Optionally consumed tap water flow into the boiler house
WOOD-DRY	WOOD	MASS	kg/h	25	C	1	bar	-1	(S)	MASSFRAC:C[0.47]&H[0.06]&O[0.45]&S[0]&ASH[0.02]	Consumed wood fuel flow into the boiler house
NATURALGAS	CH4	MASS	kg/h	5	C	1	bar	-1	(V)	MASSFRAC:CH4[1]	Consumed natural gas fuel flow into the boiler house
ELEC-50KV	MEDIUMVOLTAGE	ELECTRICITY	kW	-	-	-	bar	-	-	-	Consumed electricity 50 kV flow into the boiler house
ELEC-13KV	LOWVOLTAGE	ELECTRICITY	kW	-	-	-	bar	-	-	-	Consumed electricity 13 kV flow into the cheese production unit
FLUEGAS-M	CO2INFLUEGAS	EMISSIONS	kg/h	120	C	1	bar	-1	(V)	MOLEFRAC:CO2[0.15]&H2O[0.12]&N2[0.73]	Non-captured CO2 in flue gas flow from the boiler house
CO2-PURE	CO2	EMISSIONS	kg/h	25	C	1	bar	-1	(V)	MASSFRAC:CO2[1]	Captured pure CO2 flow from the boiler house

INTERNAL CONNECTORS

UID	NAME (ALIAS)	TYPE	FLOW UNIT	TEMPERATURE	TEMPERATURE UNIT	PRESSURE	PRESSURE UNIT	VAPOR FRACTION [MASS]	PHASE	COMPOSITION [MOLE/MASS]	DESCRIPTION
STEAM-3BAR-CHEESE	STEAM	MASS	kg/h	SAT	C	3	bar	1	(V+L)	MASSFRAC:H2O[1]	Steam vapor flow into the cheese production unit
COND-3BAR-CHEESE	CONDENSATE	MASS	kg/h	SAT	C	3	bar	0	(V+L)	MASSFRAC:H2O[1]	Condensate flow from the cheese production unit

Figure B6: External and internal connector definitions for an example of ex-ante model. Aliases are user-defined; external connectors must reference unique identifiers from the global connectors database.

VARIABLES

VARIABLE NAME	TYPE	FULL NAME	USER GRADE	DEFAULT VALUE OR FORMULA OR PATH TO SC	PHYSICAL UNIT	MIN BOUND	MAX BOUND	LATEX SYNTAX	REFERENCE	DESCRIPTION
MILK_flow	MEASURED	Milk inlet flowrate	MANDATORY	100.00	kg/h	50.00	150.00	F^{in}_{milk}	@Florez-Orrego et al	The amount of fresh milk delivered to the dairy plant
SALT_flow	MEASURED	Salt inlet flowrate	MANDATORY	1.00	kg/h	0.50	2.00	F^{in}_{salt}	@Boztas et al	The amount of salt required for the production of cheese
CHEESE_flow	MEASURED	Cheese outlet flowrate	MANDATORY	10.00	kg/h	5.00	40.00	F^{out}_{cheese}	@Ribeiro Domingos et al	The amount of cheese that is produced from the milk
WASTEWATER_flow	MEASURED	Waste water outlet flowrate	MANDATORY	91.00	kg/h	50.00	100.00	F^{out}_{WW}	@Bhatnagar and Baibhav	s further treatment before being disposed into the environment
LOWVOLTAGE_flow	MEASURED	Low voltage electricity requirement	MANDATORY	3.00	kW	1.50	4.50	E^{in}_{elecLV}	@Florez-Orrego et al	e low voltage electricity required in the dairy plant for pumping
STEAM_flow	MEASURED	Steam consumption flowrate	MANDATORY	74.69	kg/h	37.35	112.04	F^{in}_{steam}	@Boztas et al	Steam consumption of the dairy plant
CONDENSATE_flow	MEASURED	Return condensate flowrate	MANDATORY	74.69	kg/h	37.35	112.04	F^{out}_{cond}	@Ribeiro Domingos et al	The condensate return going back to the boiler house
pasteurization_heating_Tin	MEASURED	Milk inlet temperature	MANDATORY	25 (or >>/Streams/S27/TEMP)	°C	12.50	37.50	$T^{in}_{milk,past}$	@Florez-Orrego et al	The inlet temperature of the milk to be pasteurized
pasteurization_heating_Tout	MEASURED	Milk outlet temperature	MANDATORY	70.00	°C	35.00	105.00	$T^{out}_{milk,past}$	@Boztas et al	The target temperature for the pasteurization of the milk
pasteurization_heating_Hin	MEASURED	Milk inlet enthalpy flow	MANDATORY	0.00	kW	0.00	0.00	$H^{in}_{milk,past}$	@Ribeiro Domingos et al	Enthalpy content of the milk at the inlet temperature
pasteurization_heating_Hout	MEASURED	Milk outlet enthalpy flow	MANDATORY	5 (or <</Block/HEX101/QCALC)	kW	2.50	7.50	$H^{out}_{milk,past}$	@Bhatnagar and Baibhav	Enthalpy content of the milk at the outlet temperature
pasteurization_heating_cp	LITERATURE	Spec heat capacity of milk to pasteurization	OPTIONAL	4.00	kJ/kgK	2.00	6.00	$c^{milk,past}_{p}$	@Florez-Orrego et al	Specific heat capacity of the milk
pasteurization_heating_ms	MEASURED	Mass flow of milk to pasteurization	MANDATORY	100.00	kg/h	50.00	150.00	$m^{heat}_{milk,past}$	@Boztas et al	Amount of milk to be pasteurized
pasteurization_heating_dTmin	LITERATURE	Milk dTmin contribution	OPTIONAL	5.00	°C	2.50	7.50	$\Delta T_{\min}^{milk}$	@Ribeiro Domingos et al	ifference contribution to the heat exchange by the milk stream
pasteurization_heating_alpha	LITERATURE	Milk heat transfer coefficient	PRIVATE	0.50	kW/m2K	0.25	0.75	α^{milk}	@Florez-Orrego et al	oefficient contribution to the heat exchange by the milk stream
pasteurization_quenching_Tin	MEASURED	Milk quenching inlet temperature	MANDATORY	70.00	°C	35.00	105.00	$T^{in}_{milk,quench}$	@Florez-Orrego et al	et temperature of the milk to be quenched after pasteurization
pasteurization_quenching_Tout	MEASURED	Milk quenching outlet temperature	MANDATORY	25.00	°C	12.50	37.50	$T^{out}_{milk,quench}$	@Boztas et al	et temperature of the milk to be quenched after pasteurization
pasteurization_quenching_Hin	MEASURED	Milk quenching inlet enthalpy flow	MANDATORY	5.00	kW	2.50	7.50	$H^{in}_{milk,quench}$	@Ribeiro Domingos et al	Enthalpy content of the milk at the inlet temperature
pasteurization_quenching_Hout	MEASURED	Milk quenching outlet enthalpy flow	MANDATORY	0.00	kW	0.00	0.00	$H^{out}_{milk,quench}$	@Bhatnagar and Baibhav	Enthalpy content of the milk at the outlet temperature
pasteurization_quenching_cp	MEASURED	Spec heat capacity of milk to quenching	OPTIONAL	4.00	kJ/kgK	2.00	6.00	$c^{milk,quench}_{p}$	@Florez-Orrego et al	Specific heat capacity of the pasteurized milk
pasteurization_quenching_ms	MEASURED	Mass flow of milk to quenching	MANDATORY	100.00	kg/h	50.00	150.00	$m^{cool}_{milk,quench}$	@Boztas et al	Amount of milk to be quenched after pasteurization
pasteurization_quenching_dTmin	LITERATURE	Milk dTmin contribution	OPTIONAL	5.00	°C	2.50	7.50	$\Delta T_{\min}^{milk}$	@Ribeiro Domingos et al	ence contribution to the heat exchange by the pasteurized milk
pasteurization_quenching_alpha	LITERATURE	Milk heat transfer coefficient	PRIVATE	0.50	kW/m2K	0.25	0.75	α^{milk}	@Florez-Orrego et al	cient contribution to the heat exchange by the pasteurized milk
evaporation_preheating_Tin	MEASURED	Inlet temperature of milk for evaporation	MANDATORY	25.00	°C	12.50	37.50	$T^{in}_{milk,preh}$	@Florez-Orrego et al	let temperature of the pasteurized milk for the sensible heating
evaporation_preheating_Tout	LITERATURE	Boiling temperature of milk	MANDATORY	90.00	°C	45.00	135.00	$T^{out}_{milk,preh}$	@Boztas et al	get temperature of the pasteurized milk for the sensible heating
evaporation_preheating_Hin	MEASURED	Inlet enthalpy flow of milk before preheating	MANDATORY	0.00	kW	0.00	0.00	$H^{in}_{milk,preh}$	@Ribeiro Domingos et al	thalpy content of the pasteurized milk at the inlet temperature
evaporation_preheating_Hout	MEASURED	Outlet enthalpy flow of milk after preheating	MANDATORY	6.77	kW	3.39	10.16	$H^{out}_{milk,preh}$	@Bhatnagar and Baibhav	halpy content of the pasteuriezd milk at the outlet temperature
evaporation_preheating_cp	LITERATURE	Spec heat capacity of subcooled water	OPTIONAL	4.12	kJ/kgK	2.06	6.18	c^{H2O}_{p}	@Florez-Orrego et al	Specific heat capacity of the pasteurized milk
evaporation_preheating_ms	MEASURED	Mass flow of subcooled water	MANDATORY	91.00	kg/h	45.50	136.50	$m^{H2O}_{preheat}$	@Boztas et al	Amount of milk to be sensibly heated before phase change
evaporation_preheating_dTmin	LITERATURE	Milk dTmin contribution	OPTIONAL	5.00	°C	2.50	7.50	$\Delta T_{\min}^{milk}$	@Ribeiro Domingos et al	ence contribution to the heat exchange by the pasteurized milk
evaporation_preheating_alpha	LITERATURE	Milk heat transfer coefficient	PRIVATE	0.50	kW/m2K	0.25	0.75	α^{milk}	@Florez-Orrego et al	cient contribution to the heat exchange by the pasteurized milk

Figure B7: Engineering variable definitions for an example of ex-ante model, specifying type, user grade, physical units, value (default or formula or flowsheeting software path), reference, and description.

UNIT DESCRIPTION
This unit models the production of cheese, at a production capacity of 10 kg/h of cheese, with energy supply by steam at 3 bar.

UNIT NAME	MINIMUM CAPACITY MULT	MAXIMUM CAPACITY MULT
CheeseProduction	1	1

Figure B8: Definition for the name and capacity multipliers (minimum/maximum) of a unit(s) of an example of ex-ante model to establish the model implementation range.

CONNECTORS

NAME (ALIAS)	TYPE	DIRECTION	FLOW VALUE	FLOW UNIT	TEMPERATURE VALUE	TEMPERATURE UNIT	PRESSURE VALUE	PRESSURE UNIT	VAPOR FRACTION [MASS]	PHASE	COMPOSITION [MOLE/MASS]	REFERENCE STREAM	DESCRIPTION
MILK	MASS	IN	100.00	kg/h	25	C	1	BAR	-1	(L)	MASSFRAC:FAT[0.03]&H2O[0.97]	Yes	Infeed milk flow into the cheese production unit
SALT	MASS	IN	1.00	kg/h	25	C	1	BAR	-1	(S)	MASSFRAC:NACL[1]	No	Consumed salt flow into the cheese production unit
CHEESE	MASS	OUT	10.00	kg/h	25	C	1	BAR	-1	(S)	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	No	Produced cheese flow from the cheese production unit
WASTEWATER	MASS	OUT	91.00	kg/h	25	C	1	BAR	-1	(L)	MASSFRAC:H2O[0.99]&FAT[0.01]	No	Produced waste water flow from the cheese production unit
LOWVOLTAGE	ELECTRICITY	IN	3.00	kW	-	-	-	-	-	-	-	No	Consumed electricity 13 kV flow into the cheese production unit
STEAM	MASS	IN	74.69	kg/h	SAT	C	3	BAR	1	(V+L)	MASSFRAC:H2O[1]	No	Steam vapor flow into the cheese production unit
CONDENSATE	MASS	OUT	74.69	kg/h	SAT	C	3	BAR	0	(V+L)	MASSFRAC:H2O[1]	No	Condensate flow from the cheese production unit

Figure B9: Definition of the connectors (mass, electricity, emissions and information) of the unit(s) of an example of an ex-ante model.

HEAT STREAMS

NAME	TYPE	INLET TEMPERATURE VALUE	INLET TEMPERATURE UNIT	OUTLET TEMPERATURE VALUE	OUTLET TEMPERATURE UNIT	INLET ENTHALPY FLOW VALUE	INLET ENTHALPY FLOW UNIT	OUTLET ENTHALPY FLOW VALUE	OUTLET ENTHALPY FLOW UNIT
pasteurization_heating	Cold	25.0	C	70.0	C	0.00	kW	5.00	kW
pasteurization_quenching	Hot	70.0	C	25.0	C	5.00	kW	0.00	kW
evaporation_preheating	Cold	25.0	C	90.0	C	0.00	kW	6.77	kW
evaporation_latentheat	Cold	90.0	C	90.1	C	0.00	kW	18.00	kW
condensation_latent	Hot	90.1	C	90.0	C	18.00	kW	0.00	kW
condensation_subcooling	Hot	90.0	C	25.0	C	2.00	kW	0.00	kW
wastewater_cooling	Hot	90.0	C	25.0	C	4.77	kW	0.00	kW
steam_condensation	Hot	133.1	C	133.0	C	50.00	kW	0.00	kW
cheese_curation	Cold	15.0	C	25.0	C	0.00	kW	0.07	kW

SPEC HEAT CAP VALUE	SPEC HEAT CAP UNIT	MASS FLOW VALUE	MASS FLOW UNIT	DTMIN CONTR VALUE	DTMIN CONTR UNIT	HEAT TRANSFER COEF VALUE	HEAT TRANSFER COEF UNIT	HEAT CASCADE	REFERENCE STREAM	DESCRIPTION
4.00	kJ/kgK	100.00	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Milk pasteurization heating demand
4.00	kJ/kgK	100.00	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Milk pasteurization quenching demand
4.12	kJ/kgK	91.00	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Milk evaporation preheating demand
2410.00	kJ/kgK	26.89	kg/h	2.5	K	1.00	kW/m2K	CHEESEPLANT	No	Evaporation enthalpy of milk concentration
2410.00	kJ/kgK	26.89	kg/h	2.5	K	1.00	kW/m2K	CHEESEPLANT	No	Condensation enthalpy of water from milk concentration
4.12	kJ/kgK	26.89	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Subcooling of water separated in the milk concentration
4.12	kJ/kgK	64.11	kg/h	5	K	0.50	kW/m2K	CHEESEPLANT	No	Waste water cooling demand
2410.00	kJ/kgK	74.69	kg/h	2.5	K	1.00	kW/m2K	CHEESEPLANT	No	Heating demand provided by the steam condensation
2.53	kJ/kgK	10.00	kg/h	8	K	0.10	kW/m2K	DEFAULT	No	Cheese curation heating demand (maintain)

Figure B10: Definition of the heat streams (hot/cold) of the unit(s) of an example of an ex-ante model featuring specific heat cascades that characterize the waste heat recovery potential, temperature profiles and restricted heat exchanges.

EQUIPMENTS						
NAME	TYPE	SUBTYPE	REFERENCE	PARAMS	VALUE	UNIT
MilkPump	Pump	Single-stage centrifugal	@Towler and Sinnott	Power	2	kW
				Material	Stainles Steel	-
				Lifetime	20	years
				Date	2025	year
MilkEvaporator	Evaporators	Agitated falling film	@Towler and Sinnott	Area	3000	m2
				Pressure	1	bar
				Material	Stainles Steel	-
				Lifetime	20	years
				Date	2025	year
CheeseProdSys	Asset	-	@Florez-Orrego et al	Capital	75000	EUR
				Lifetime	20	years
				Date	2025	year

Figure B11: Definition of the equipment within the unit(s) of an example of ex-ante model, used for estimating the investment cost. Reference sizes are important to observe the costing intervals and account for the economies of scale.

FIXED OPERATING COST [EUR/y]	VARIABLE OPERATING COST [EUR/h]
0	0

Figure B12: Definition of the fixed and variable operating costs of the unit(s) of an example of ex-ante model for estimating the ancillary operating costs (labour, catalyst, etc.).

Although primarily designed for steady-state process integration analyses, the template architecture offers opportunities for extension. Consistent boundary specifications may accommodate multiscale and multi-time modelling involving different levels of detail, as well as batch and pseudo-transient operating regimes [29]. Typical days defined by temporal variation profiles, occurrence frequencies, and sequences for selected variables (e.g., in CSV data files) enable handling process integration problems leveraging, for example, storage solutions for intermittent renewable energy management (power-to-gas, batteries, pumped hydro storage). This versatile structure, separating metadata, data, and model interfacing/flowsheeting specifications, provides a foundation for further developments while preserving automated validation, indexing, and interoperability as Task XXIV activities evolve.

Task XXIV also aims to catalogue modelling approaches for energy infrastructure, waste treatment and recycling, distribution networks, and storage technologies (Figure B13), building upon previous and ongoing research projects and collaboration with researchers from ETSAP using MARKAL, TIMES, Energy Scope, and related frameworks. The goal is to examine the balance between local and regional energy production, consumption, and storage, and to share experiences on the impact of decentralized versus centralized renewable energy on decarbonization scenarios. The simulation of seasonal or limited capacity for supply and absorption by markets of feedstock, produced goods, and services, together with the allocation of impact indicator coefficients to documented engineering flows, will support strategies to quantify the social cost of carbon, linking industrial impacts to social and ecosystem effects for informed policy-making.

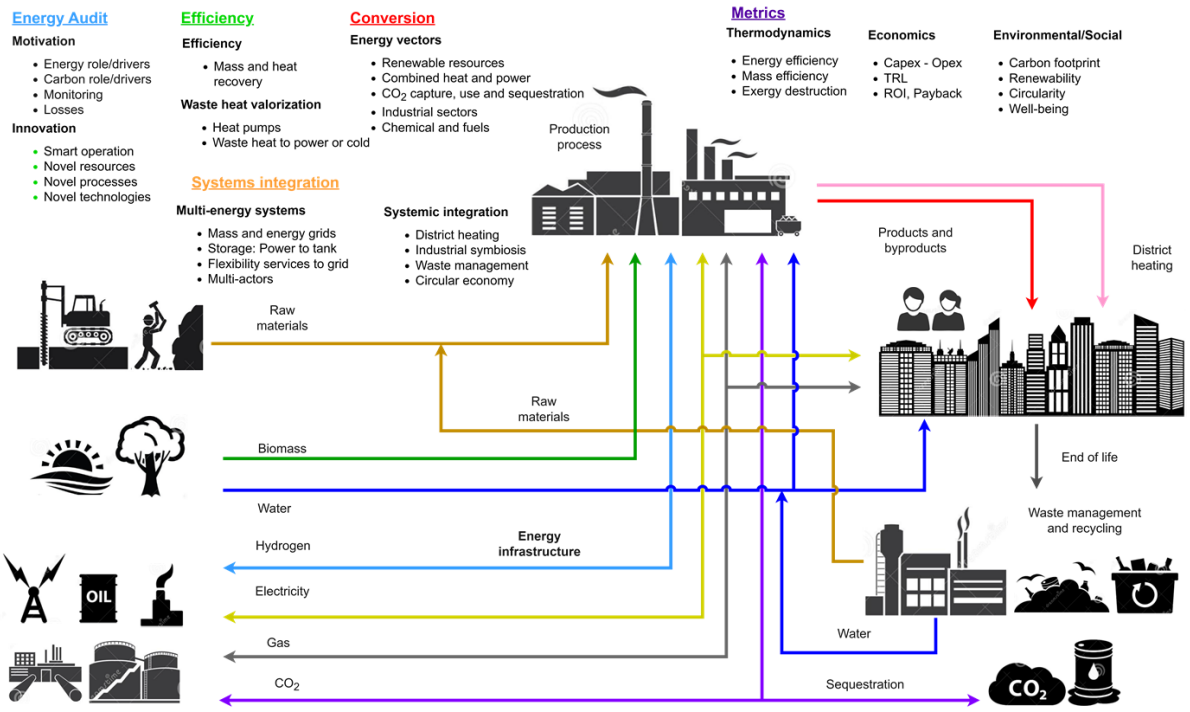


Figure B13: Exchanges between industrial processes and energy technologies with the energy infrastructure, waste treatment and recycling, distribution networks, urban systems and storage solutions [13].

Further discussions will embrace life cycle perspectives and material and carbon circularity to formulate preliminary conclusions on feasibility, cost-effectiveness, and the effects of net-zero technology integration on burden-shifting and broader sustainability goals (planetary limits, carbon footprint, externalities, health, resource depletion). These metrics can also serve to benchmark the performance of the different process integration and optimization frameworks used by Task participants. The goal is not to identify an optimal "correct" solution, but rather to gather and document diverse perspectives on potential symbiotic exchanges between industries and energy technologies and systems. It is also important to characterize the technical and environmental constraints that currently hinder integration strategies in real-world applications. To this end, a systematic framework will be collaboratively developed to capitalize on the database of ex-ante models and operationalize data analysis and visualization tools for generating comprehensively documented reports (Figure B14).

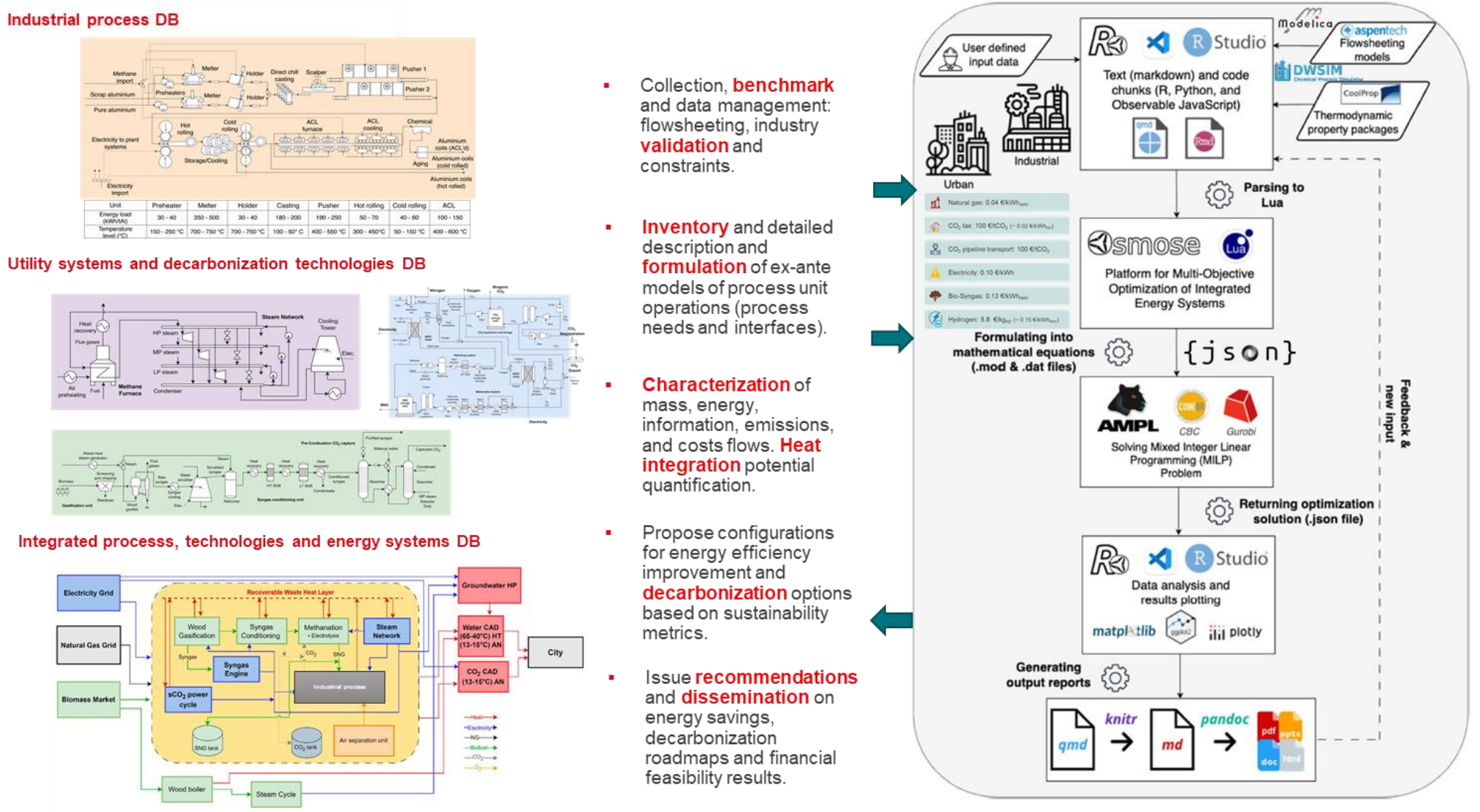


Figure B14: Proposed systematic framework to capitalize on the database of ex-ante models and operationalize data analysis and visualization tools for generating comprehensively documented reports [30–32].

Appendix C. Coordination meetings and participants' survey: details

This appendix provides the detailed account of the coordination meetings and of the participants' survey summarized in Section 2.4.

C.1 Kick-off and intra-task plenary meetings

The kickoff meeting on March 14th 2025 marked the official launch of IETS Task XXIV activities, bringing together participants from over ten countries to establish a framework for collaborative model sharing and documentation. Task managers introduced the structure of Subtask 1, which aims to enable a shared database of ex-ante models of industrial processes, decarbonization technologies, and energy systems featuring sustainability metrics. This session outlined the different ways participants could contribute to the Task, spanning from development of ontologies and sharing protocols to dissemination through workshops and educational programs (Figure C1).

Participants were familiarized with the administrative framework of IETS Technology Collaboration Programme (TCP), including the reporting activities comprising bi-annual status reports for Executive Committee meetings, annual report contributions, and final report with executive summary, synthesis of findings, and compilation of contributions. The meeting also clarified the engagement procedure and Letter of commitment (Figure C2), in which Participant Organizations (POs) provide information about the date of notification and the organizations and enterprises composing their national consortia. Based on this template, the Task Managers (TMs) notify IETS Secretariat, which in turn contact the Contracting Party (CP, either a country or a sponsor) to formalize PO participation. Participants committed to dedicating at least 13 person-weeks per year, with the understanding that this time is shared with their own national research projects. It was also emphasized that POs must secure funding and availability to attend working meetings throughout the operating period (Figure C3).

Following the dissemination guidelines of the IETS TCP Secretariat, the kick-off meeting was announced on professional networks (LinkedIn[11]) and the recording is available on the YouTube channel[12].

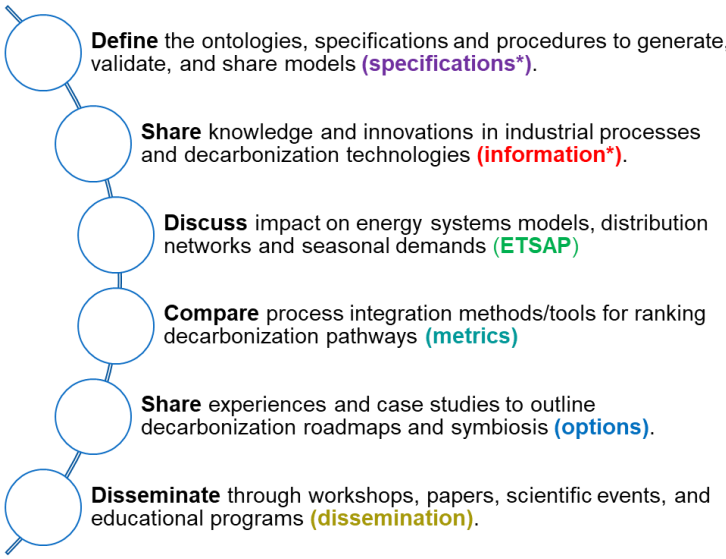


Figure C1: Contributions to Subtask 1 of Task XXIV by the participant countries.

[11] https://www.linkedin.com/posts/daniel-florez-orrego_process-integration-for-industry-decarbonization-activity-7281451644796469248-3RPV

[12] https://youtu.be/sFwzXBfyAgU

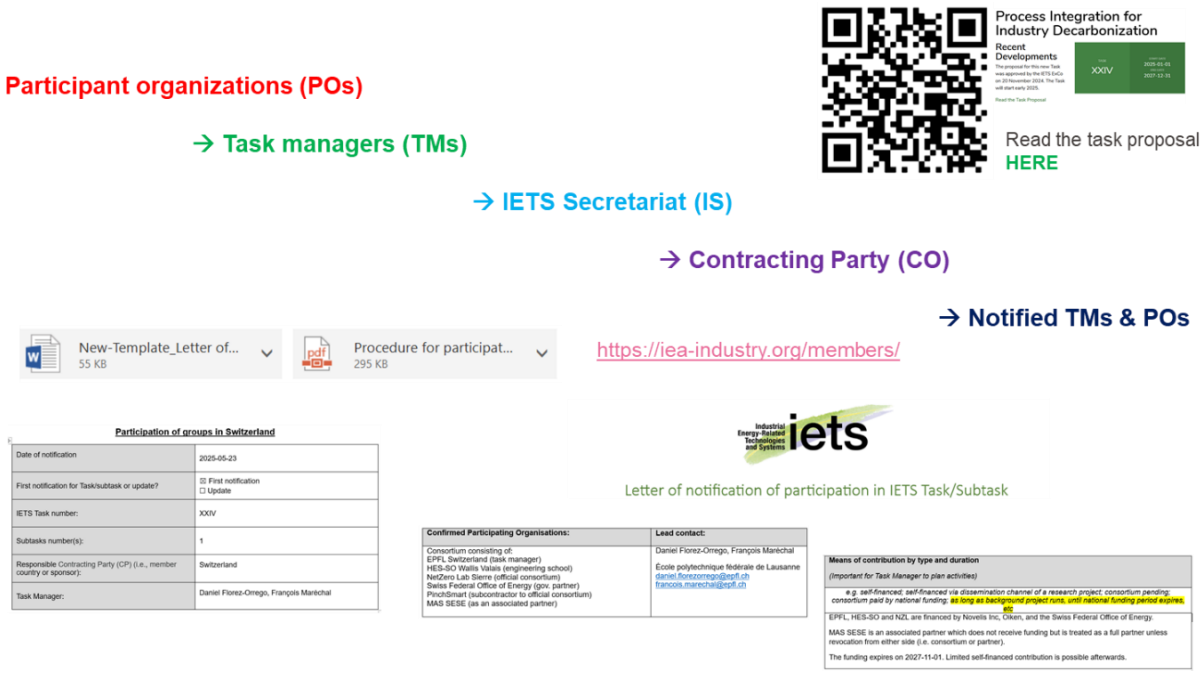


Figure C2: Engagement procedure and Letter of Commitment required for Participating Organizations to officially join Subtask 1 of Task XXIV.



Figure C3: Snapshots of the Task XXIV kick-off meeting on March 14th 2025.

The intratask discussions during the April and May meetings revealed the complex challenges surrounding model sharing in process integration. Participants openly discussed potential tensions between the aim for open collaboration and the need to protect intellectual property and confidential industrial data. Preliminary black, grey, and white box model classifications emerged as a practical solution, allowing

contributors to share models at different levels of detail depending on confidentiality constraints. These exchanges laid the groundwork for the layered disclosure protocols and anonymization strategies that would later be formalized in the template structure. During the meeting on April 11th 2025, Task XXIV participants presented in detail their areas of expertise, their experience with process modelling, integration and decarbonization assessments, and indicated their willingness to contribute across the following thematic areas:

- Industrial sectors and industrial process models
- Decarbonization technologies
- Process integration models
- Energy systems models
- Life cycle sustainability assessment models
- Roadmaps generation and dissemination

The EPFL presentation as Participant Organization highlighted existing databases and projects that could feed into a shared repository, including AIDRES, EPOS, MASBOOK, TRILATE, METHAREN, IEA ANNEX 58, NZL, and SCCR databases. This inventory of available models, spanning industrial processes from aluminium and steel to dairy and watchmaking, alongside decarbonization technologies, such as carbon capture, power-to-gas, heat pumps, and biomass conversion units, evidenced the substantial foundation upon which Task XXIV could build.

The Task XXIV participants' meeting on May 23rd 2025 transitioned from conceptual to concrete template development discussions, elaborating on ontologies and procedures for sharing models, the time required to translate and adapt models, and the number of shareable models used as starting points for identifying the connections (interfaces) that participants' models can offer to one another. Features such as metadata (authorship, software, sharing layer); mass, energy and emissions and information flows ("ports"); documented and referenced parameters and assumptions; description of main decision (input parameters) and derived (calculated) variables; simplified block flow diagrams; and supplementary material (flowsheets, publications, and other files) were identified as minimum information required to populate and validate the ex-ante models content. Those requirements were expanded in the consolidated model's template through the subsequent bilateral meetings and the participants' survey.

On April 10th 2025, Task XXIV participated in a joint online workshop co-organized by the IETS TCP and the Energy Technology Systems Analysis Programme (ETSAP), focusing on industrial sector modelling in energy system models[13]. The workshop, moderated by Etienne Bernier (Natural Resources Canada) and Anna Krook-Riekkola (ETSAP), explored the role of the energy systems in transitioning towards a decarbonized industry and promoted collaboration on modelling data collection. Task Manager Daniel Florez-Orrego, together with Task XXIV participants from Switzerland (Oktay Boztas, Meire Ribeiro-Domingos, Pullah Bhatnagar) and Belgium (Muhammad Salman) presented a talk titled "Process Integration for Industry Decarbonization: Enabling a Shared Database of Ex-Ante Models". The presentation showcased selected case studies on aluminium, olefins, sugar, and synthetic natural gas production, demonstrating how ex-ante models can support system-wide analyses considering energy system security, carbon management, electrical grid balancing, and advanced decarbonization pathways. The workshop concluded with a discussion on how ETSAP's economy-wide energy system modelling frameworks and IETS's process-level models could complement each other to improve industrial decarbonization modelling.

These meetings and events helped establish the procedural foundation for sustained international collaboration and underscore the Task's value proposition: not merely gathering industrial data but developing reliable models and transparent sustainability metrics that enable objective and reproducible assessments. By addressing participation procedures, clarifying responsibilities, mapping expertise, introducing the template structure, and planning dissemination activities, Task XXIV managers created an environment where participants could contribute despite their diverse institutional contexts and research

[13] https://iea-industry.org/events/event/joint-workshop-with-etsap/

approaches. These discussions also served as preparation for presenting Task progress to the IETS Executive Committee in May 2025. Moreover, they enriched the dissemination strategy for reaching broader audiences through subsequent plenary lectures and invited talks at major international conferences (EUBCE in Valencia, Spain; ESCAPE in Ghent, Belgium; and IBBCE in Lisbon, Portugal). These dissemination activities will be detailed in the following sections of this report.

C.2 Bilateral meetings

Between September and November 2025, Task managers conducted bilateral meetings with representatives from each participating country to clarify remaining questions about the Task procedures, shared template, and confidentiality concerns. These discussions provided deeper insights into participants' expectations, concerns, opportunities, and challenges, which informed the survey designed for the in-person meeting.

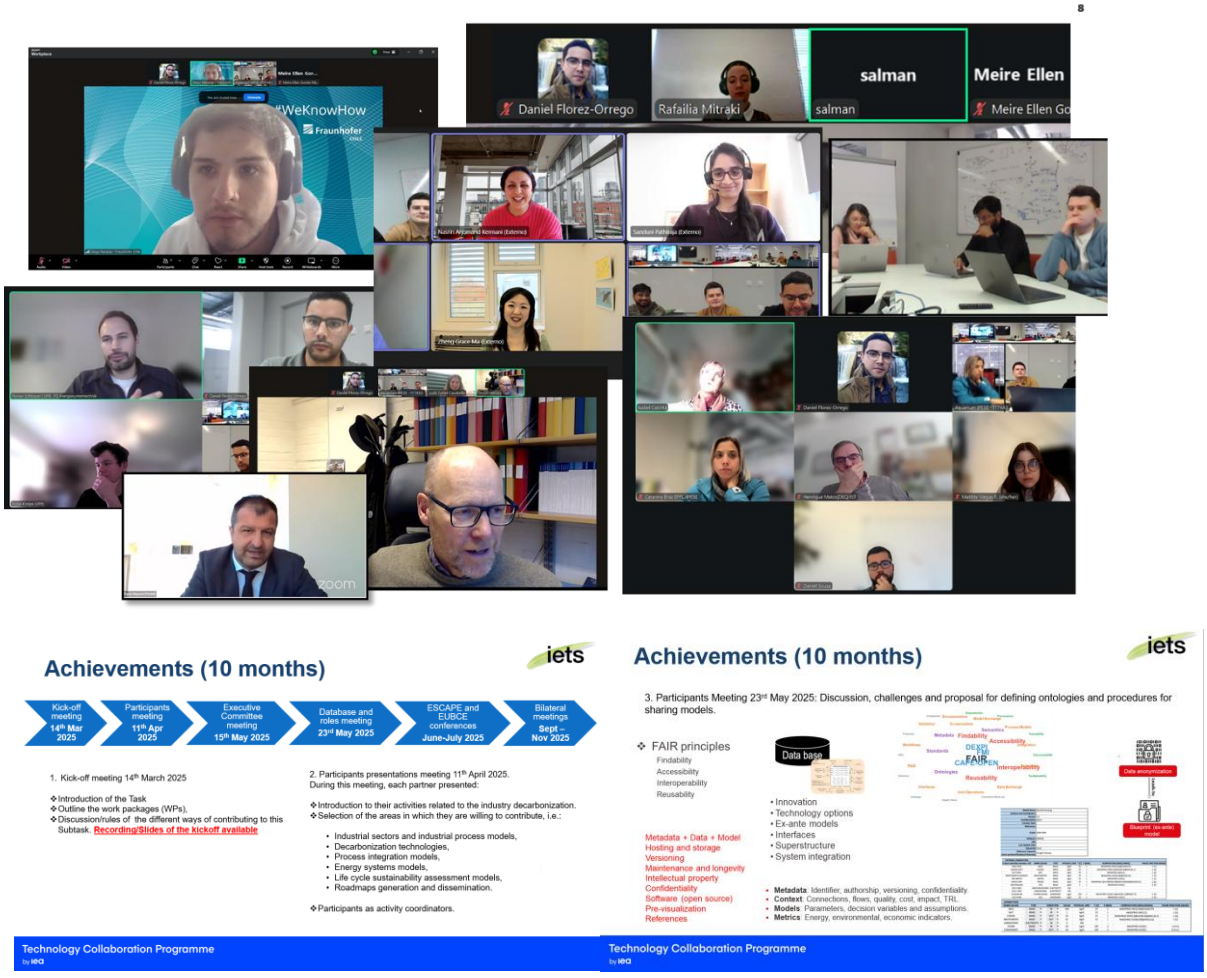


Figure C4: Snapshots of bilateral meetings with each Task XXIV participant country consortium held from September to November 2025.

Each country articulated distinctive perspectives: Chile emphasized data exchange policies between researchers and industrial partners; Belgium highlighted the need for different database-access interfaces for various stakeholders; Sweden addressed the impact of project confidentiality on model sharing between universities; Italy focused on model portability and compatibility across scales; Denmark questioned the added value and costs of additional documentation compared to journal publications; Portugal proposed validation first by sub-teams then by the broader community; Germany sought to achieve data anonymization while maintaining real-world relevance; Switzerland advocated maximizing open-access

data for public information; Netherlands carefully evaluated open sharing options for models built on proprietary software; and Colombia promoted model dissemination in broader educational platforms. These bilateral exchanges proved invaluable for understanding the diverse institutional contexts and refining the model-sharing framework

According to Figure C5, confidentiality and intellectual property protection are among the most prominent concerns of Task XXIV participants. Concepts such as non-disclosure agreements, embargo period, patents, intellectual property protection, proprietary software restrictions and licensing dominated the discussions, reflecting the conflict between the desire for open collaboration and the need to safeguard sensitive industrial data. Participants proposed practical solutions including layered access mechanisms, controlled access protocols, and anonymization strategies to balance transparency with data protection. The emphasis on approval processes and privacy considerations underscored the importance of establishing clear "rules of the game" that respect institutional and industrial constraints, while enabling meaningful model sharing.

A second focus is on validation and quality assurance of shared models. Participants stressed the need for peer-review, benchmarking, and certification to ensure that the ex-ante models accurately represent real-world data and industrial performance. Technical interoperability also featured with discussions addressing proprietary or open-source software, libraries and tools (Aspen, DWSIM, Python, Excel, Coolprop), modelling and simulation approaches (mathematical programming: linear, non-linear, integer; sequential modular or equation-oriented), and scientific foundations (thermodynamics, kinetics, heat integration, pinch analysis, multi-scale). The coexistence of black-box, grey-box, and white-box model classifications reflected the diversity of modelling paradigms across participant institutions and the need for flexible interfaces that accommodate different levels of detail and computational frameworks.

The bilateral meetings also highlighted the opportunities arising from international collaboration and knowledge dissemination. Participants emphasized the value of engaging industrial partners, stakeholders, policymakers, and sub-teams in building a shared repository that extends beyond academic research. Documentation emerged as a critical enabler, with metadata, parameters, and literature references ensuring reproducibility and traceability, building trust about the future results about decarbonization roadmaps. Additionally, the pedagogical and educational dimensions were recognized as important outcomes, with participants envisioning the database as a resource for students and continuing education programs. These discussions reinforced that Task XXIV's value lies not only in technical model sharing but also in fostering a collaborative network that bridges research, industry, and policymaker.

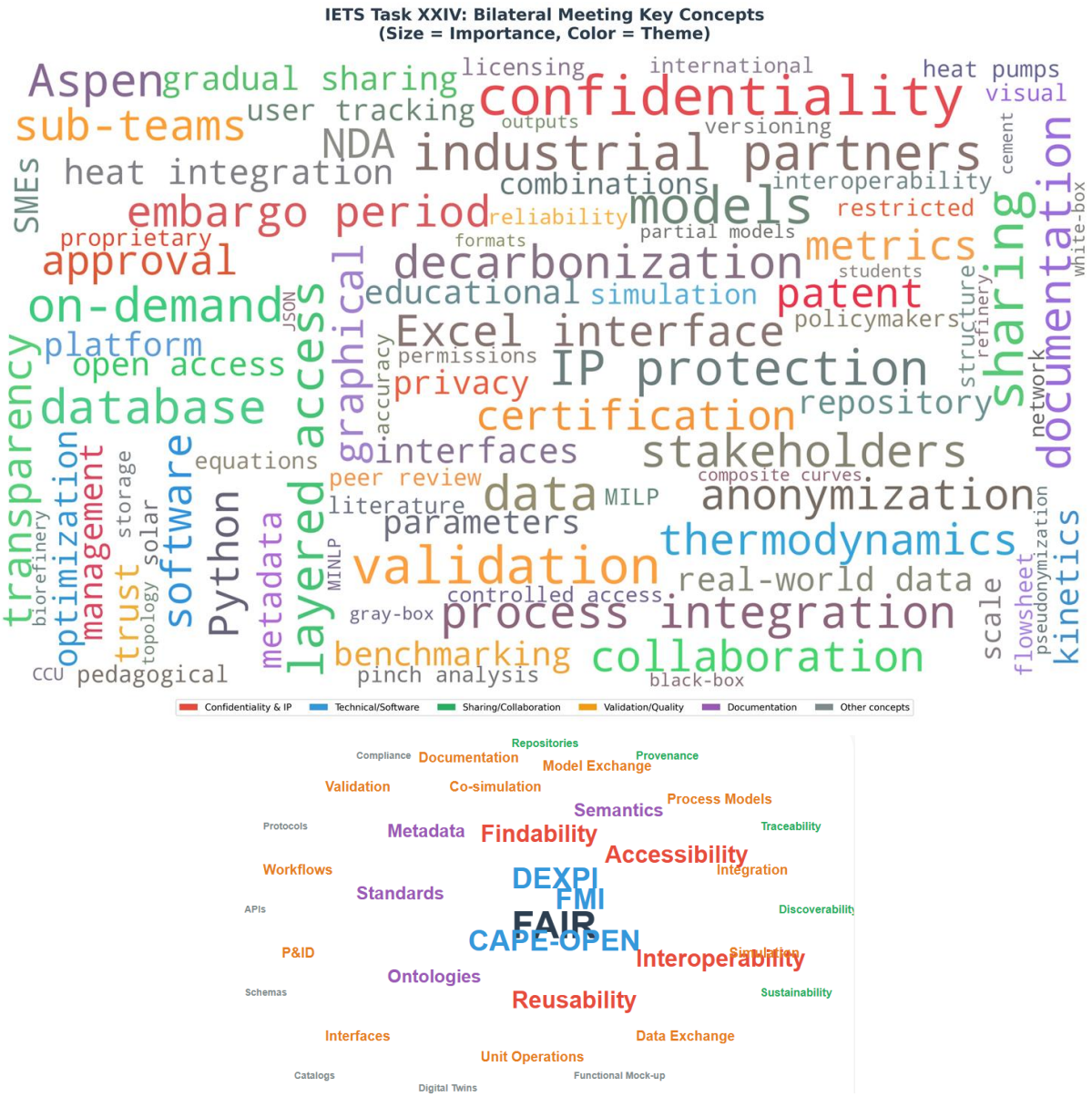


Figure C5: Key concepts emerging from bilateral meetings with Task XXIV participant countries, illustrating major concerns, expectations, and proposed approaches for the model-sharing framework.

C.3 Detailed results of the participants' survey

A comprehensive survey was designed to maintain momentum and gather structured feedback on the proposed template ahead of the first in-person intratask meeting. The survey (20-30 min.) comprised 57 questions organized into eleven thematic sections: (A) Participants and Institutional Information, (B) Models Inventory and Characteristics, (C) Software Platforms and Modelling Paradigms, (D) Feedback on Preliminary Template, (E) Confidentiality, Conflict of Interests, and Intellectual Property, (F) Attribution, Citation, and Co-authorship, (G) Data Sharing Layers and Access Levels, (H) Validation and Peer Review, (I) Database Features and Functionalities, (J) Incentives and Concerns, and (K) Closing Questions. This survey assessed expectations, motivations, and concerns regarding contributions to shared model databases and captured an overview of participants' modelling experiences about paradigms, structural approaches, and optimization methodologies. Practical questions addressed key implementation aspects, such as number of models used as a starting point, compatibility of current model structure and translation effort into a proposed 'simplified' template to accommodate diverse model practices

across participating countries [14]. Ten responses from institutions in nine countries (Switzerland, Denmark, Belgium, Italy, Sweden, Colombia, Chile, Portugal, and Germany) were received before 4 December 2025. All the responses were processed and summarized to be discussed in the first in-person meeting at the International Conference on Bioenergy and Bioeconomy (IBBCE 2025) held in December 5th 2025 in Lisbon, Portugal. The complete survey template is presented in the Annex of this report, and the main takeaways are presented next [15].

The industrial sectors and process ex-ante models emerged as the dominant research focus, followed by decarbonization technologies and process integration frameworks (Figure C6). Energy systems models and roadmap generation also featured prominently, confirming that Task XXIV community is expert in areas relevant to the shared database initiative. The participants can contribute detailed, sector-specific models covering manufacturing, chemicals, food, fuels production, waste treatment and heavy industry. Table 7 summarizes the models to be used as the starting point by the participant countries. Life cycle sustainability assessment, although less frequently mentioned, still represents a complementary domain for rational environmental impact evaluation of the shared models.

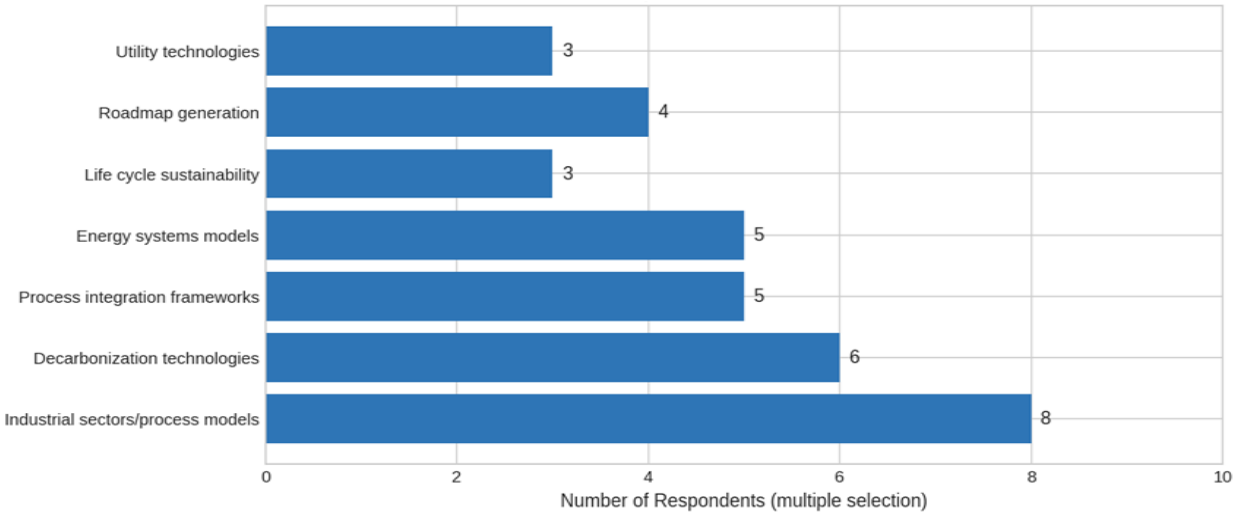


Figure C6: Primary research focus areas.

Regarding software and programming ecosystems (Figure C7), Python dominates software landscape, followed by Aspen® Suite, reflecting the importance of commercial process simulators for thermodynamic modelling. MATLAB® and Simulink and Excel/VBA-based tools also remain relevant. DWSIM, which is an open-source alternative to commercial simulators, and TRNSYS, for thermal energy systems analysis, round out these ecosystems. More specialized in-house tools, such as OSMOSE engine written in Lua language and using AMPL/GLPK optimization suites, were developed by EPFL consortia. This heterogeneity presents both challenges and opportunities for the database generation, which must accommodate multiple file formats and paradigms. The variety of optimization frameworks (AMPL, GAMS, GLPK), and solvers (CBC, CPLEX, Gurobi, Baron, Bonmin, Glpsol) also indicates the need for solver-agnostic model representation.

[14] Survey template https://forms.office.com/e/XdQ6CfqDRc
[15] Summary of survey results https://forms.office.com/Pages/AnalysisPage.aspx?AnalyzerToken=rU65F81etH5Re4w3HDUU4T3T2x6BPfDC&id=alXC9vvEsUqix54iDfEcQ0Uk5UPaNjdLgnY5gpa-MulUM0owNzQzSVNaUTNQWU85UVRORTBVVFVRRC4u

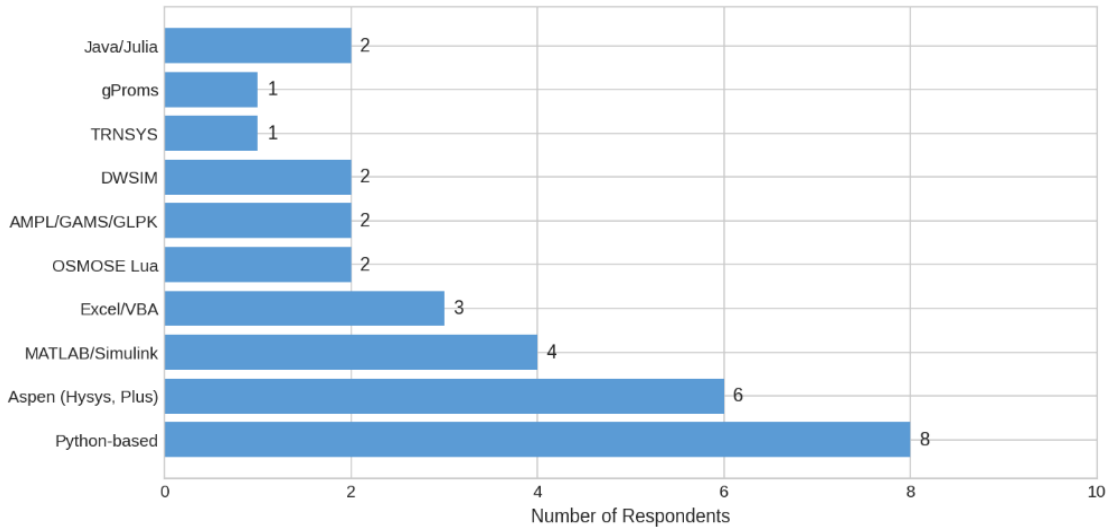


Figure C7: Software and programming ecosystems used by the participant countries.

The licensing distribution reveals a hybrid approach, as 70% of participants use both licensed and open-source software, while 30% rely mainly on licensed tools. Notably, no respondent operates exclusively with open-source solutions. This pattern reflects a reality of academic research, where commercial tools like Aspen® still provide specialized thermodynamic databases and validated unit operation models that remain difficult to replicate in open-source alternatives. DWSIM, Python and other related open tools offer flexibility for custom optimization and data processing. For the shared database, this hybrid reality necessitates strategies for handling proprietary file formats alongside open standards. The absence of purely open-source users suggests that license-free accessibility cannot be guaranteed for all shared models, reinforcing the importance of the multi-layered access framework and the potential need for developing surrogate models.

Version control practices also show variations across participants (Figure C8). Git-based system is used either as a standalone solution or combined with cloud storage (OneDrive, Google Drive), while a large fraction of the respondents relies solely on cloud storage for managing file history. A smaller fraction of participants uses spreadsheets as catalogs or do not track file changes using any tool. This fragmentation may pose challenges for model provenance and reproducibility. Documentation practices also vary, with embedding documentation as plain text within flowsheets or Python code files being the most common practice, followed by using separate Word/ODT documents, and fewer using Jupyter notebooks or advanced tools, like Quarto and Obsidian. The relatively low adoption of integrated documentation approaches (e.g., Jupyter, Quarto) suggests an opportunity for the database initiative to promote literate programming practices that would enhance model transparency and reusability.

Among survey participants, OneDrive dominates model storage practices, likely reflecting institutional Microsoft® agreements at European universities. GitLab and GitHub adoption indicates that certain participants are already comfortable with automated version-controlled repositories. Google Drive and institutional Network Attached Storage (NAS) solutions ranked among the least adopted storage options. Most respondents use multiple platforms, suggesting no single solution meets all needs. Moreover, in view of some participants' constraints, a centralized registry linking to distributed institutional repositories may align existing practices with other considerations, such as the physical location of the model database. A hybrid approach allows a centralized platform to standardize model descriptions and host the latest shareable versions, while external institutional repositories still could manage more detailed model development, confidential data, versioning, and history.

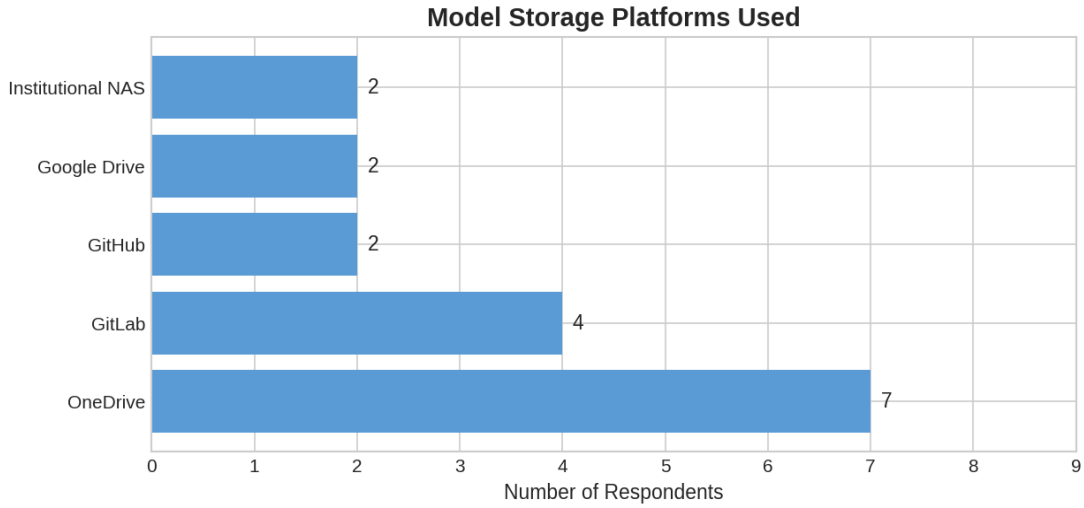


Figure C8: Model storage platforms currently in use by the participant countries.

In fact, confidential industrial data (e.g., NDAs, patents, and co-owned models) emerges as the most critical factor and significant barrier to open model sharing, reflecting the reality of industry-funded academic research (Figure C9). Unpublished research data follows closely, explained by academic publication pressures and intellectual property concerns. Provided that data originates from open sources and literature, the sharing of commercial technology specifications, model details (e.g., kinetics, equipment sizing), and economic assumptions faces relatively low barriers. Accordingly, the database should prioritize mechanisms for protecting industrial partnerships and preserving publication priority, while other modelling data may be easily shareable with appropriate anonymization. Other notable concerns include the financial burden of documentation effort, sharing process complexity, risk of model misuse, and quality validation. In contrast, the fear of exposing errors in one's own models and loss of competitive advantage ranked lowest, indicating that the research community is relatively comfortable with peer scrutiny and competence. The predominance of concerns around industrial partnerships and time-resource constraints suggests these areas require priority attention in the database governance framework.

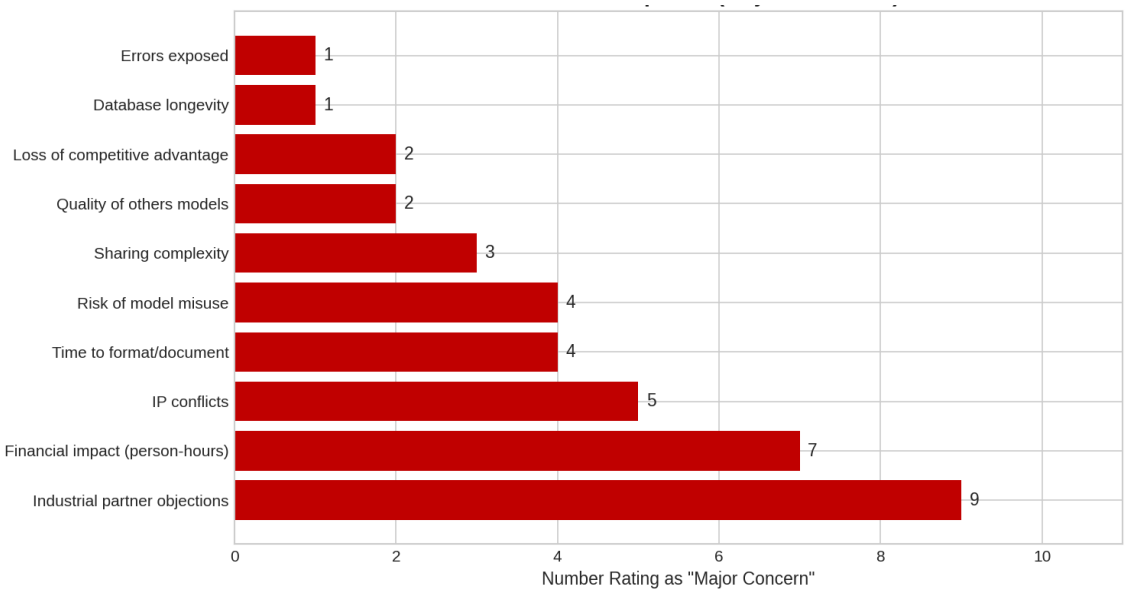


Figure C9: Major concerns and barriers to fully open participation.

Citation in academic publications that use shared models arises as the primary recognition expectation, consistent with established academic incentive structures. Acknowledgment within the models database and the mention in studies drawing upon open access initiatives were rated as moderately important. Notably, the expectations around co-authorship inclusion in derivative publications suggest this matter should be negotiated case-by-case, rather than mandated. Divided views on commercial compensation reflect differing perspectives on whether academic model sharing should generate revenue. Several respondents advocated for a recognition based on the level of contribution of a given model of theirs to any other derivative work (vital, adapted, or ancillary contribution).

Regarding intellectual property (IP) protection mechanisms, Creative Commons (CC) licensing, embargo periods aligned with journal publication timelines, tiered access levels, and patent protection before sharing were considered important by participating countries (Figure C10). In contrast, the application of non-disclosure agreements (NDAs) and academic use-only restrictions for model use and adaptation were rated as less important. The limited support for full open access confirms that unrestricted sharing, while philosophically appealing, does not align with the practical constraints most participants face. Support for embargo periods directly addresses the publication priority concerns identified among confidentiality factors. The comparable support for Creative Commons licensing and tiered access suggests the relevance of a hybrid approach, in which standardized licensing for basic access levels is combined with customized agreements for detailed model sharing.

In this regard, a multi-layered sharing framework was proposed to accommodate varying levels of data sensitivity (Table 8). Layer 0 (public metadata), Layer 1 (structure and topology), and Layer 2 (mass and energy balances) received nearly unconditional support from respondents (Figure 5). Disclosure of detailed thermodynamic data (Layer 3) and kinetic and reaction data (Layer 4) garnered conditional support. Detailed industrial, economic, and financial data (Layer 5) proved more sensitive, while full model details (Layer 6) emerged as the most contentious, with divided opinions on disclosure. The high conditional support rates for Layers 4–6 suggest that many participants would share this sensitive information under specific arrangements, such as tied to bilateral agreements, embargo periods, or access restricted to academic users. Critically, some participants (DTU, SDU, Fraunhofer CL) indicated stricter institutional or legal constraints. As an example, some partners “are not permitted to share code or executable models outside without a formal agreement reviewed and approved their legal office” and “under the Danish Copyright Act (Section 59), software and models developed at public research institutions belong to the employer institution”. Moreover, some models in Fraunhofer could be "distributed as compiled binaries or functional mock-up units (FMUs), rather than open-source code" to protect IP.

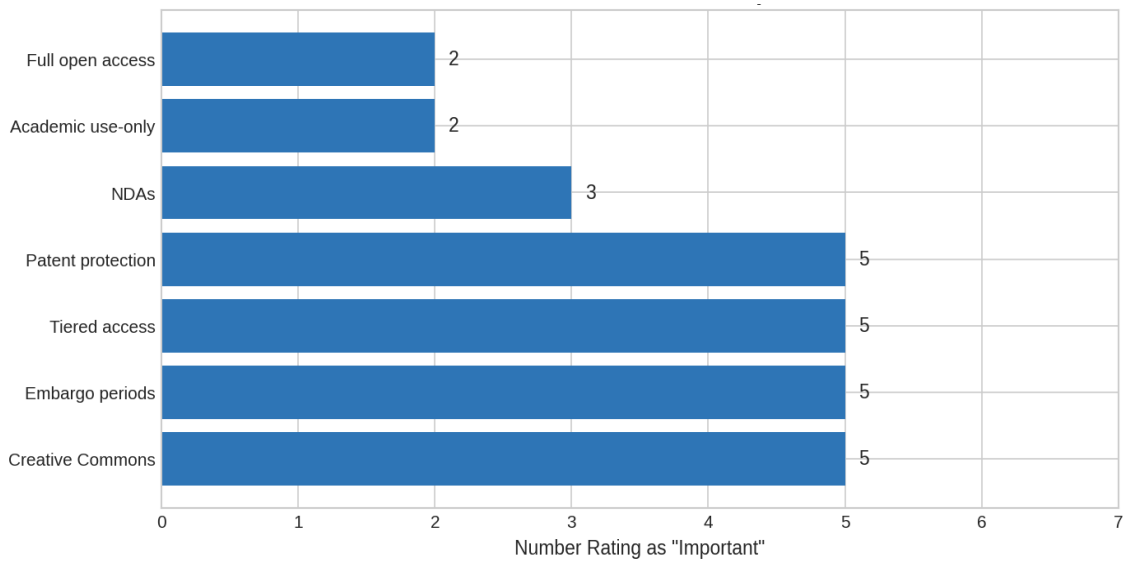


Figure C10: Intellectual property protection mechanisms regarded as important.

Sharing existing published models shows the strongest commitment, as published work has already cleared confidentiality and quality hurdles. Nevertheless, helping translate models to standard templates and validating other participants' models highlight the interest for the standardization work and the existence of peer review capacity (Figure C11). Developing new models specifically for the database receives more cautious response, recognizing the significant effort required. Sharing unpublished models and providing industrial validation data show the most hesitation, confirming that the database should initially prioritize published model incorporation, while building trust and frameworks to eventually incorporate unpublished and industrial data.

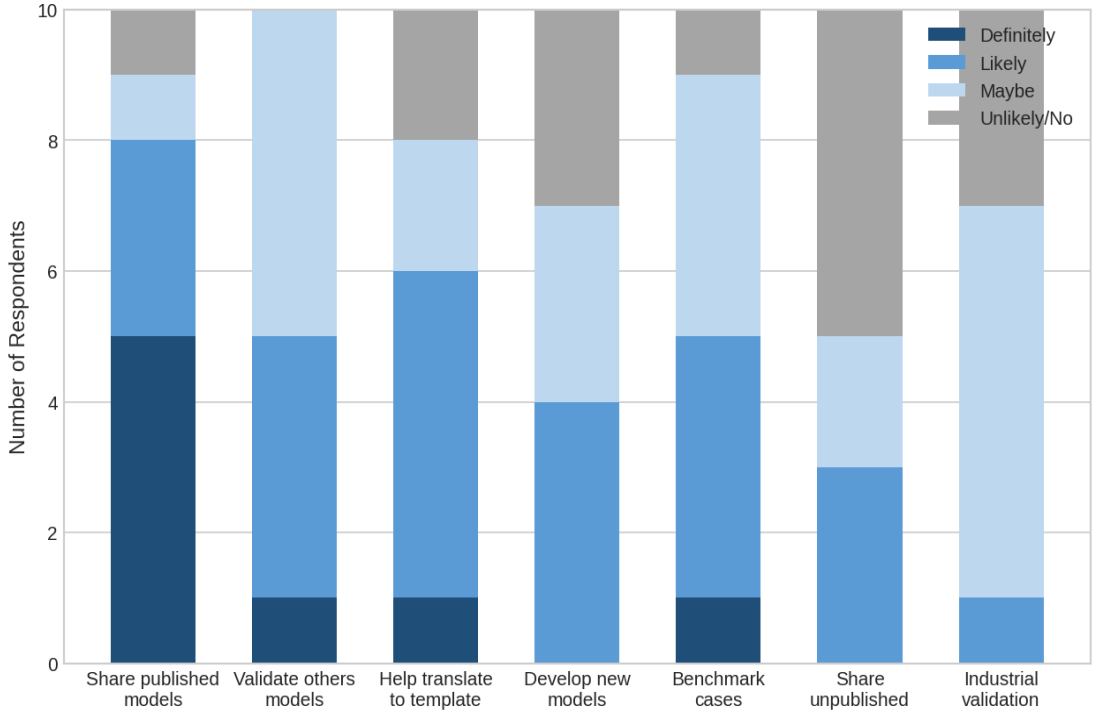


Figure C11: Breakdown of the contribution interests of the Task participants.

Regarding the adaptation of sensitive data for model sharing, the use of representative or synthetic data based on real data is the preferred anonymization technique, allowing models to retain realistic behavior, while obscuring actual plant-specific values and identifiers. Normalizing data to a reference capacity (per ton of mass or energy output) is perceived as a scale-independent representation, although operative ranges and economies of scale (for investment figures) must be carefully preserved to maintain model validity. Rounding numbers to reduce precision and providing uncertainty ranges (instead of absolute values) offer simpler anonymization steps suited for less sensitive data. Aggregating data from multiple sources (e.g., literature, specialists, suppliers) is also a viable statistical approach to blend plant-specific information, while preserving overall trends.

Currently, literature benchmarking stands as the most consistently applied validation method among participants, highlighting the barriers that the academic researchers face in accessing up-to-date industrial measurements. The absence of a documented and centralized model database for direct cross-validation leaves them with a handful of alternatives, including peer-review prior to publication, and comparison with available experimental, prototype or industrial plant data, or revision by industrial experts. These findings underscore an opportunity for Task XXIV to address this gap by establishing a framework that supports multiple anonymization and validation approaches. Model quality certification procedures received strong support, with participants favouring either open comparison among all Task participants or the creation of specialized sub-teams for validating specific model categories under stricter confidentiality precautions. Additional suggestions include maintaining validated and non-validated model versions, as well as defining standard validation metrics for automated quality and consistency testing.

These findings were presented and discussed during the first in-person Task XXIV meeting in Lisbon on 5 December 2025, providing an opportunity for participants to collectively interpret and prioritize implementation actions.

C.4 First in-person meeting and Annual Report preparation

The first in-person meeting of Task XXIV took place on December 5th 2025 at Higher Institute of Education and Sciences (ISEC), located at Alameda das Linhas de Torres 179, Room 1.2, Lisbon, Portugal, from 14:30 – 16:45 (CET), at the same venue of the International Conference on Bioenergy and Bioeconomy (IBBCE 2025). The agenda of this annual in-person meeting (Table C1) included a summary of key findings from bilateral meetings, an overview of survey results on the model-sharing framework, and short presentations from each participant country covering their shareable models, feedback on simplified data template, and concerns regarding confidentiality, citation, layered sharing proposal, and anonymization. The primary goal was to achieve a common understanding and agree on the "rules of the game" encompassing simplified ontologies, model structure, sharing processes, maintenance and peer-review procedures, expectations/concerns and confidentiality protocols. The meeting concluded with an explanation of the Annual Report contribution requirements and outlined next steps for the coming year's activities. For this meeting, it was not required yet to fill any template with participants' detailed modelling data, as it will be part of the activities in 2026. The recordings of the meeting are available in the YouTube Channel [16]. The session preparation and moderation were led by co-manager Dr. Daniel Florez-Orrego and EPFL's Task XXIV core support team Oktay Boztas (assistant engineer), Dr. Meire Ribeiro-Domingos (post-doc), Pullah Bhatnagar (PhD student), as well as co-manager Prof. François Maréchal.

Table C1: Agenda of the first in-person intratask meeting of Task XXIV participant countries.

Item	Duration	Responsible
Welcome	2 min	Daniel Florez-Orrego
Summary of key findings from bilateral meetings	5 min	Daniel Florez-Orrego
Overview of the survey's results on the model-sharing framework definition:		EPFL's Task XXIV core support team
• Template discussion and feedback (ontologies and shared "models" structure)	8 min	Daniel Florez-Orrego
• Models inventory and documentation framework (database features and deployment)	8 min	Oktay Boztas
• Procedures ("rules of the game", expectations/concerns and confidentiality issues)	8 min	Meire Ribeiro Domingos
Short presentations per participant country (6 min. each) Questions and open discussion	60 min	Diego Naranjo, Matilde Viegas Ferreira, Fernando Martins, Daniel Sousa, Pedro Mendes, Henrique Mattos, Carmen Bacariza, Rafailia Mitraki, Robin Kreipe, Florian Schlosser, Judit Fortet, Simon Harvey, Pablo Silva Ortiz, Lijun Wu, Frederik Wagner.
Break	5 min	
Explanation of the Annual Report contribution (5-page per country incl. dissemination activities). Proposed deadline: 19th Dec	5 min	Daniel Florez-Orrego
Conclusions	20 min	Daniel Florez-Orrego and François Maréchal
Next steps, future bilateral meetings, photos	5 min	Task XXIV participants
Any other business (AOB)	5 min	Task XXIV participants

Figures C12 and C13 show pictures of the First in-person meeting of Task XXIV participants and snapshots of their selected presentations.

[16] Recordings of the annual in-person meeting: https://youtu.be/IWhMKEcKnFQ?si=YHJ_bsOfy2EaWft8

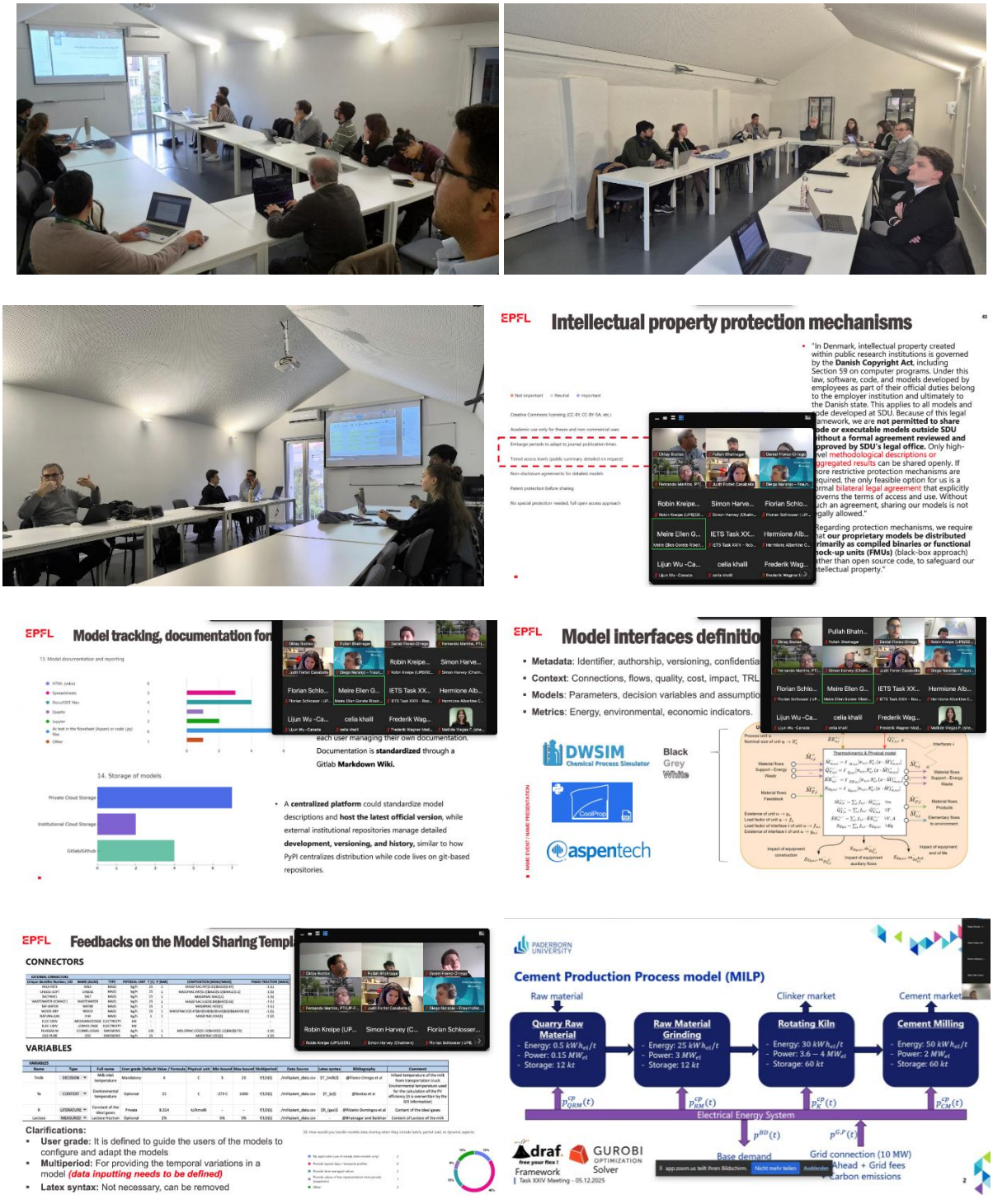


Figure C12: First in-person meeting of Task XXIV in December 5th 2025 carried out in the context of the International Conference on Bioenergy and Bioeconomy (IBBCE 2025) in Lisbon, Portugal.

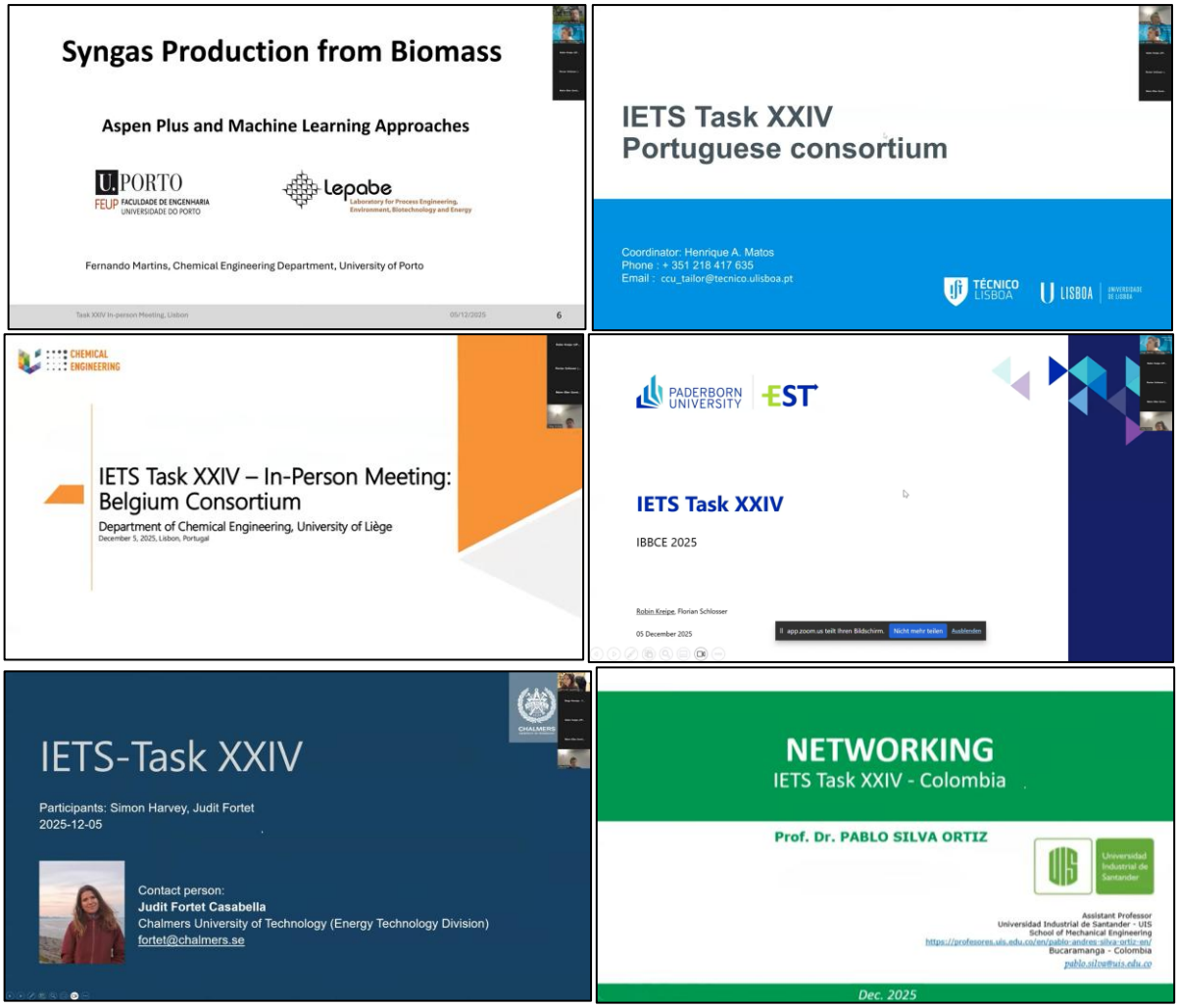


Figure C13: Snapshots of Task XXIV participants' presentations in the First in-person meeting in Lisbon, Portugal on December 5th 2025.

Participants were requested a brief contribution (minimum 5 and maximum 10 pages per participant country) to the Task XXIV Annual Report for 2025. They can share editable slides, document files, or LaTeX files, as well as PDFs containing (i) A brief description of the country's consortium (experience, research focus, collaboration with other Tasks, etc.); (ii) A list of conferences and events attended, as well as technical and journal publications related to Task XXIV objectives; (iii) Exemplary diagrams or snapshots illustrating their model structures, including links to model repositories (internal or external); (iv) Simplified workflow descriptions and illustrative screenshots of their optimization framework(s) (in-house or proprietary software); (v) Examples of output reports, documentation excerpts, and other relevant content from their modelling experiences, database management or layouts. Suggested deadline for this contribution was December 19th 2025.